

**Genomic and Molecular Analysis of a 17p13.3 Microdeletion
and Its Candidate Visual-Neurological Mechanisms**

Complete First-Pass In Silico Research Record, Phases 1-6

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Frozen scientific compilation, Version 1.0

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Abstract

This first-pass in silico project evaluated a heterozygous 17p13.3 microdeletion, GRCh37/hg19 chr17:1,411,408-1,841,103, for molecular and visual-neurological relevance. Phase 1 defined a 429,696-bp interval affecting at least 17 annotated loci, including 11 fully deleted protein-coding genes and major coding disruption of INPP5K and RTN4RL1. Phase 2 showed that the interval affects multiple molecular systems, with the strongest convergence in membrane, phosphoinositide, endosomal, retinal trophic, RNA-processing, and genome-maintenance biology; heterozygous deletion was not treated as uniform 50% functional loss. Phase 3 identified the clearest retinal convergence in an RPE-centered INPP5K-PITPNA-SERPINF1 module, with PRPF8 broadly represented across neural retina. Human dLGN data did not support magnocellular specificity, and the interval did not emerge as a major V1-specific developmental program. Phase 4 found the CNV biologically consequential, with strongest support for retinal/RPE maintenance, photoreceptor integrity, endolysosomal handling, stress response, and neural-development/network effects. A distributed functional-reserve model fit the evidence better than a single-gene or hyperresponsive-retina model. Phase 5 used deterministic integration and simulation to show that modest heterozygous effects may be missed with inadequate biological replication, multiple-frequency testing can produce false positives, and sparse data can misrepresent distributed mechanisms. Phase 6 integrated and froze Version 1.0. The final conclusion is that the deletion is a credible distributed visual-neural reserve lesion, while exact-CNV retinal physiology, thalamic transfer, V1 flicker responses, cortical synchronization, photoparoxysmal response, and visually triggered seizure causation remain unresolved.

Editorial and Provenance Note

This unified edition reformats the project's frozen scientific record into one APA 7-style document. The scientific conclusions of the individual phases are not reopened by this compilation. Original per-phase page headers, page numbers, and duplicate Works Cited sections are omitted; the consolidated APA reference list appears once near the end. Phase 3 is represented by its abstract/methods summary, research-through-3D record, and integrated expression/spatiotemporal analysis because those are distinct preserved components of the phase. For Phase 5, the available frozen project artifact is the Phase 5 v1.0 casual-reader summary; a separate full narrative Phase 5 master report is not present in the archived project files used for this compilation. The Phase 5 section therefore reproduces the available frozen summary without inventing missing results prose. The reader-facing Vocabulary and Formulas Appendix is included before the consolidated References. All coordinates, genotype states, evidence boundaries, and frozen conclusions retain their original project meaning.

Phase 1

Locus Characterization and Structural Definition of the 17p13.3 Microdeletion

1. Research Objective

Phase 1 establishes the structural genomic foundation for the larger investigation of an intersti-

tial 17p13.3 microdeletion and its possible downstream molecular, biochemical, neurological, and visual-system effects. The index deletion was reported as:

GRCh37/hg19 chr17:1,411,408–1,841,103

The objectives of this phase were to determine:

- 1. the exact genomic interval being studied;**
- 2. all identifiable genes and other annotated loci intersected by that interval;**
- 3. which protein-coding genes are fully versus partially deleted;**
- 4. whether the two breakpoint-intersected genes, INPP5K and RTN4RL1, lose actual protein-**
- coding sequence;
- 5. the relationship of each breakpoint to the current clinically standardized transcript;**
- 6. the approximate percentage of coding sequence removed from each breakpoint gene;**
- 7. which noncoding RNAs and pseudogenes are contained within the interval;**
- 8. which nearby loci should explicitly be excluded;**
- 9. and which structural questions remain unresolved because the reported clinical coordinates may**

not represent nucleotide-level chromosome junctions. This phase is deliberately restricted to structural genomic characterization. No visual, neurological, developmental, or other

phenotype is inferred from the presence of a gene inside the deletion. The final product of the phase is designated:

17p13.3 Deletion Map v1.0

2. Reproducibility Standard

A researcher reproducing this analysis should maintain a record of:

- database name;
- database or annotation release when available;
- access date;
- genome assembly;
- chromosome;
- genomic coordinates;
- transcript accession;
- transcript version;
- protein accession;
- strand;
- source URL or database accession;
- calculations performed;
- interpretation;
- and unresolved discrepancies.

The project should preferably use a directory structure such as:

```
Phase_1_Locus_Characterization/ | |——01_raw_coordinates/ |——02_gene_annotations/
|——03_transcripts/ |——04_clinvar_landmarks/ |——05_ccds/ |——06_non_coding_loci/
```

|—07_calculations/ |—08_figures/ |—09_source_archive/ |—README.txt A master

spreadsheet should contain at minimum:

Field

Description

Source

NCBI, Ensembl, ClinGen, etc.

Assembly

GRCh37, GRCh38

Chromosome

chr17

Start

Genomic start

End

Genomic end

Strand

+ or -

Gene

HGNC gene symbol

Gene type

Protein coding, ncRNA, pseudogene

Transcript

Accession.version

CNV state

Full / partial / outside

Coding affected

Yes / No / unresolved

Evidence

Source supporting classification

Access date

Date data were retrieved

Notes

Annotation discrepancies or caveats This prevents later phases from relying upon browser screenshots or undocumented manual interpretation.

3. Coordinate System and Genome Assembly

3.1 Freeze the genome assembly

The first requirement is to preserve the genome assembly exactly as reported. The index finding uses:

GRCh37 / hg19

For NCBI, the corresponding reference assembly is:

GRCh37.p13 — GCF_000001405.25

NCBI maintains GRCh37.p13 annotation resources, while Ensembl maintains a dedicated GRCh37 archive. The Ensembl GRCh37 gene annotation is based on an older frozen gene build, whereas some variation and phenotype information continues to receive updates. Consequently, Ensembl GRCh37 and current RefSeq/MANE annotations should not be expected to contain identical transcript versions. (ncbi.nlm.nih.gov) The working rule for this study is therefore: Do not mix GRCh37 and GRCh38 coordinates within a single structural calculation. If a source

provides only GRCh38 coordinates, those coordinates must either be excluded from the GRCh37 calculation or deliberately converted using a documented assembly-conversion method.

3.2 Preserve the original clinical coordinates

Create a source record containing: Assembly: GRCh37/hg19 Chromosome: 17 Start: 1,411,408 End: 1,841,103 Variant: deletion If the coordinates are treated as 1-based inclusive, the nominal interval length is:

$$1,841,103 - 1,411,408 + 1 = 429,696 \text{ bp}$$

or approximately:

429.7 kb

The clinical description of >430 kb should be preserved as reported rather than silently changed, because clinical microarray boundaries may reflect probe resolution, reporting conventions, or approximate interval designation. For calculations in this project, the explicit genomic coordinates take precedence over the rounded verbal size.

4. Coordinate-Convention Control

Coordinate conventions are a potential source of one-base replication errors. NCBI GFF3 annotation uses 1-based coordinates, with the start and end positions representing genomic feature boundaries. (ncbi.nlm.nih.gov) BED-based formats used by UCSC and many bioinformatics programs use a 0-based, half-open convention. (genome.ucsc.edu) Therefore the 1-based inclusive interval: chr17:1,411,408-1,841,103 would be represented in BED format as: chr17

17p13.3_index_deletion The original clinical coordinates should never be overwritten by this conversion. Instead preserve both: Clinical/1-based: chr17:1411408-1841103 BED/0-based

half-open: chr17 1411407 1841103 A peer attempting to reproduce the analysis should verify which convention is expected by every software package before entering coordinates.

5. Initial Browser Verification

Step 1 — Ensembl GRCh37

Open the Ensembl GRCh37 Archive and select the human GRCh37 assembly. Search: 17:1411408-1841103 The GRCh37 archive supports direct browsing of genes, transcripts, noncoding RNAs, structural variants, and regulatory annotations on the older assembly. (grch37.ensembl.org) Record:

- every visible protein-coding locus;
- every noncoding locus;
- strand orientation;
- apparent breakpoint-intersected genes;
- transcript IDs;
- and any loci lying very close to either boundary.

Do not yet interpret phenotype relevance. The purpose is to produce an initial candidate locus list.

Step 2 — NCBI Gene / Genome Data Viewer

Repeat the interval inspection using NCBI resources. For each candidate locus:

1. search the official gene symbol in NCBI Gene;
2. locate the Genomic context section;
3. select or identify the GRCh37.p13 row;
4. record the genomic start and end;
5. record strand;
6. record exon count where shown;
7. record RefSeq transcript and protein accessions.

The same gene should be checked independently in at least two annotation resources when it lies close to a deletion boundary.

6. Classifying Gene-Level CNV Overlap

For each gene with genomic interval:

$[Gs, Ge]$ and the deletion:

$[Ds, De]$ use the following rules.

Full deletion

A gene is completely encompassed if:

$Gs \geq Ds$ and

$Ge \leq De$

Partial deletion

A gene is partially affected if the two intervals overlap but at least one gene boundary extends outside the deletion. Formally:

$\max(Gs, Ds) \leq \min(Ge, De)$ while the full-deletion condition is not satisfied.

Outside

A gene is outside the deletion if:

$$Ge < Ds$$

or

$$Gs > De$$

These rules should be applied to genomic coordinates before any functional interpretation.

7. Identification of the Two Boundary Genes

Using this interval method, two protein-coding genes intersect the deletion boundaries:

- INPP5K at the lower-coordinate boundary;
- RTN4RL1 at the upper-coordinate boundary.

The remaining confirmed internal protein-coding genes fall completely inside the deletion. The next analytical problem is therefore not merely whether INPP5K and RTN4RL1 overlap the deletion, but whether the overlap includes coding sequence.

8. Selecting the Reference Transcript

8.1 Why MANE Select is used

Transcript selection is necessary because one gene may have multiple isoforms with different exons and coding boundaries. The Matched Annotation from NCBI and EMBL-EBI project (MANE) provides a standardized transcript pair in which the RefSeq and Ensembl/Gencode representatives are matched in exon structure and coding sequence. MANE Select is intended to provide one representative transcript per locus and is widely useful for clinical reporting and genomic research. (ncbi.nlm.nih.gov) For Phase 1, the rule is:

Use the current MANE Select transcript as the primary transcript-level reference

whenever one is available. Alternative transcripts should be preserved separately rather than substituted silently.

8.2 INPP5K reference transcript

Record: Gene: INPP5K RefSeq MANE Select: NM_016532.4 Ensembl MANE Select: ENST00000421807.7 Protein: NP_057616.2 CCDS: CCDS11004.1 Protein length: 448 aa The CCDS database confirms that NM_016532.4 and ENST00000421807.7 are accepted MANE Select members of CCDS11004.1 and encode a 448-amino-acid protein. (ncbi.nlm.nih.gov)

8.3 RTN4RL1 reference transcript

Record: Gene: RTN4RL1 RefSeq MANE Select: NM_178568.4 Ensembl MANE Select: ENST00000331238.7 Protein: NP_848663.1 CCDS: CCDS45569.1 Protein length: 441 aa NCBI identifies RTN4RL1 as a two-exon minus-strand gene and reports NM_178568.4 / NP_848663.1 as the validated RefSeq transcript/protein pair, with CCDS45569.1 as its consensus coding sequence. (ncbi.nlm.nih.gov)

9. INPP5K Breakpoint Mapping

9.1 Establish genomic orientation

NCBI places INPP5K on the minus strand. This is critical because transcript direction is opposite increasing chromosome coordinates. For a minus-strand gene: higher genomic coordinates = transcript 5' side lower genomic coordinates

= transcript 3' side The deletion begins at: 1,411,408 inside INPP5K. The deleted portion runs from that point toward the higher-coordinate end of the gene. Therefore: The index deletion removes the 5' portion of INPP5K.

10. Using ClinVar as Transcript Landmarks

ClinVar variant reports provide genomic locations, HGVS expressions, transcript information, protein changes, and assembly-specific mappings. ClinVar itself distinguishes calculated molecular consequence from experimentally demonstrated functional consequence,

which is important when interpreting these data. (ncbi.nlm.nih.gov) The method used here is not to infer pathogenicity from the variants. Instead, known variant coordinates are used as landmarks linking transcript position to GRCh37 genomic position.

10.1 Search procedure

For INPP5K:

1. open ClinVar;

2. search INPP5K NM_016532.4;

3. verify that the result uses transcript NM_016532.4;

4. locate the GRCh37 genomic mapping;

5. record both:

- HGVS cDNA coordinate;
- GRCh37 genomic coordinate;

6. compare that genomic coordinate with the deletion boundary.

The following landmarks were obtained:

Classification for structural Transcript position

GRCh37 position

mapping c.165

1,416,843

Deleted side c.277

1,413,078

Deleted side c.419

~1,412,607

Deleted side

~1,412,457

Deleted side c.554+16 c.793

~1,401,400

Retained side The scientific interpretation is independent of the clinical classification of those individual variants. The only question being asked is: Does this known transcript coordinate map inside or outside the deletion?

11. INPP5K Breakpoint Bracketing

Because: c.554+16 → deleted while: c.793 → retained the deletion breakpoint lies between those transcript regions. This establishes that the deletion extends beyond coding position c.554 but does not reach the region containing c.793. Direct CCDS/exon reconciliation further places approximately the first 666 coding nucleotides of NM_016532.4 on the deleted chromosome segment under the reported-coordinate model. Because the INPP5K protein contains 448 amino acids:

$$448 \times 3 = 1344$$

amino-acid-coding nucleotides. The fraction removed is therefore approximately:

$$1344 = 0.4955$$

or:

49.6%

Reproducible INPP5K result

Gene: INPP5K CNV state: Partial Strand: - Affected side: 5' Coding sequence affected: Yes Estimated amino-acid-coding sequence removed: ~49.6% Transcript: NM_016532.4 Model limitation: based on reported CNV coordinates

12. RTN4RL1 Breakpoint Mapping

12.1 Establish orientation

RTN4RL1 is also located on the minus strand. Its GRCh37 gene span is approximately: chr17:1,837,971-1,928,628 while the deletion ends at: 1,841,103 Therefore only the low-coordinate portion of the genomic gene span overlaps the deletion. Because the gene is on the minus strand, the low-coordinate segment corresponds to the 3' side of the transcript. Thus: The deletion removes the 3' portion of RTN4RL1.

13. Confirming RTN4RL1 Coding Loss

Repeat the ClinVar landmark procedure using: NM_178568.4 A useful landmark is: NM_178568.4:c.1056G>A p.Pro352= which maps to approximately: GRCh37 chr17:1,840,060 Because:

$$1,411,408 < 1,840,060 < 1,841,103$$

this coding position lies inside the deleted interval. Therefore: RTN4RL1 coding sequence is definitively intersected. This conclusion does not depend upon whether the variant itself is pathogenic, benign, or synonymous. The variant is simply being used as a coordinate marker.

14. RTN4RL1 Coding-Overlap Calculation

CCDS45569.1 contains the consensus coding sequence corresponding to the 441-amino-acid RTN4RL1 protein. Under the reported-coordinate model, approximately: 1,313 of 1,326 CCDS nucleotides fall on the deleted side of the breakpoint. The estimated CCDS loss is therefore:

$$1326 = 0.9902$$

or approximately:

99.0%

Reproducible RTN4RL1 result

Gene: RTN4RL1 CNV state: Partial Strand: - Affected side: 3' Coding sequence affected: Yes Estimated CCDS nucleotide sequence removed: ~99.0% Transcript: NM_178568.4 Model limitation: based on reported CNV coordinates This illustrates why the percentage of the entire genomic gene span deleted is not a reliable proxy for the fraction of coding sequence disrupted.

15. Direct GFF3 Validation

The browser and ClinVar procedure should be independently checked against an assembly-matched annotation file. NCBI provides GFF3 annotations for annotated genome assemblies, and GFF3 start/end positions are represented as 1-based genomic coordinates. (ncbi.nlm.nih.gov) For GRCh37, use an annotation corresponding to: Assembly: GRCh37.p13 Accession: GCF_000001405.25 The NCBI GRCh37 annotation report can be used to document the annotation release associated with the downloaded file. (ncbi.nlm.nih.gov)

15.1 Manual GFF3 procedure

Download the GRCh37.p13 GFF3 annotation and search for: NM_016532.4 and: NM_178568.4 Extract every row where feature type is: exon or: CDS Record:

- chromosome accession;
- feature type;
- start;
- end;
- strand;
- phase;
- transcript accession.

Create one table per transcript. Example:

Transcript

Feature

Start

End

Strand

Overlap with CNV

NM_016532.4

CDS

X

Y

–

Full/Partial/None

NM_016532.4

CDS

X

Y

–

Full/Partial/None The exact rows should be preserved in the project archive.

16. Computational CDS-Overlap Calculation

For each coding segment:

$[Cs, Ce]$ and deletion interval:

$[Ds, De]$ calculate:

$Overlap = \max(0, \min(Ce, De) - \max(Cs, Ds) + 1)$ Then sum across all CDS segments:

$$L_{deleted} = \sum$$

$$Overlap_i$$

i Calculate total coding length:

$$LCDS = \sum$$

$$(Ce_i - Cs_i + 1)$$

i Then:

$$\%CDS \text{ deleted} = L_{deleted}$$

$$\times 100$$

$LCDS$ If reporting amino-acid-coding sequence, exclude the stop codon from the denominator. If reporting CCDS nucleotide sequence, explicitly state whether the stop codon is included. This distinction explains why the INPP5K result is reported as approximately 49.6% of amino-acid-

coding sequence, whereas the RTN4RL1 result is described as approximately 99% of CCDS se-

quence.

17. Optional BEDTools Replication

A researcher using BEDTools may create the deletion BED file: chr17

index_deletion and convert the relevant CDS annotations into BED coordinates. Then

use: `bedtools intersect \ -a transcript_CDS.bed \ -b index_deletion.bed \ -wao` The output

provides the number of overlapping bases for each CDS feature. Important: GFF3 coordinates

must be converted correctly from 1-based inclusive to 0-based half-open BED coordinates before

performing this analysis. Failure to convert coordinate conventions consistently will produce boundary errors.

18. IGV Visual Quality-Control Procedure

IGV may be used as an independent visual check.

Procedure

1. open IGV;

2. select the hg19/GRCh37 human genome;

3. navigate to:

chr17:1,350,000-1,950,000

4. load the deletion as a BED track;

5. display gene/transcript annotation;

6. zoom into the lower breakpoint around:

chr17:1,411,408

7. zoom into the upper breakpoint around:

chr17:1,841,103

8. confirm visually that:

- INPP5K crosses the lower deletion boundary;
- RTN4RL1 crosses the upper deletion boundary;
- the internal genes lie within the deletion;
- RTN4RL1-AS1 lies outside the interval.

IGV should be treated as visual quality control, not as the sole source of numeric coordinates. Numeric calculations should derive from exported annotation data.

19. Census of Fully Deleted Protein-Coding Genes

After establishing the breakpoint genes, query every locus between the two boundaries.

For each candidate:

1. retrieve its GRCh37 gene coordinates;
2. apply the full/partial/outside equations;
3. independently confirm boundary-near genes;
4. record gene type;
5. avoid phenotype interpretation at this stage.

The final confirmed protein-coding inventory is:

Gene

Structural classification

INPP5K

Partial; 5' coding disruption

Fully encompassed

PITPNA

SLC43A2

Fully encompassed

SCARF1

Fully encompassed

RILP

Fully encompassed

PRPF8

Fully encompassed

TLCD2

Fully encompassed

Fully encompassed

WDR81**SERPINF2**

Fully encompassed

SERPINF1

Fully encompassed

SMYD4

Fully encompassed

RPA1

Fully encompassed

RTN4RL1

Partial; 3' coding disruption Thus:

11 fully encompassed protein-coding genes

2 partially coding-disrupted protein-coding genes

13 affected protein-coding loci total

20. Noncoding RNA Census

A protein-coding-only search is insufficient for complete structural characterization.

Repeat the interval query while including:

- lncRNA;
- antisense RNA;
- microRNA;

- host RNA genes;
- pseudogenes;
- other annotated RNA loci.

20.1 PITPNA-AS1

NCBI identifies PITPNA-AS1 as a validated antisense noncoding RNA locus. Relevant validated transcript: NR_028514.1 Coordinate comparison places the locus completely inside the deletion. Classification: FULL

20.2 MIR22HG

Record: Gene: MIR22HG GRCh37: chr17:1,614,798-1,619,504 Strand: - Type: noncoding RNA host gene Validated transcripts include: NR_028503.2 NR_028504.2 NR_028505.2 All genomic coordinates fall inside the deletion. Classification: FULL

20.3 MIR22

Record: Gene: MIR22 GRCh37: chr17:1,617,197-1,617,281 Strand: - Type: microRNA RefSeq: NR_029494.1 Classification: FULL No phenotype interpretation is assigned at this stage.

21. Pseudogene Census

An expanded interval search identified: RN7SL105P with GRCh37 coordinates: chr17:1,507,668-1,507,966 The locus lies completely inside the deletion. Classification: FULL Its biological relevance to the study is currently: UNSUPPORTED / UNKNOWN It should nevertheless be retained in the structural map because Phase 1 records what is physically deleted, not merely what is presently believed to be clinically significant.

22. Explicit Negative Boundary Check: RTN4RL1-AS1

Every boundary analysis should also document the nearest loci that are not deleted. NCBI places: RTN4RL1-AS1 at approximately: GRCh37 chr17:1,921,010-1,924,000 The index deletion ends at: 1,841,103 Therefore:

$$1,921,010 > 1,841,103$$

and RTN4RL1-AS1 lies approximately 80 kb beyond the index interval. Classification: OUTSIDE This negative result is important because larger 17p13.3 deletions can encompass RTN4RL1-AS1. A peer should therefore not copy gene lists from broader “17p13.3 deletion” reports without performing coordinate-level comparison.

23. Minimum Structural Inventory

After combining protein-coding, noncoding, and pseudogene results, the minimum confirmed affected locus inventory is:

Protein-coding

Partial 1. INPP5K 2. RTN4RL1

Full 3. PITPNA 4. SLC43A2 5. SCARF1 6. RILP 7. PRPF8 8. TLCD2 9. WDR81 10. SERPINF2 11. SERPINF1 12. SMYD4 13. RPA1

Noncoding

14. PITPNA-AS1

15. MIR22HG

16. MIR22

Pseudogene

17. RN7SL105P

Explicitly excluded

RTN4RL1-AS1 Therefore:

At least 17 structurally confirmed loci are affected by the index deletion under the

present annotation scope. The phrase at least is necessary. This inventory does not yet constitute a complete regulatory map.

24. Transcript-Version Quality Control

The project identified discrepancies between current MANE annotations and the historical Ensembl GRCh37 gene build. This is expected. Ensembl states that the GRCh37 gene annotation is based on its older frozen annotation rather than the continuously updated main human gene annotation. (grch37-mart.ensembl.org) For example:

INPP5K

Current: NM_016532.4 ENST00000421807.7 Historical GRCh37 archive representation:
NM_016532.3 ENST00000421807.2

RTN4RL1

Current: NM_178568.4 ENST00000331238.7 Historical GRCh37 archive representation:
NM_178568.2 ENST00000331238.6 MANE itself notes that MANE transcripts are defined against GRCh38, while mappings are made available for GRCh37 and other assemblies for

clinical interpretation. (ncbi.nlm.nih.gov) The replication rule is therefore: Record transcript accession and version exactly.

Do not assume that the tran-

script displayed by an older GRCh37 browser is the latest clinically standardized

transcript.

25. ClinGen Dosage Data: Record but Do Not Interpret Yet

Once the structural map is frozen, a researcher may record ClinGen dosage-sensitivity status for each protein-coding gene. ClinGen evaluates evidence for haploinsufficiency and triplosensitivity at gene and genomicregion levels. (clinicalgenome.org) For Phase 1, however, dosage information should be recorded as metadata only. Example fields:

Gene

HI score

TS score

Curation date

Evidence status Functional prioritization based on those scores belongs to Phase 2. This prevents Phase 1 from selecting genes merely because they fit an anticipated phenotype.

26. Breakpoint-Resolution Limitation

The strongest limitation in Phase 1 concerns the meaning of the clinical CNV endpoints. The calculations treat: chr17:1,411,408-1,841,103 as the operational deleted interval. If these values originated from chromosomal microarray analysis, they may represent the positions of the last informative probes or another assay-defined boundary rather than the exact DNA junction. Therefore the analysis establishes:

feature overlap at the resolution of the reported CNV

but not necessarily: the exact nucleotide at which the chromosome physically broke and rejoined. Consequently, statements such as: ~49.6% INPP5K coding loss ~99.0% RTN4RL1 CCDS loss must be qualified: under the reported-coordinate model. Nucleotide-resolution confirmation would require an assay capable of identifying the physical breakpoint, such as appropriate whole-genome sequencing, targeted breakpoint sequencing, long-read sequencing, or another validated breakpoint-resolution method.

27. Independent Peer-Replication Checklist

A peer should be able to reproduce Phase 1 without seeing the project's conclusions beforehand. The recommended blinded replication sequence is:

A. Freeze input

☐Record GRCh37 ☐Record chr17:1411408-1841103 ☐Record original clinical size description separately

B. Verify assembly

☐Confirm GRCh37.p13 / GCF_000001405.25 ☐Do not substitute GRCh38 coordinates

C. Census loci

☐Search Ensembl GRCh37 ☐Search NCBI ☐Create independent gene list ☐Include ncRNAs and pseudogenes ☐Record strand orientation

D. Classify overlap

☐Full ☐Partial ☐Outside

E. Resolve breakpoint transcripts

☐Identify MANE Select ☐Preserve accession.version ☐Record CCDS ☐Record protein length

F. Map INPP5K

- ☐ Confirm minus strand ☐ Verify ClinVar transcript landmarks ☐ Extract CDS coordinates
- ☐ Calculate CDS overlap ☐ Compare with ~49.6%

G. Map RTN4RL1

- ☐ Confirm minus strand ☐ Verify c.1056 landmark ☐ Extract CDS coordinates ☐ Calculate CDS overlap ☐ Compare with ~99%

H. Negative controls

- ☐ Confirm RTN4RL1-AS1 is outside ☐ Check other loci immediately outside both boundaries

I. Annotation reconciliation

- ☐ Compare current MANE against GRCh37 Ensembl ☐ Record version discrepancies rather than replacing them

J. Archive

- ☐ Save GFF/GTF files ☐ Save tables ☐ Record database release ☐ Record access date ☐ Save calculation sheet or script
- If the peer's result differs, the discrepancy should first be investigated for:

- 1. assembly mismatch;**
- 2. transcript-version mismatch;**
- 3. coordinate-convention mismatch;**
- 4. inclusive versus half-open interval treatment;**
- 5. alternative transcript selection;**
- 6. annotation-release differences.**

Only after these are excluded should the biological interpretation itself be questioned.

28. Replication Output Expected

A successful independent replication should approximately recover:

Feature

Expected result

Assembly

GRCh37/hg19

CNV

chr17:1,411,408–1,841,103

Nominal span

429,696 bp

INPP5K

Partial, 5' coding disruption

INPP5K coding loss

~49.6% under coordinate model

RTN4RL1

Partial, 3' coding disruption

RTN4RL1 CCDS loss

~99% under coordinate model

Full coding genes

Total affected coding genes

Confirmed ncRNA genes

Confirmed pseudogene

Minimum loci

RTN4RL1-AS1

Outside A replication does not need to reproduce every historical transcript ID identically if annotation databases have changed after the original analysis. It must instead document the annotation version used and explain any discrepancy.

29. Evidence Classification

Structurally Established

The following findings are supported at the resolution of the reported CNV:

- INPP5K is partially deleted.
- INPP5K protein-coding sequence is affected.
- The deleted INPP5K sequence lies on the transcript's 5' side.
- Approximately 49.6% of the MANE amino-acid-coding sequence lies within the operational deletion interval.
- RTN4RL1 is partially deleted.
- RTN4RL1 protein-coding sequence is affected.
- The deleted RTN4RL1 sequence lies on the transcript's 3' side.
- Approximately 99% of its CCDS nucleotide sequence lies within the operational deletion interval.
- Eleven internal protein-coding genes are completely encompassed.
- PITPNA-AS1, MIR22HG, and MIR22 are completely encompassed.
- RN7SL105P is encompassed.
- RTN4RL1-AS1 is outside the deletion.
- At least 17 annotated loci are structurally affected.

Structurally Unresolved

Phase 1 does not establish:

- nucleotide-exact chromosome junctions;
- whether INPP5K produces a truncated RNA;
- whether RTN4RL1 produces a truncated RNA;
- nonsense-mediated decay;
- transcript stability;
- altered protein production;
- promoter activity;
- enhancer effects;
- compensatory expression from the intact allele;
- tissue-specific dosage effects.

Phenotypically Unresolved

Phase 1 does not establish:

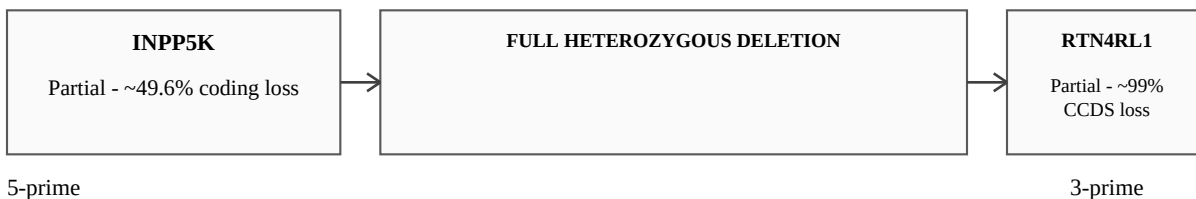
- visual stress;
- photophobia;
- magnocellular dysfunction;
- retinal dysfunction;
- cortical hyperexcitability;
- seizures;
- hearing loss;
- developmental delay;
- intellectual disability;

- congenital heart disease causality;
- or any other genotype–phenotype relationship.

These are deliberately deferred.

30. Phase 1 Final Structural Model

The deletion can now be represented as:



This diagram describes the affected chromosome only. The homologous chromosome is not represented and may retain intact copies unless additional variation is present.

31. Phase 1 Conclusion

Phase 1 establishes the index CNV as a structurally defined interstitial 17p13.3 deletion rather

than a generic 17p13.3 syndrome interval. The most important structural findings are that both breakpoint genes lose coding sequence and that their functional coding disruption is substantially different from what genomic-span overlap alone would imply. **INPP5K** loses approximately half of its MANE protein-coding sequence from the affected chromosome under the reported-coordinate model. **RTN4RL1** presents an even more pronounced discrepancy between genomic-span overlap and coding disruption: although the deletion intersects only a small proportion of the total genomic locus, it encompasses approximately 99% of the relevant CCDS nucleotide sequence. The internal interval completely encompasses at least eleven additional protein-coding genes, three established noncoding RNA genes, and one RNA pseudogene. The resulting minimum map contains:

17 affectedannotatedloci

No phenotype has been inferred from this finding. The purpose of Phase 1 is to establish a reproducible genomic substrate upon which later biochemical and functional analyses can operate.

32. Phase Status

PHASE 1 — COMPLETE

Frozen Structural Reference

Designation: 17p13.3 Deletion Map v1.0

Assembly: GRCh37/hg19

Interval: chr17:1,411,408–1,841,103

Future work must use this structural reference unless:

- 1. an amended clinical report changes the CNV coordinates;**
- 2. breakpoint-resolving sequencing identifies the physical junction;**
- 3. a demonstrable calculation error is identified;**
- 4. or a future annotation update materially changes transcript interpretation.**

If any of those occur, the reference should not be silently edited. It should instead become:

Deletion Map v1.1 or v2.0

with the change documented.

Phase 2

Molecular and Chemical Characterization of the Interstitial 17p13.3 Microdeletion

Abstract

An interstitial 17p13.3 microdeletion spanning GRCh37/hg19 chr17:1,411,408–1,841,103 removes approximately 429.7 kb and disrupts a minimum of 17 annotated loci. Phase

1 established that the deletion fully encompasses eleven protein-coding genes—PITPNA,

SLC43A2, SCARF1, RILP, PRPF8, TLCD2, WDR81, SERPINF2, SERPINF1, SMYD4, and RPA1—while partially disrupting INPP5K and RTN4RL1. Exon-level analysis demonstrated that the partial genomic designations substantially underestimate coding disruption: approximately 49.6% of the INPP5K amino-acid-coding sequence and approximately 99.0% of the RTN4RL1 CCDS are deleted under the reported-coordinate model. Four additional noncoding loci—PITPNA-AS1, MIR22HG, MIR22, and RN7SL105P—are completely encompassed.

Phase 2 systematically characterized the molecular functions, biochemical mechanisms,

structural evidence, loss-of-function biology, heterozygous dosage evidence, neural relevance, retinal/visual relevance, and experimental tractability of all affected loci. The principal finding is not the identification of a single candidate disease gene. Instead, the interval contains several independently characterized regulators of membrane lipid composition, phosphoinositide metabolism, endosomal maturation, intracellular neuronal transport, extracellular neuronal signaling, retinal trophic support, RNA processing, post-transcriptional regulation, epigenetic modification, DNA replication and repair, amino-acid transport, efferocytosis, and fibrinolysis.

The strongest multi-gene convergence involves INPP5K, PITPNA, TLCD2, WDR81,

RILP, and SERPINF1, which affect different levels of membrane and phospholipid

homeostasis. RTN4RL1 extends this system into cell-surface neuronal structural signaling. PRPF8 provides an independent spliceosomal mechanism with definitive evidence for PRPF8-related autosomal-dominant retinopathy, although the established retinal disease mechanism cannot be assumed to be equivalent to whole-gene haploinsufficiency. RPA1 provides the strongest direct evidence that heterozygous loss within the interval can produce a measurable human cellular phenotype: RPA1-containing deletions have been associated experimentally with reduced RPA1 abundance and impaired ATR-dependent DNA-damage responses, including functional rescue following RPA1 restoration (Clinical Genome Resource, “PRPF8-Related Retinopathy”).

The noncoding analysis identifies the MIR22HG/MIR22 unit as the most consequential regulatory locus outside the protein-coding set. Complete Mir22 loss exacerbates experimental epilepsy in mice, demonstrating neurological functionality while not establishing a human heterozygous seizure phenotype (Silva et al.).

Collectively, Phase 2 supports a model of multisystem molecular dosage disruption **with a prominent membrane/phosphoinositide/endosomal convergence, but does not**

establish a definitive clinical phenotype for the specific heterozygous deletion. The principal unresolved question is whether one-copy loss produces sufficiently large, tissue-specific and developmentally coincident perturbations to alter retinal or neurological physiology. This question provides the basis for subsequent spatiotemporal functional-genomic and pathway analyses.

Keywords: 17p13.3; microdeletion; copy-number variation; phosphoinositides; endosome;

membrane homeostasis; PRPF8; RPA1; SERPINF1; WDR81; MIR22; dosage sensitivity; retinal biology; neurodevelopment.

1. Introduction

Copy-number variants can affect biological systems in ways that are not adequately represented by simple gene-counting approaches. A deletion may remove entire genes, disrupt only selected exons, eliminate regulatory RNAs, alter dosage of multimeric complexes, or simultaneously perturb several mechanistically related pathways. Consequently, interpretation of a rare CNV requires more than identification of disease-associated genes located within the interval.

The deletion evaluated in this project is an interstitial 17p13.3 microdeletion reported on the GRCh37/hg19 assembly as:

chr17:1,411,408–1,841,103

corresponding to a nominal span of:

429,696 bp, or approximately 429.7 kb.

Phase 1 established a minimum inventory of 17 affected loci. Thirteen are protein-coding genes:

1. INPP5K
2. PITPNA
3. SLC43A2
4. SCARF1
5. RILP
6. PRPF8
7. TLCD2
8. WDR81
9. SERPINF2
10. SERPINF1
11. SMYD4
12. RPA1
13. RTN4RL1

Four additional affected loci are noncoding:

14. PITPNA-AS1
15. MIR22HG
16. MIR22
17. RN7SL105P.

Eleven of the protein-coding genes are completely encompassed. INPP5K and RTN4RL1 lie at the CNV boundaries. Detailed coding analysis showed that INPP5K loses approximately 49.6% of its amino-acid-coding sequence, including a large portion of the catalytic domain, while RTN4RL1 loses approximately 99.0% of its CCDS. These findings demonstrate why genomic overlap percentage alone is an inadequate measure of functional gene disruption.

The purpose of Phase 2 was therefore to move from structural annotation to molecular **characterization. The central question was:**

What molecular functions are disrupted by heterozygous dosage loss of the affected loci, and which of these functions provide experimentally testable biochemical mechanisms for subsequent visual-system and neurological analysis?

The analysis was deliberately phenotype-neutral at the outset. Genes were not prioritized because they appeared to fit a proposed clinical phenotype. Instead, every affected protein-coding gene was characterized independently, followed by parallel evaluation of the noncoding loci.

The resulting analysis reveals a deletion that affects multiple independent cellular systems but also contains a notable concentration of genes involved in membrane lipid **regulation, phosphoinositide biology, endosomal maturation, intracellular trafficking, and neuronal membrane signaling. The distinction between these two observations—multisystem**

disruption and local mechanistic convergence—is central to interpretation.

2. Analytical Framework

2.1 Structural state

Each locus was first classified according to the Phase 1 structural model.

The eleven fully encompassed protein-coding genes were treated as complete heterozygous gene losses.

INPP5K was classified as a major partial coding disruption because approximately 666 of 1,344 amino-acid-coding nucleotides are deleted under the reported-coordinate model.

RTN4RL1 was classified as a functionally near-complete coding disruption because approximately 1,313 of 1,326 CCDS nucleotides are deleted.

All four noncoding loci are completely encompassed.

This classification describes the affected chromosome only. The homologous chromosome 17 retains the second genomic copy unless another independent variant affects it.

2.2 Evidence categories

Each protein-coding gene was evaluated independently across the following dimensions:

- structural disruption;
- biochemical characterization;
- experimental structural evidence;
- experimental loss-of-function evidence;
- human heterozygous dosage evidence;
- neural relevance;
- direct retinal/visual relevance;
- chemical and experimental tractability.

These dimensions were intentionally maintained separately.

A protein may have exceptionally well-characterized biochemical activity but little evidence that heterozygous loss affects humans. Conversely, a gene may have comparatively modest mechanistic characterization while possessing direct human dosage evidence.

The noncoding loci were assessed using a modified framework emphasizing:

- RNA class;
- experimentally demonstrated regulatory activity;
- perturbation evidence;
- heterozygous dosage evidence;
- neural relevance;

- visual relevance;
- experimental tractability.

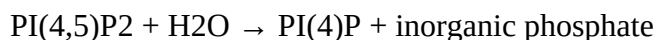
Protein-centered metrics such as catalytic domains, three-dimensional protein structures, pLI, and LOEUF were not mechanically transferred to lncRNAs, miRNAs, or pseudogenes.

3. Protein-Coding Locus Characterization

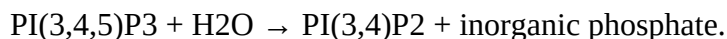
3.1 INPP5K: phosphoinositide 5-phosphate hydrolysis

INPP5K encodes an inositol polyphosphate 5-phosphatase. Its principal membrane substrates include PI(4,5)P₂ and PI(3,4,5)P₃.

The major reactions can be represented as:



and:



The reaction is Mg²⁺-dependent.

Phase 1 showed that the deletion removes approximately the first 666 coding nucleotides

of the canonical INPP5K coding sequence on the affected allele. Because INPP5K lies on the minus strand, this removes the 5' coding region. The deleted portion intersects the catalytic EEP domain and removes annotated catalytic residues, including residues involved in metal binding and proton-transfer chemistry.

A 2026 human INPP5K crystal structure resolved the catalytic domain at 1.9 Å in complex with the noncompetitive inhibitor CPD-1, establishing a structurally defined catalytic and allosteric system (Nomura et al.).

The deletion therefore cannot produce an intact canonical INPP5K catalytic domain from the affected allele under the reported-coordinate model. Whether the rearranged allele produces no transcript, an unstable transcript, or an aberrant protein remains unresolved.

INPP5K represents a strong biochemical candidate but weakly established human haploinsufficiency candidate.

3.2 PITPNA: phosphatidylinositol and phosphatidylcholine transfer

PITPNA encodes phosphatidylinositol transfer protein alpha, a soluble lipid-binding protein capable of transporting phosphatidylinositol and phosphatidylcholine between membrane environments.

PITPNA does not catalyze phosphate transfer or phospholipid hydrolysis. Its chemistry is instead based on reversible sequestration of a single phospholipid within a hydrophobic binding cavity.

Conceptually:

$\Rightarrow$

PI in membrane A + PITPNA



with an analogous cycle for phosphatidylcholine.

PITPNA therefore affects the availability of phosphatidylinositol from which phosphorylated phosphoinositides are generated.

The neurological importance of PITPNA is strongly supported in the vibrator mouse. Reduction of PITPNA expression to approximately 20% of normal causes tremor, brainstem and spinal-cord neurodegeneration, and premature death (Hamilton et al.).

The index CNV completely deletes one PITPNA copy.

The important distinction is that the severe vibrator phenotype represents much greater reduction than an ordinary heterozygous deletion. Nonetheless, the model demonstrates that PITPNA possesses dose-dependent neurological biology.

3.3 SLC43A2: essential neutral amino-acid transport

SLC43A2 encodes LAT4, a plasma-membrane facilitated-diffusion transporter for neutral amino acids including:

- leucine;
- isoleucine;
- valine;
- methionine;
- phenylalanine.

Transport is not directly ATP driven and is largely sodium independent.

The protein therefore introduces a nutrient-transport mechanism distinct from the membrane-signaling cluster.

Potential downstream consequences of altered SLC43A2 activity include changes in:

- essential amino-acid availability;
- leucine-sensitive mTORC1 signaling;
- methionine-dependent S-adenosylmethionine metabolism;
- general cellular anabolic state.

However, these are downstream possibilities rather than established consequences of heterozygous SLC43A2 loss.

Experimental loss-of-function models establish important placental, intestinal and renal transport roles, but current evidence for a direct heterozygous neurological or retinal phenotype is weak.

SLC43A2 should therefore remain an independent nutrient-transport axis rather than being forced into the principal neural convergence model.

3.4 SCARF1: scavenger recognition and efferocytosis

SCARF1 encodes a type-I scavenger receptor involved in recognition and removal of apoptotic material.

One important mechanism involves C1q acting as a molecular bridge between phosphatidylserine-rich apoptotic membranes and SCARF1.

The functional sequence is approximately:

apoptotic membrane

↓

C1q recognition

↓

SCARF1 engagement

↓

efferocytosis.

SCARF1-deficient mice develop defective apoptotic-cell clearance and lupus-like autoimmunity, establishing strong loss-of-function biology.

However, SCARF1 currently contributes little direct evidence to the visual-neurological hypothesis.

Its principal Phase 2 significance is the demonstration that the deletion also affects an immune-clearance and damage-recognition pathway.

3.5 RILP: RAB7-dependent late-endosomal transport

RILP encodes Rab-interacting lysosomal protein, an effector of active RAB7A.

The relevant interaction is:

$\Rightarrow$

RAB7A-GTP + RILP

RAB7A-GTP:RILP complex.

RILP does not hydrolyze GTP. Instead, it recognizes the active GTP-bound state of RAB7A and connects late endosomes and lysosomes to:

- dynein;
- dynactin;
- HOPS;
- additional trafficking machinery.

This enables minus-end-directed intracellular transport and perinuclear positioning of late endosomal compartments.

Recent neuronal work has strengthened the relevance of this system substantially. Interference with RILP-RAB7A-mediated late-endosome transport impairs dendritic cargo movement and dendrite arborization (Yap et al.).

RILP therefore provides direct mechanistic evidence that the deletion affects neuronal intracellular trafficking, although human haploinsufficiency remains weakly characterized.

3.6 PRPF8: spliceosomal RNA processing

PRPF8 encodes one of the central structural and regulatory proteins of the spliceosome.

It contacts pre-mRNA, U5 snRNA, U6 snRNA, and multiple spliceosomal proteins and contributes to organization of the catalytic center.

The two principal pre-mRNA splicing reactions are transesterification reactions.

First:

branch-point adenosine 2'-OH attacks the 5' splice site

forming the intron lariat.

Second:

the released exon 1 3'-OH attacks the 3' splice site

joining the two exons.

PRPF8 itself is not accurately described as a conventional phosphodiesterase or ligase.

Catalysis is largely RNA based, with PRPF8 organizing the molecular environment in which the reactions occur.

The index deletion removes one complete PRPF8 copy.

PRPF8 is exceptional because its retinal relevance is established independently of this CNV. ClinGen classifies autosomal-dominant PRPF8-related retinopathy as Definitive (Clinical Genome Resource, "PRPF8-Related Retinopathy").

Recent experimental work on retinitis-pigmentosa-associated PRPF8 variants showed reduced pre-mRNA binding, slower intron splicing and altered expression of hundreds of genes in retinal pigment epithelial cells (Zimmann et al.).

Heterozygous PRPF8 variants have also been reported in individuals with neurodevelopmental disorders (O'Grady et al.).

A critical limitation remains: classic PRPF8-associated retinopathy is usually linked to specific pathogenic variants, particularly C-terminal variants, and is not established as a simple whole-gene haploinsufficiency disorder.

Consequently:

PRPF8 has definitive retinal disease relevance, but the phenotype associated specifically with heterozygous whole-gene deletion remains incompletely defined.

3.7 TLCD2: phospholipid acyl-chain composition and membrane fluidity

TLCD2 is a small multi-pass membrane protein belonging to the TLC-domain family.

Its experimentally supported function involves regulation of which fatty-acid species are incorporated into membrane phospholipids.

Reduction of TLCD1/2 activity increases incorporation of long-chain polyunsaturated fatty acids into phospholipids and alters membrane fluidity (Ruiz et al.).

The mechanism is not equivalent to fatty-acid synthesis or bulk uptake. TLCD proteins appear to influence intracellular phospholipid composition, particularly the balance among saturated, monounsaturated and polyunsaturated fatty-acid-containing phospholipids.

This is important because membrane acyl-chain composition affects:

- fluidity;
- curvature;
- protein mobility;
- organelle interactions;
- susceptibility to lipid peroxidation.

The index deletion removes one complete TLCD2 copy.

Human haploinsufficiency is not established, and TLCD1 may provide substantial functional redundancy.

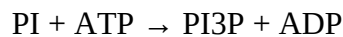
TLCD2 is therefore a strong biochemical membrane candidate but weak dosage candidate.

3.8 WDR81: PI3P regulation and endosome maturation

WDR81 is a large endosomal regulatory protein with particularly important consequences for the emerging membrane model.

WDR81, together with WDR91, negatively regulates the class III phosphatidylinositol 3kinase complex involved in formation of phosphatidylinositol-3-phosphate.

The relevant enzymatic conversion is:



performed by VPS34/PIK3C3.

WDR81 does not catalyze the reaction. It restrains the PI3K complex.

Loss of WDR81/WDR91 activity therefore produces:

excess PI3P

↓

excessive early-endosome fusion

↓

defective early-to-late endosome conversion

↓

abnormal receptor trafficking.

This regulatory mechanism has been directly demonstrated experimentally (Liu et al.).

WDR81 loss also disrupts EGFR trafficking and MAPK signaling in neural progenitor cells and reduces progenitor proliferation and brain growth in experimental models (Carpentieri et al.).

WDR81 additionally participates in selective autophagic removal of ubiquitinated protein aggregates.

Thus WDR81 links:

phosphoinositide chemistry

to:

endosomal identity

to:

neural developmental signaling.

Established human WDR81 disorders are predominantly recessive, so heterozygous dosage consequences remain much less certain.

3.9 SERPINF2: plasmin inhibition and fibrinolysis

SERPINF2 encodes alpha-2-antiplasmin, the principal physiological inhibitor of plasmin.

Plasmin degrades fibrin during fibrinolysis.

Alpha-2-antiplasmin inhibits plasmin using the classical serpin suicide-inhibition mechanism:

plasmin + alpha-2-antiplasmin

↓

reactive-center-loop cleavage

↓

covalent serpin-protease intermediate

↓

conformational trapping

↓

inactive plasmin-antiplasmin complex.

SERPINF2 is additionally crosslinked into fibrin by factor XIIIa, placing the inhibitor directly within the clot.

Severe biallelic deficiency causes hyperfibrinolytic bleeding.

Importantly, heterozygous carriers can demonstrate approximately half-normal biochemical activity and may range from asymptomatic to clinically significant bleeding following surgery, trauma or dental procedures. A review of 123 reported cases confirms this variable heterozygous phenotype (Matrane et al.).

SERPINF2 therefore provides meaningful evidence that one-copy loss can produce a **measurable biochemical dosage effect, although it does not materially strengthen the principal**

visual-neural model.

3.10 SERPINF1: PEDF neurotrophic and retinal signaling

SERPINF1 encodes pigment epithelium-derived factor, or PEDF.

Despite being a serpin-family protein, PEDF is not a conventional protease inhibitor. Its serpin fold has instead been adapted to extracellular signaling.

PEDF has established:

- neurotrophic activity;
- antiangiogenic activity;
- extracellular-matrix binding;

- retinal protective effects;
- roles in bone mineralization.

One important receptor is PNPLA2/PEDF-R, which possesses phospholipase activity.

PEDF therefore connects an extracellular trophic signal to membrane-lipid metabolism.

Experimental ablation of PEDF-R in mice causes altered retinal phospholipid composition, photoreceptor abnormalities, apoptosis and attenuation of electroretinographic responses. Even heterozygous Pnpla2 animals demonstrated electrophysiological attenuation in that experimental system (Bernardo-Colón et al.).

Heterozygous human SERPINF1 mutation carriers provide especially useful dosage evidence. Eighteen carriers showed reduced SERPINF1 transcript levels and significantly reduced circulating PEDF but no detectable abnormalities in bone density or related skeletal parameters (Al-Jallad et al.).

This demonstrates:

heterozygous loss

↓

measurable biochemical dosage reduction

without necessarily producing:

a penetrant clinical phenotype.

SERPINF1 is therefore one of the strongest direct retinal-biological candidates in the interval, while remaining an incompletely established heterozygous retinal disease candidate.

3.11 SMYD4: protein lysine methylation and epigenetic regulation

SMYD4 encodes a SET- and MYND-domain lysine methyltransferase.

The fundamental methyltransferase reaction is:

protein lysine + S-adenosylmethionine

↓

methylated lysine + S-adenosylhomocysteine.

SMYD4 therefore directly performs a covalent chemical modification.

Experimental evidence implicates SMYD4 in:

- H3K4 methylation;
- interaction with HDAC1;
- regulation of histone acetylation;
- developmental transcription.

More recent human-cell work has demonstrated nonhistone activity. SMYD4 promotes lysine monomethylation of MYH9, increasing its ubiquitination and degradation and thereby influencing WNT/beta-catenin signaling (Yang et al.).

Despite this increasingly strong biochemical characterization, human heterozygous dosage sensitivity remains weak.

SMYD4 is therefore best treated as an epigenetic and post-translational regulatory component rather than a major direct neural or retinal candidate.

3.12 RPA1: single-stranded DNA protection and ATR signaling

RPA1 encodes the 70-kDa subunit of the heterotrimeric Replication Protein A complex.

RPA binds exposed single-stranded DNA during:

- replication;
- stalled replication forks;
- homologous recombination;
- DNA repair;

- telomere replication.

Binding is noncovalent and mediated through multiple OB-fold DNA-binding domains.

RPA-coated ssDNA also recruits ATRIP and contributes to activation of the ATR DNAdamage checkpoint.

The index deletion completely removes one RPA1 copy.

RPA1 is particularly important because human cellular haploinsufficiency has already been experimentally demonstrated in the context of contiguous gene deletions. Patient-derived cells containing heterozygous RPA1 deletions demonstrated impaired ATR-dependent DNAdamage responses (O’Driscoll et al.).

The original experiments further demonstrated correction after RPA1 restoration, strengthening attribution of the cellular phenotype to RPA1 dosage.

Current ClinGen dosage sensitivity remains formally Awaiting Review, emphasizing that a molecular haploinsufficiency phenotype does not automatically constitute a fully curated clinical haploinsufficiency syndrome (Clinical Genome Resource, “RPA1 Dosage Sensitivity Curation”).

RPA1 therefore provides the strongest evidence in the interval that:

heterozygous gene loss itself can produce a directly measurable molecular

defect.

Its strongest consequence is genome-maintenance vulnerability rather than established visual-system dysfunction.

3.13 RTN4RL1: neuronal extracellular signaling and axonal plasticity

RTN4RL1 encodes NgR3, a GPI-anchored neuronal receptor.

Although only a small percentage of the genomic RTN4RL1 span overlaps the deletion, approximately 99% of the CCDS is removed under the Phase 1 model.

The affected allele therefore cannot encode an intact canonical NgR3 receptor.

NgR3 contains extracellular leucine-rich repeats and recognizes glycosaminoglycan components of chondroitin sulfate proteoglycans.

NgR1 and NgR3 jointly mediate CSPG-associated restriction of axonal growth. Combined receptor loss enhances regeneration following optic-nerve injury, whereas isolated single-receptor loss produces substantially weaker effects (Dickendesher et al.).

This establishes two important points.

First, RTN4RL1 is directly relevant to neuronal structural signaling.

Second, substantial Nogo-receptor-family redundancy exists.

RTN4RL1 therefore provides strong neural and direct visual-pathway experimental relevance, but relatively weak evidence that heterozygous loss alone produces a human phenotype.

4. Noncoding Loci

4.1 PITPNA-AS1

PITPNA-AS1 is an antisense long noncoding RNA.

Experimental cancer-cell studies demonstrate that it can participate in:

- miRNA sequestration;
- RNA-protein interactions;
- mRNA stabilization;
- regulation of EGFR/PI3K-AKT and other signaling pathways.

However, effects are strongly context dependent, and evidence remains dominated by cancer models.

Critically, there is insufficient evidence to assume that PITPNA-AS1 loss directly reduces PITPNA expression merely because the genes are antisense neighbors.

PITPNA-AS1 should therefore remain in transcriptomic analysis but currently represents a low-to-moderate-priority regulatory candidate.

4.2 MIR22HG

MIR22HG is a long noncoding host transcript containing the MIR22 precursor.

This creates two molecular layers:

MIR22HG lncRNA function

and:

miR-22 production.

The host RNA has experimentally demonstrated regulatory functions independent of mature miR-22, including RNA-protein interactions and effects on major signaling pathways.

The direction of MIR22HG activity is highly context dependent, particularly across different malignancies.

The index deletion removes one MIR22HG copy and simultaneously removes one genomic MIR22 precursor.

This structural relationship is important:

MIR22HG and MIR22 are distinct functional annotations but not

independent deletion events.

4.3 MIR22

MIR22 encodes a regulatory microRNA processed into mature miR-22 species that associate with Argonaute-containing RISC complexes.

The simplified mechanism is:

miR-22 + Argonaute

↓

target-mRNA base pairing

↓

translation suppression and/or increased target-RNA degradation.

MicroRNAs are capable of regulating many transcripts simultaneously.

Consequently, MIR22 loss has the potential to affect regulatory networks rather than a single biochemical reaction.

Direct neurological evidence exists.

Genetic deletion of Mir22 in mice resulted in earlier onset, increased frequency and increased duration of spontaneous seizures following experimental status epilepticus (Silva et al.).

Importantly, baseline brain structure and behavior were comparatively normal in the knockout model before the epileptogenic challenge.

This suggests that miR-22 may influence response or resilience to neural stress rather than being required for basic nervous-system formation.

The finding does not establish that one-copy MIR22 loss causes seizures in humans.

MIR22 nevertheless becomes the most compelling noncoding neurological candidate in the interval.

4.4 RN7SL105P

RN7SL105P is an inferred 7SL-related pseudogene.

Functional 7SL RNA is an important component of the signal recognition particle, but this ancestral relationship cannot be transferred automatically to RN7SL105P.

No convincing locus-specific evidence currently establishes:

- functional RNA production;
- SRP participation;
- neural activity;
- retinal activity;
- clinically significant dosage sensitivity.

RN7SL105P is therefore retained for structural completeness but presently receives no major mechanistic weight.

5. Integrated Molecular Characterization Matrix

Principal

Structural

Heterozygous

Locus

Neural relevance

Visual relevance

molecular system

consequence

dosage evidence

~49.6% coding loss;

Phosphoinositide INPP5K

catalytic region

Weak

Moderate

Limited–moderate

hydrolysis

disrupted PITPNA

PI/PC transfer

Full gene loss

Moderate

Strong

Limited

Essential amino-acid SLC43A2

Full gene loss

Weak

Limited

Minimal

transport

Efferocytosis/ SCARF1

Full gene loss

Weak

Limited

Minimal

immune clearance

Late-endosome RILP

Full gene loss

Weak

Strong

Limited

transport

Moderate/uncertain PRPF8

Pre-mRNA splicing

Full gene loss

Strong

Very strong

mechanism

Membrane TLCD2

phospholipid

Full gene loss

Weak

Limited

Minimal

composition

PI3P/endosome WDR81

Full gene loss

Weak

Strong

Limited

maturation

Fibrinolysis/plasmin

Moderate–strong SERPINF2

Full gene loss

Limited

Minimal

inhibition

biochemical

PEDF trophic/retinal

Moderate SERPINF1

Full gene loss

Strong

Very strong

lipid signaling

biochemical

Protein SMYD4

methylation/epigenet

Full gene loss

Weak

Limited

Minimal

ic regulation

DNA

Strongest direct RPA1

replication/repair/A

Full gene loss

Moderate

Minimal

molecular evidence

TR response

Axon/synapse RTN4RL1

~99% CCDS loss

Weak-moderate

Strong

Moderate

signaling PTPNA-AS1

lncRNA regulation

Full locus loss

Unknown

Limited

None established

lncRNA/miRNA- MIR22HG

Full locus loss

Unknown

Limited–moderate

Limited

host regulation

miRNA post-

Moderate–strong MIR22

transcriptional

Full locus loss

Unknown in humans

Limited

experimental

regulation

7SL-related RN7SL105P

Full locus loss

None established

None

None

pseudogene

6. Principal Mechanistic Convergence: Membrane, Phosphoinositide and Endosomal Homeostasis

The strongest Phase 2 finding is the presence of several independently characterized genes affecting different layers of membrane biology.

This cluster consists principally of:

PITPNA

→ phosphatidylinositol availability and lipid transfer

INPP5K

→ phosphoinositide 5-phosphate hydrolysis

WDR81

→ regulation of endosomal PI3P production

TLCD2

→ phospholipid fatty-acyl composition and membrane fluidity

SERPINF1

→ extracellular trophic signaling linked to phospholipid metabolism

RILP

→ RAB7-dependent late-endosomal transport.

These genes do not perform the same biochemical reaction.

Their importance lies in their positions at distinct regulatory layers of a common cellular

These genes do not perform the same biochemical reaction. Their importance lies in their positions at distinct regulatory layers of a common cellular system.

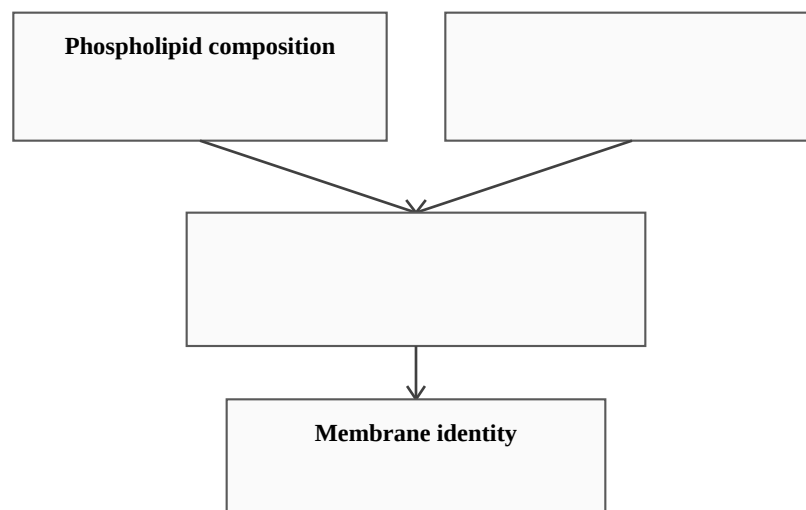
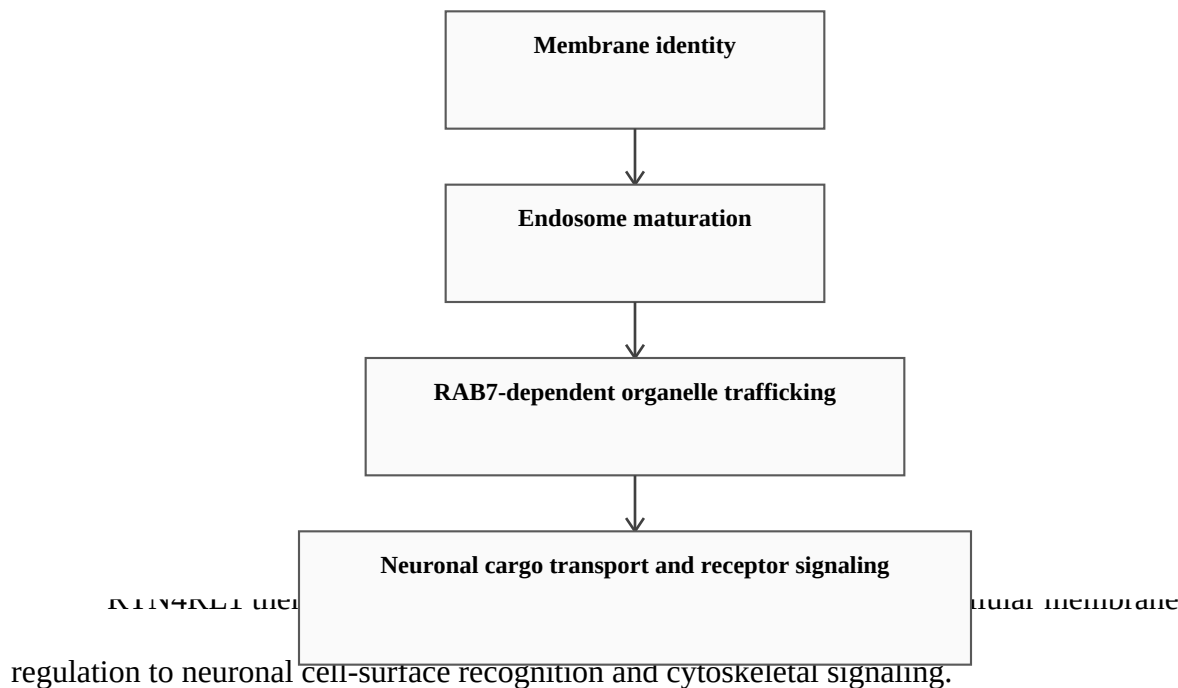
Simplified integrated model

Figure continues on the next page.



7. The Convergence Is Not Necessarily Directionally Additive

The observation that multiple genes affect the same broad system does not mean their deletion drives every molecular variable in the same direction.

This is particularly important for phosphoinositides.

Reduced PITPNA could potentially decrease efficient phosphatidylinositol supply.

Reduced WDR81 could increase endosomal PI3P because normal WDR81 restrains class III PI3K activity.

Reduced INPP5K could impair degradation of PI(4,5)P₂ and PI(3,4,5)P₃.

These effects occur:

- on different phosphoinositide species;
- in different cellular compartments;
- through different reactions.

Consequently, the current evidence supports:

dysregulated phosphoinositide homeostasis

rather than:

uniformly increased or uniformly decreased phosphoinositide signaling.

The possibility of opposing and compensatory effects is one of the most important reasons that transcriptomic and pathway analysis must precede causal interpretation.

8. Independent Regulatory Axes

8.1 RNA processing

PRPF8 creates a potentially broad regulatory mechanism because altered spliceosomal function can affect many transcripts simultaneously.

A later testable question is whether PRPF8 dosage perturbation preferentially affects transcripts involved in:

- phosphoinositide metabolism;
- endosomal trafficking;
- neuronal signaling;
- retinal maintenance.

Such an observation would connect PRPF8 to the membrane cluster through functional convergence.

It cannot currently be assumed.

8.2 Post-transcriptional RNA regulation

MIR22HG/MIR22 introduces a second RNA regulatory layer.

Unlike PRPF8, which regulates intron removal, miR-22 controls mRNA abundance and translation through sequence-specific RISC targeting.

Thus the deletion simultaneously affects:

pre-mRNA processing

and:

post-transcriptional gene regulation.

MIR22's experimental influence on epileptogenesis makes this mechanism relevant to later neurological analysis (Silva et al.).

8.3 Epigenetic regulation

SMYD4 contributes covalent protein methylation and chromatin-associated regulation.

This introduces a third level of information control:

chromatin/protein modification

↓

RNA processing

↓

post-transcriptional miRNA regulation.

Whether these three regulatory systems converge on common target pathways remains unknown.

8.4 Genome maintenance

RPA1 establishes a separate genome-integrity axis.

The deletion therefore affects not only membrane signaling and RNA processing but also:

- replication-fork protection;
- ssDNA stabilization;
- ATR checkpoint activation;
- homologous recombination;
- telomere-associated DNA metabolism.

The direct human cellular haploinsufficiency evidence for RPA1 makes this mechanism especially important even though it is not primarily visual (O'Driscoll et al.).

9. Additional Systemic Axes

9.1 Amino-acid transport

SLC43A2 contributes an essential-amino-acid transport mechanism.

This potentially affects nutrient sensing and metabolism but currently has limited direct evidence for visual-neural involvement.

9.2 Immune clearance

SCARF1 contributes apoptotic-cell clearance and immune homeostasis.

Its inclusion demonstrates that the deletion can influence extracellular damage recognition and immune regulation independently of the neural mechanisms.

9.3 Hemostasis

SERPINF2 contributes fibrinolytic regulation.

Its heterozygous carrier literature is important because it demonstrates that reduced gene dosage can result in measurable biochemical loss of function even when clinical penetrance is incomplete (Matrane et al.).

10. Heterozygous Dosage Is the Principal Unresolved Variable

The most important lesson from Phase 2 is that gene importance and haploinsufficiency are not equivalent.

For most loci, severe experimental loss establishes biological importance.

It does not establish that reducing gene copy number from two to one produces a clinically significant effect.

Three loci provide particularly useful dosage examples.

RPA1

Heterozygous genomic loss has been associated directly with:

- reduced RPA1 abundance;
- impaired ATR signaling;
- replication-stress abnormalities;
- molecular rescue after RPA1 restoration.

This constitutes the strongest direct one-copy molecular evidence in the interval (O'Driscoll et al.).

SERPINF1

Heterozygous carriers demonstrate reduced PEDF but no clearly penetrant skeletal phenotype.

This demonstrates:

molecular dosage effect without corresponding overt clinical disease. (Al-

Jallad et al.)

SERPINF2

Heterozygous carriers can show approximately half-normal biochemical activity, but clinical expression ranges from absent to substantial bleeding under physiological challenge (Matrane et al.).

These examples argue against a simplistic model in which:

one deleted allele = exactly 50% function = predictable disease.

Biological output depends on:

- transcriptional compensation;
- protein stability;

- pathway reserve;
- paralogous proteins;
- complex stoichiometry;
- tissue expression;
- developmental stage;
- environmental or physiological stress.

A more defensible concept is:

reduced functional reserve.

The deletion may reduce the capacity of multiple systems without eliminating them.

11. Neural and Visual-System Evidence

Phase 2 identifies several genes with strong neural relevance but only a smaller subset

with direct retinal or visual-system evidence.

11.1 PRPF8

PRPF8 provides the strongest established human retinal disease relationship.

ClinGen currently classifies PRPF8-related autosomal-dominant retinopathy as Definitive (Clinical Genome Resource, “PRPF8-Related Retinopathy”).

The relevant limitation is mechanism: known pathogenic retinal variants cannot automatically be equated with whole-gene deletion.

11.2 SERPINF1

SERPINF1 provides strong direct retinal biology through:

- PEDF neurotrophic signaling;
- photoreceptor survival;
- retinal ganglion-cell support;

- angiogenesis regulation;
- retinal phospholipid metabolism.

PEDF-R deficiency alters both retinal lipid composition and electrophysiological function experimentally (Bernardo-Colón et al.).

SERPINF1 is therefore one of the strongest mechanistic retinal candidates even though heterozygous SERPINF1 deletion is not an established dominant retinal disorder.

11.3 RTN4RL1

RTN4RL1 provides direct visual-pathway evidence through retinal ganglion-cell axon biology and optic-nerve regeneration.

Combined NgR1/NgR3 loss enhances regeneration after optic-nerve crush (Dickendesher et al.).

This supports an optic-pathway mechanism but not a normal visual-perception phenotype.

11.4 WDR81, RILP and PITPNA

These genes provide strong neuronal mechanisms involving:

- neural-progenitor proliferation;
- endosomal maturation;
- dendritic cargo transport;
- neuronal survival.

WDR81 loss alters endosomal EGFR trafficking and neural-progenitor proliferation (Carpentieri et al.).

RILP-RAB7A late-endosomal transport is directly relevant to dendrite arborization (Yap et al.).

PITPNA reduction produces severe neurodegeneration in mouse models (Hamilton et al.).

These findings justify neurological investigation but do not establish a specific human visual phenotype.

11.5 MIR22

MIR22 adds a neural regulatory mechanism associated experimentally with response to epileptogenic stress.

Complete knockout exacerbates spontaneous seizure development after status epilepticus (Silva et al.).

This evidence is mechanistically important but cannot establish seizure susceptibility from heterozygous MIR22 loss.

12. What Phase 2 Supports

The combined evidence supports several conclusions with relatively high confidence.

First, the deletion removes or severely disrupts multiple biologically functional loci.

Second, several affected genes independently regulate components of the same broader membrane/endosomal system.

Third, the convergence is chemically substantive rather than merely semantic. The affected functions include:

- lipid binding and transfer;
- phosphate-ester hydrolysis;
- PI3P synthesis regulation;
- phospholipid acyl-chain composition;
- phospholipase-linked extracellular signaling;

- RAB7-dependent organelle transport.

Fourth, several affected genes have direct experimental neural or retinal functions.

Fifth, heterozygous molecular dosage effects are demonstrable for at least some genes, most clearly RPA1 and biochemically for SERPINF1 and SERPINF2.

Sixth, the deletion also affects independent systems that should not be forced into the primary neural model.

The most defensible overall characterization is therefore:

a multisystem heterozygous dosage lesion containing a prominent

membrane/phosphoinositide/endosomal-neuronal convergence.

13. What Phase 2 Does Not Establish

Phase 2 does not establish that the deletion causes a specific clinical phenotype.

It does not demonstrate that all affected genes experience exactly 50% reduction in transcript or protein abundance.

It does not demonstrate that all affected pathways are impaired in the same direction.

It does not establish synergy or epistasis among deleted genes.

It does not establish that the membrane/endosomal convergence is active at pathological magnitude in retinal or neural tissue.

It does not establish that PRPF8 whole-gene deletion produces the same phenotype as established PRPF8 retinal variants.

It does not establish that heterozygous WDR81 loss reproduces recessive WDR81 disorders.

It does not establish that heterozygous SERPINF1 loss causes retinal degeneration.

It does not establish that heterozygous MIR22 loss causes seizures.

It does not establish:

- photophobia;
- pattern glare;
- visual stress;
- magnocellular dysfunction;
- cortical visual hyperexcitability;
- epilepsy;
- hearing abnormalities;
- congenital cardiac disease

as consequences of this deletion.

These remain hypotheses or phenotype variables for later investigation.

14. Limitations

14.1 CNV breakpoint resolution

The reported array coordinates may represent probe-defined boundaries rather than nucleotide-exact physical breakpoints.

Coding overlap is therefore established at CNV resolution, but exact breakpoint-junction biology remains unknown.

14.2 Transcript fate

For the partially deleted genes INPP5K and RTN4RL1, genomic disruption does not establish the exact RNA product.

Potential outcomes include:

- reduced transcription;
- aberrant splicing;

- unstable RNA;
- nonsense-mediated decay;
- unusual truncated products.

These require direct transcript evidence.

14.3 Dosage compensation

Complete deletion of one genomic allele does not prove a 50% reduction in cellular protein.

The intact allele may:

- compensate transcriptionally;
- fail to compensate;
- overcompensate;
- show tissue-specific regulation.

14.4 Paralogous redundancy

Several genes have functionally related family members.

Examples include:

RTN4RL1

with RTN4R and RTN4RL2,

and:

TLCD2

with TLCD1.

Such redundancy can reduce the phenotypic impact of heterozygous loss.

14.5 Evidence derived from severe loss

Many mechanistic studies use:

- homozygous knockout;
- severe hypomorphic alleles;
- RNA knockdown;
- cancer-cell overexpression;
- experimental injury.

These conditions cannot automatically be extrapolated to constitutional heterozygous deletion.

14.6 Tissue specificity

A molecular effect observed in:

- liver;
- kidney;
- tumor cells;
- fibroblasts;
- cultured neurons

may not occur identically in:

- retina;
- retinal pigment epithelium;
- retinal ganglion cells;
- thalamic visual nuclei;
- visual cortex.

This is the principal reason Phase 3 requires tissue- and cell-type-specific expression analysis.

14.7 Network inference

Pathway overlap does not establish direct interaction.

The terms:

- functional convergence;
- pathway overlap;
- network proximity

are currently appropriate.

Terms such as:

- synergistic interaction;
- epistasis;
- causal pathway

require additional evidence.

15. Phase 2 Working Model

The integrated molecular model can be represented as follows:

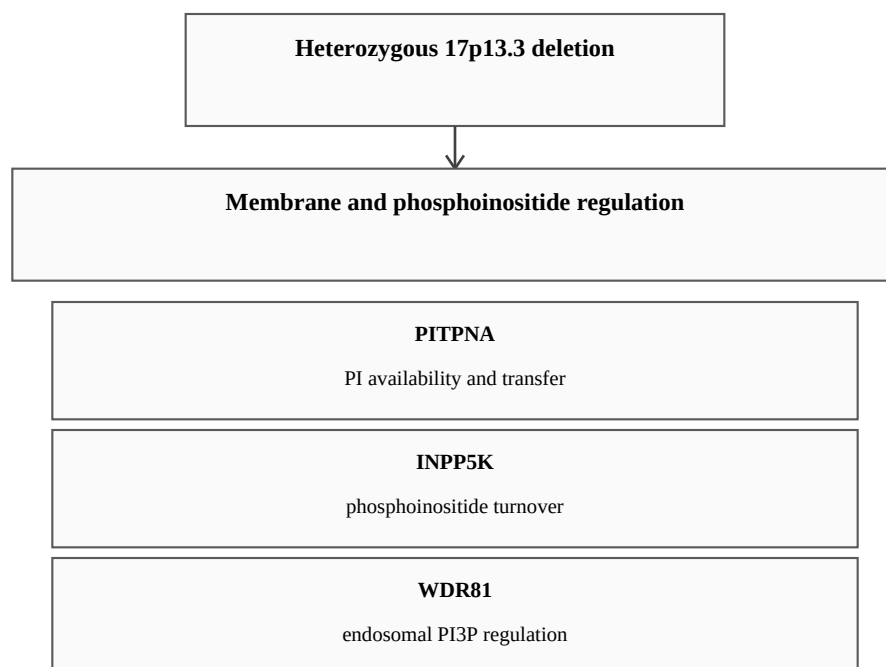
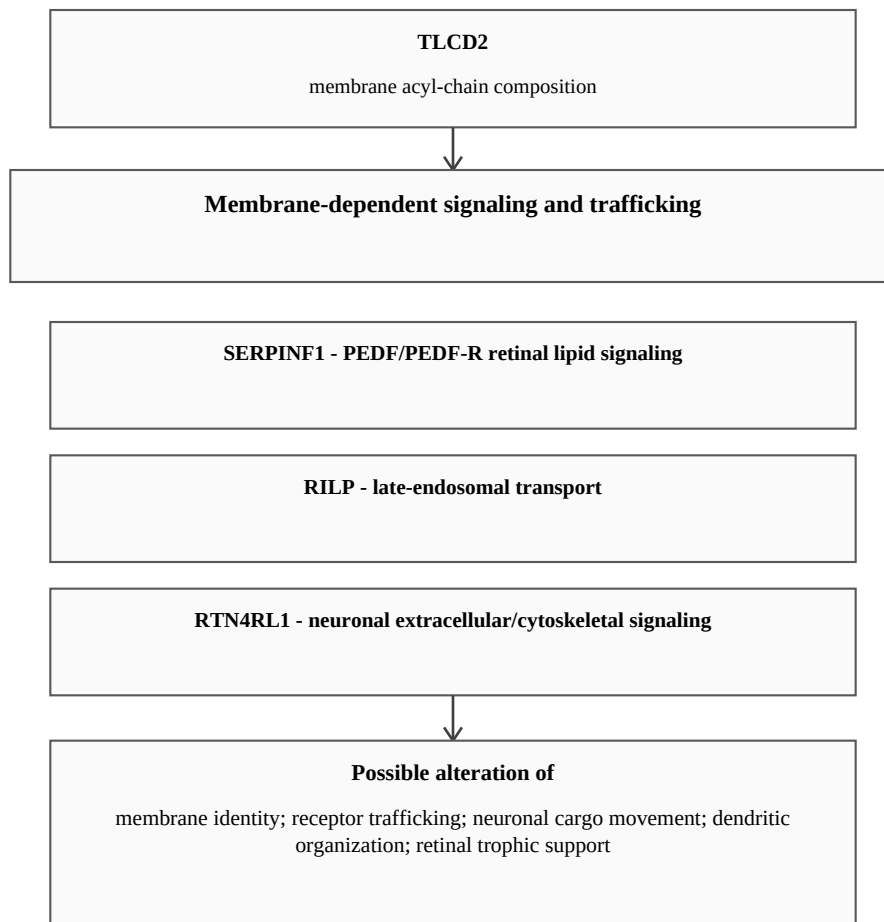
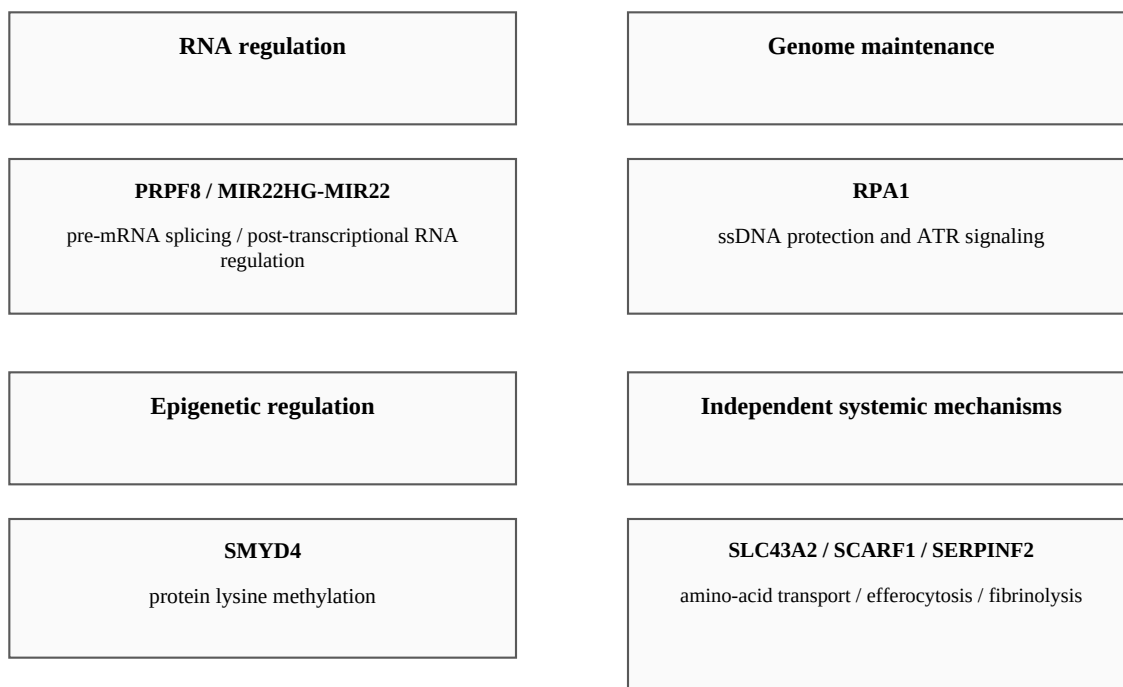


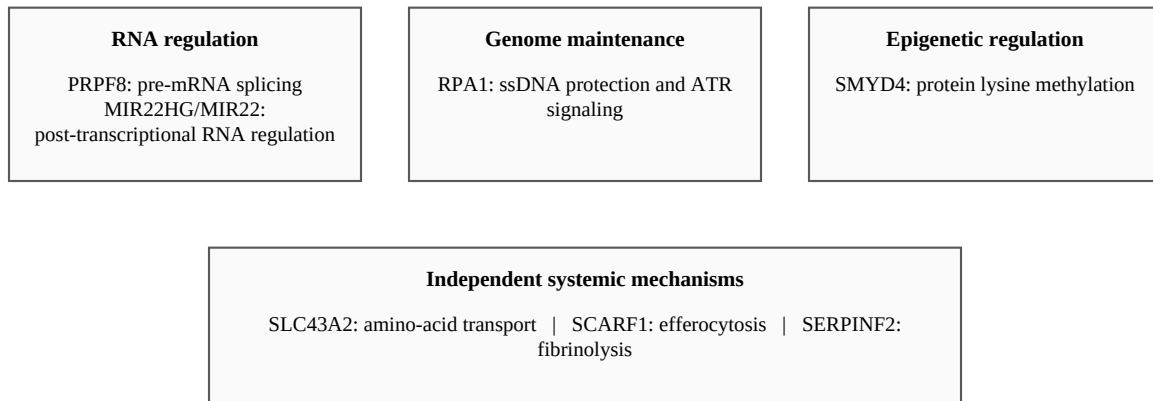
Figure continues on the next page.



In parallel with:



Phase 2 working model - independent parallel axes



16. Implications for the Next Analytical Phase

Phase 2 has reduced the research problem substantially.

The next question is no longer:

Which genes are deleted?

Nor is it simply:

Which deleted genes have neurological annotations?

The appropriate next question is:

Are the molecularly strongest candidate genes actually expressed together in the relevant human retinal and neurological cell populations, and do their expression patterns overlap at developmentally relevant times?

Phase 3 should therefore test:

1. tissue expression of all affected genes;
2. retinal cell-type expression;
3. neuronal cell-type expression;
4. developmental timing;
5. coexpression among the membrane/endosomal candidates;
6. whether PRPF8, MIR22 and other regulatory loci intersect those same expression programs;

7. whether candidate genes are expressed within visual system pathways strongly enough to make the Phase 2 convergence biologically relevant in vivo.

The analysis should remain genome-first rather than phenotype-first.

17. Conclusion

Phase 2 demonstrates that the chr17:1,411,408–1,841,103 microdeletion is molecularly complex.

The deletion does not merely remove a collection of unrelated genes. It simultaneously eliminates or severely disrupts multiple regulators of:

- phosphoinositide metabolism;
- phospholipid transport;
- membrane lipid composition;
- endosomal identity;

- intracellular trafficking;
- retinal trophic signaling;
- neuronal structural plasticity;
- RNA splicing;
- post-transcriptional RNA regulation;
- protein methylation;
- DNA replication and repair;
- amino-acid transport;
- immune clearance;
- fibrinolysis.

The strongest mechanistic convergence involves INPP5K, PITPNA, TLCD2, WDR81, RILP and SERPINF1, with RTN4RL1 extending the system into neuronal surface signaling.

PRPF8 independently provides definitive evidence for a retinal disease relationship under established pathogenic variant mechanisms, while RPA1 provides the strongest direct evidence that heterozygous genomic loss within the interval can produce a measurable human molecular defect (Clinical Genome Resource, “PRPF8-Related Retinopathy”; O’Driscoll et al.).

The MIR22HG/MIR22 unit contributes an additional regulatory mechanism with experimentally established effects on neural response to epileptogenic stress (Silva et al.).

The central conclusion is therefore not that this deletion has been assigned a phenotype.

Rather:

The deletion produces a credible multi-gene molecular architecture in which reduced functional reserve across membrane, endosomal, retinal, RNA-

regulatory and genome-maintenance systems is biologically plausible.

The magnitude, tissue specificity, developmental timing and clinical consequences of those reductions remain unresolved.

Accordingly, Phase 2 establishes mechanistic plausibility and research priorities, not causality.

The results provide a sufficiently developed molecular foundation to proceed to spatiotemporal functional genomics, where the principal hypothesis can be tested without assuming the clinical conclusion in advance.

Phase 3

Spatiotemporal Functional Genomics

Phase 3A - Abstract and Methods Summary

Abstract-Ready Phase 3 Summary

Phase 3 was organized into two analytical parts to determine whether the molecular

mechanisms identified in Phase 2 have the spatial opportunity to operate within the human visual system. Part I — retinal and ocular localization combined adult whole-tissue screening with adult human retinal cell-type analysis. Bulk retinal data showed the strongest expression among affected protein-coding loci for PRPF8, SERPINF1, PITPNA, RPA1, INPP5K, and TLCD2. Cell-type resolution then identified the clearest spatial convergence in retinal pigment epithelium, where INPP5K, PITPNA, and SERPINF1 converge on phosphoinositide regulation, phospholipid handling, and PEDF-associated trophic/lipid signaling. PRPF8 was broadly represented across neural retina, including photoreceptors, while TLCD2 showed a preliminary cone-photoreceptor signal. In contrast, the predicted WDR81-RILP trafficking pair was not demonstrated as a shared adult-retinal neuronal module. Part II — central visual-relay

specificity tested whether the affected interval preferentially mapped onto magnocellular,

parvocellular, or koniocellular transcriptional programs in the human dorsal lateral geniculate nucleus. None of the 17 affected loci appeared among the reproducible human M-versus-P marker genes or principal K-cell markers reported in the reference dataset. Phase 3 therefore provides evidence about where affected loci are expressed, which visual-system cell

populations can plausibly experience their molecular effects, and whether the deletion preferentially maps onto a specific visual relay stream. The evidence to date supports a

distributed retinal and neural functional-reserve model, with the strongest cell-specific signal centered on RPE membrane/lipid biology, but does not support a selectively magnocellular lesion or establish a definitive visual-neurological phenotype.

What Phase 3 Provides

Phase 2 established mechanistic plausibility by determining what the affected loci

normally do. Phase 3 adds spatial opportunity: it tests whether those loci are normally active in the tissues and cell populations required for the proposed mechanisms to matter in the human visual system. Through the work completed to date, Phase 3 distinguishes broad retinal expression from cell-specific localization, tests whether molecularly related deleted genes are actually co-located in the same retinal compartment, and directly evaluates whether the deletion preferentially corresponds to one classical thalamic visual pathway. It therefore narrows the hypothesis without treating expression as proof of dysfunction.

Step-by-Step Method Used Through Phase 3D

Step 1 — Freeze the candidate set and hypotheses before examining expression results

All 17 structurally affected loci identified in Phase 1 were retained. No gene was removed because Phase 2 had assigned it weak visual or neurological relevance. The primary

Phase 3 hypotheses were frozen in advance, including membrane/endosomal co-localization,

direct retinal localization, WDR81-RILP overlap, developmental enrichment, magnocellular specificity, and regulatory overlap. This was done to reduce phenotype-driven selection and confirmation bias.

Step 2 — Separate loci according to assay compatibility

Protein-coding genes were treated as generally measurable by conventional bulk and single-cell RNA sequencing. PITPNA-AS1 and MIR22HG were retained with the recognition that lncRNAs can be poorly captured in some single-cell platforms. Mature MIR22 was not interpreted from ordinary poly(A)-selected RNA-seq because mature microRNAs require small-RNA or miRNA-specific measurement. RN7SL105P was treated cautiously because pseudogene homology can cause ambiguous short-read mapping.

Step 3 — Screen adult whole-body tissue expression

Adult expression resources including GTEx and NCBI Gene summaries were used to determine whether each affected protein-coding locus was broadly expressed, brain biased, peripheral-tissue enriched, or strongly tissue specific. This established general biological context before any retina-specific ranking was performed. Cross-platform expression magnitudes were not treated as directly interchangeable.

Step 4 — Perform the initial adult human retinal screen

Human ocular resources, especially eyeIntegratation and the Human Protein Atlas retinal series, were used to determine which affected genes were detectably expressed in adult retina. The strongest bulk retinal signals among the affected protein-coding genes were PRPF8, SERPINF1, PITPNA, RPA1, INPP5K, and TLCD2. Genes showing weak whole-retina abundance were not discarded because bulk measurements can dilute expression from rare cell populations.

Step 5 — Resolve the retinal signal to individual cell classes

Large human retinal single-cell atlases and independent adult human retinal datasets were used to test localization among photoreceptors, bipolar cells, amacrine cells, horizontal cells,

retinal ganglion cells, Müller glia, astrocytes, microglia, vascular populations, and RPE.

Published marker frameworks were used as quality-control references for identifying major retinal populations.

Step 6 — Test whether the Phase 2 molecular convergence survives cell-type resolution

The pre-specified membrane/endosomal candidate set was examined for shared localization. The strongest supported adult-retinal convergence became an RPE-centered **INPP5K-PITPNA-SERPINF1 module. PRPF8 was broadly present across neural retina, and**

TLCD2 showed a preliminary cone-associated signal. Conversely, WDR81 and RILP were not demonstrated as a shared mature retinal neuronal module. Negative results were retained rather than redefining the candidate set after inspection.

Step 7 — Move from retina to the central visual relay

Human dorsal lateral geniculate nucleus single-nucleus RNA-seq data from the Allen/Bakken comparative LGN resource were used to test the pre-specified magnocellular hypothesis. The analysis treated magnocellular, parvocellular, and koniocellular identities symmetrically rather than defining magnocellular enrichment as the desired result.

Step 8 — Compare affected loci with published human M/P/K transcriptional markers

The human dLGN study showed that M and P projection neurons largely form one transcriptomic MP population with a graded M-to-P continuum. The published reproducible M/P marker set and principal K-cell markers were compared against all 17 affected loci. None of the affected loci was a defining M, P, or K marker. The result was therefore recorded as a meaningful negative finding: current adult human dLGN transcriptomics does not support preferential magnocellular targeting by the deletion.

Step 9 — Update the model without converting expression into causality

The evidence from the retinal and dLGN analyses was integrated with the Phase 2 mechanisms. Expression was interpreted only as opportunity for a mechanism, not proof that the deletion produces dysfunction. The working visual-system model therefore shifted away from a selectively magnocellular lesion toward a distributed retinal and neural functional-

reserve model, with the strongest present spatial convergence in RPE. No definitive phenotype

was assigned.

Current Phase 3 Interpretation

Through Phase 3D, the study has established that several molecularly important deleted genes are normally positioned within biologically relevant human visual-system compartments. The strongest cell-type-specific convergence is INPP5K + PITPNA + SERPINF1 in RPE, while PRPF8 has broad neural-retinal availability and TLCD2 may contribute a cone-associated lipid component. At the same time, the data weaken two earlier possibilities: WDR81 and RILP have not been demonstrated as a shared adult-retinal trafficking module, and none of the affected loci demonstrates published magnocellular-specific transcriptional enrichment in adult human dLGN. The Phase 3 evidence therefore supports continued investigation of a distributed retinal/neural mechanism while rejecting, at the present level of evidence, a specifically magnocellular transcriptomic explanation.

Phase 3 remains in progress. Human V1 and developmental brain/retinal analyses have

not yet been completed and must not be represented as results.

Phase 3B - Research Through Phase 3D

Document scope. This document reproduces the Phase 3 research conducted through Phase 3D. The wording of the research sections is preserved; prior hyperlink-style source notes

have been converted to MLA-style parenthetical citations. Phase 3B was defined in the frozen protocol but was not executed as a separate results section before Phase 3C, so its protocol language is included with an explicit scope note rather than with invented results.

Phase 3A — Adult Whole-Tissue and Initial Ocular Expression Screen

1. Objective

Phase 3A asks whether the 17 loci affected by the 17p13.3 microdeletion are

transcriptionally active in adult human tissues relevant to the project's visual-neurological hypothesis.

The purpose is not to infer disease from expression.

Instead, Phase 3A establishes the first requirement for biological plausibility:

A deleted locus must normally be expressed in the relevant tissue before reduced dosage can plausibly perturb that tissue through its canonical
molecular function.

All 17 loci remain under consideration.

2. Data Sources

The current GTEx adult bulk-expression release is GTEx Analysis V11. The portal provides gene-level TPM and median tissue-expression matrices, including the current V11 median gene-level TPM file. ("GTEx Portal")

For ocular-specific analysis, eyeIntegration v2.12 integrates publicly available healthy human ocular RNA-seq datasets with a substantial subset of GTEx samples processed through a common pipeline. It provides downloadable gene- and transcript-level raw-count matrices and includes multiple ocular tissue categories. ("eyeIntegration")

The initial retinal screen below additionally uses current Human Protein Atlas retinal RNA-expression pages. These provide a 105-sample adult human retinal series and consensus normalized transcript measurements. These values are useful for screening but will not substitute for the independent eyeIntegration and single-cell replication planned later in Phase 3.

Current NCBI Gene expression summaries were used as an independent whole-body localization check.

3. Whole-Body Expression Pattern

The first result is that the deleted interval does not consist predominantly of retinaspecific or brain-specific genes.

Most affected protein-coding genes are broadly or ubiquitously expressed.

This is methodologically important.

A visual-neurological mechanism, if present, will therefore probably depend on:

cell type + developmental context + dosage sensitivity

rather than simply: brain-specific gene expression.

4. Whole-Tissue Classification

Gene

Current broad tissue pattern

Phase 3A interpretation

Ubiquitous; NCBI highlights adipose

Broadly available molecular

INPP5K

and lung

mechanism

Ubiquitous; brain is among highest-

PITPNA

Strong initial neural localization signal

expression tissues

Broad; strongest in placenta and

Primarily systemic/epithelial bulk

SLC43A2

kidney

pattern

Broad; enriched toward

Supports immune/endothelial rather

SCARF1

spleen/lung/lymphoid-endothelial

than primary neuronal axis

tissues

Bulk tissue does not discriminate

RILP

Ubiquitous

neural function

Expected for core spliceosomal

PRPF8

Ubiquitous

protein

TLCD2

Broad; adipose and colon prominent

No bulk brain enrichment

Broad in current expression summary,

Requires cell-type and regional

WDR81

although RefSeq specifically describes

resolution

predominant brain expression

Strongly biased toward liver and

Supports systemic fibrinolytic rather

SERPINF2

kidney

than neural role

Systemic expression does not preclude

SERPINF1

Broad, very high in liver/gallbladder

strong retinal expression

Ubiquitous, generally modest

SMYD4

No major adult tissue specificity

expression

Consistent with fundamental genome-

RPA1

Ubiquitous

maintenance role

Broad but brain is the highest reported

RTN4RL1

Strong adult neural localization signal

human tissue

Representative current NCBI values include PITPNA at approximately 31.8 RPKM in

brain, RTN4RL1 at approximately 5.0 RPKM in brain, SLC43A2 at approximately 24.5

RPKM in placenta and 14.0 in kidney, and SERPINF2 at approximately 275.1 RPKM in

liver

and 42.8 in kidney. ("PITPNA Gene"; "RTN4RL1 Gene"; "SLC43A2 Gene"; "SERPINF2 Gene")

INPP5K, RILP, PRPF8, TLCD2, SMYD4, and RPA1 are currently described as ubiquitous or broadly expressed. ("INPP5K Gene"; "RILP Gene"; "PRPF8 Gene"; "TLCD2 Gene"; "SMYD4 Gene"; "RPA1 Gene")

SCARF1 shows a lymphoid/endothelial-biased pattern, while SERPINF1 remains broadly expressed despite very strong expression in several peripheral tissues. ("SCARF1 Gene"; "SERPINF1 Gene")

WDR81 is an interesting discrepancy: the current RefSeq functional summary calls it **predominantly expressed in brain, whereas NCBI's aggregated expression line labels it**

ubiquitous. This should not be forcibly reconciled; instead it becomes a reason to inspect regional and cell-level brain datasets directly. ("WDR81 Gene")

5. Adult Human Retina Screen

The current Human Protein Atlas retinal series provides an initial quantitative screen for the 13 protein-coding genes.

Table 3A.1 — Adult Human Retina Expression

Gene

Retina RNA expression

Initial classification

PRPF8

76.4 nTPM

High

SERPINF1

56.9 nTPM

High

PITPNA

43.9 nTPM

High

RPA1

20.8 nTPM

Moderate-high

INPP5K

14.7 nTPM

Moderate

TLCD2

10.1 nTPM

Moderate

RTN4RL1

7.9 nTPM

Moderate-low

SMYD4

7.2 nTPM

Moderate-low

SLC43A2

1.3 nTPM

Low

WDR81

0.7 nTPM

Low in whole-retina bulk

SERPINF2

0.5 nTPM

Very low

Gene**Retina RNA expression****Initial classification****RILP****0.4 nTPM**

Very low in whole-retina bulk

SCARF1**0.2 nTPM**

Very low

Current source values are directly reported by the Human Protein Atlas retinal pages.

("PRPF8 Retina"; "SERPINF1 Retina"; "PITPNA Retina"; "RPA1 Retina"; "INPP5K Retina"; "TLCD2 Retina"; "RTN4RL1 Retina"; "SMYD4 Retina"; "SLC43A2 Retina"; "WDR81 Retina"; "SERPINF2 Retina"; "RILP Retina"; "SCARF1 Retina")

These values should be interpreted as screening measurements, not as direct measures of functional importance.

6. First Major Phase 3 Result: PRPF8 Is Strongly Represented in Adult Retina

PRPF8 is the highest-expressed affected protein-coding gene in the initial retinal screen at approximately:

76.4 nTPM.

("PRPF8 Retina")

This substantially strengthens the Phase 2 observation that PRPF8 is biologically relevant to retinal tissue.

However, the significance is specific:

PRPF8 has both strong retinal expression and independently established

retinal disease biology.

It still does not establish that whole-gene heterozygous deletion produces the same mechanism as known pathogenic PRPF8 variants.

7. SERPINF1 Is Also Strongly Expressed in Retina

SERPINF1 is expressed at approximately:

56.9 nTPM

in the adult retinal series. ("SERPINF1 Retina")

This is particularly meaningful because Phase 2 independently established PEDF as a:

- retinal neurotrophic factor;
- antiangiogenic factor;
- regulator of retinal survival;
- participant in PEDF-R-linked phospholipid signaling.

The Phase 3A result therefore provides direct spatial opportunity for the Phase 2 mechanism.

SERPINF1 is not merely a systemic secreted gene with incidental retinal literature.

It is substantially expressed within adult retinal tissue.

8. PITPNA Provides a Strong Link Between the Phase 2 Lipid Cluster and Retina

PITPNA is expressed at approximately:

43.9 nTPM

in adult retina. ("PITPNA Retina")

NCBI also identifies brain as one of PITPNA's highest-expression tissues. ("PITPNA Gene")

This makes PITPNA one of the first genes for which Phase 3 supports both:

adult neural expression

and: substantial retinal expression.

Because PITPNA was already one of the strongest Phase 2 phosphoinositide-homeostasis candidates, this finding materially strengthens the membrane-lipid hypothesis.

9. INPP5K Is Clearly Present in Adult Retina

INPP5K averages approximately:

14.7 nTPM

in the retinal series. ("INPP5K Retina")

More importantly, current Human Protein Atlas single-cell annotations classify INPP5K as group enriched in retinal pigment epithelial cells and late spermatids, providing an early indication that its retinal bulk expression may contain a strong RPE component. ("INPP5K Single Cell")

This observation is significant because it begins to localize one of the central phosphoinositide enzymes from Phase 2 to a specific ocular cell population.

It will require confirmation in the dedicated retinal single-cell phase.

10. RPA1 Is Substantially Expressed in Adult Retina

RPA1 shows approximately:

20.8 nTPM

with relatively limited variation across the 105 retinal samples. ("RPA1 Retina")

This fits its ubiquitous genome-maintenance role.

The result establishes that the tissue contains substantial RPA1 transcript and therefore that the Phase 2 haploinsufficiency mechanism is molecularly available in retinal cells.

It does not establish that retina is preferentially vulnerable to RPA1 loss.

11. RTN4RL1 Is Detectable in Retina and Brain

RTN4RL1 shows approximately:

7.9 nTPM

in adult retina. ("RTN4RL1 Retina")

Its whole-body pattern is additionally brain biased, with NCBI listing brain as its highest reported human tissue. ("RTN4RL1 Gene")

This supports the Phase 2 interpretation of RTN4RL1 as a legitimate neural/visualpathway candidate rather than a receptor whose optic-nerve biology is disconnected from human tissue expression.

The next decisive question is:

Which retinal population expresses RTN4RL1?

If it localizes primarily to retinal ganglion cells, that would align closely with the opticnerve regeneration mechanism described in Phase 2.

If it instead localizes to another population, the working model must be revised accordingly.

12. TLCD2 Is Detectable at Moderate Levels

TLCD2 shows approximately:

10.1 nTPM

in adult retina. ("TLCD2 Retina")

This is an important result because TLCD2 had weak direct retinal evidence in Phase 2.

Phase 3A therefore upgrades one specific proposition:

The TLCD2 phospholipid-composition mechanism has an opportunity to

operate in adult retinal tissue.

It does not yet establish which retinal cells express TLCD2 or whether one-copy reduction meaningfully alters retinal membrane composition.

13. WDR81 and RILP Produce an Important Bulk-Tissue Warning

WDR81 and RILP were among the strongest neural trafficking candidates in Phase 2.

Yet whole-retina expression is low:

WDR81 $\approx$ 0.7 nTPM

RILP $\approx$ 0.4 nTPM.

("WDR81 Retina"; "RILP Retina")

This does not currently falsify the WDR81–RILP hypothesis.

Instead, there are at least three possibilities:

1. They are genuinely weakly expressed in adult retina.

That would weaken a retinal mechanism.

2. They are expressed in a small retinal population.

Bulk retinal RNA would dilute such a signal.

3. Their principal relevance lies outside retina, for example in dLGN, cortex, developmental

progenitors, or other neural tissues.

Phase 3C–G will discriminate among these possibilities.

This is exactly why bulk tissue analysis was performed before cell-type interpretation.

14. SCARF1 and SERPINF2 Behave as Predicted from Phase 2

SCARF1 and SERPINF2 show only approximately:

0.2 nTPM

and:

0.5 nTPM

respectively in adult retina. ("SCARF1 Retina"; "SERPINF2 Retina")

Their whole-body patterns are much more compatible with:

SCARF1 → immune/endothelial biology

and: SERPINF2 → hepatic/systemic fibrinolysis.

This is a useful negative result.

Phase 3A currently provides no reason to elevate either gene as a direct retinal candidate.

They remain in subsequent unbiased analyses but should not receive retinal priority merely because they fall inside the deletion.

15. SLC43A2 Remains Weak as a Direct Retinal Candidate

SLC43A2 shows only approximately:

1.3 nTPM

in adult retina. ("SLC43A2 Retina")

Combined with its placenta/kidney-biased whole-body pattern, this further supports its

Phase 2 classification as primarily an epithelial nutrient-transport gene rather than a leading

direct visual-system candidate. ("SLC43A2 Gene")

A specialized retinal cell population could still express it, so it remains in the single-cell analysis.

16. SMYD4 Is Detectable but Not Retinally Exceptional

SMYD4 averages approximately:

7.2 nTPM

in adult retina. ("SMYD4 Retina")

Its whole-body expression is otherwise ubiquitous and relatively modest. ("SMYD4 Gene")

Thus Phase 3A establishes that SMYD4 is available to operate in retinal tissue but does not provide evidence of retinal enrichment.

This is consistent with its Phase 2 classification as a broad regulatory gene rather than a direct retinal candidate.

17. Initial Assessment of the Phase 2 Membrane/Endosomal Cluster

The pre-specified Phase 2 cluster was:

INPP5K + PITPNA + TLCD2 + WDR81 + RILP + SERPINF1

The adult retinal bulk screen gives:

Gene

Retina nTPM

Phase 3A status

SERPINF1

56.9

Strongly supported

PITPNA

43.9

Strongly supported

INPP5K

14.7

Supported

TLCD2

10.1

Supported

WDR81

0.7

Weak in bulk

RILP

0.4

Weak in bulk

The first four genes therefore satisfy the most basic spatial criterion:

their Phase 2 molecular mechanisms are clearly represented in adult human

retinal tissue.

The two endosomal-trafficking genes do not yet satisfy that criterion strongly at the whole-retina level.

Consequently, the Phase 2 convergence is partially supported rather than fully supported by Phase 3A.

That distinction should be preserved.

18. Unexpectedly Strong Early Result: INPP5K–PITPNA–TLCD2–SERPINF1

Before single-cell analysis, four components of the membrane/lipid model already show measurable adult retinal expression:

PITPNA

PI/PC transfer

+

INPP5K

phosphoinositide turnover

+

TLCD2

phospholipid acyl-chain regulation

+

SERPINF1

PEDF-linked retinal lipid signaling.

This provides a preliminary spatial bridge across four molecular levels:

lipid availability → phosphoinositide chemistry → membrane composition →

extracellular retinal lipid signaling.

The result is stronger than Phase 2 alone because these genes are not simply related in abstract molecular databases.

Their transcripts are all present within the same gross tissue compartment:

adult human retina.

However, whole-retina co-presence does not prove they occur in the same cells.

That becomes the central Phase 3C test.

19. PRPF8 Creates an Independent High-Expression Retinal Axis

PRPF8's adult retinal expression is substantially higher than the membrane candidates except SERPINF1/PITPNA.

This means the deletion contains, within the same retinal tissue:

highly expressed RNA-splicing machinery

and: multiple expressed membrane/lipid regulators.

The later coexpression question is therefore particularly important:

Does PRPF8 occupy the same retinal cell classes that express the membrane-homeostasis genes?

If so, there would be spatial opportunity for splicing perturbation and membrane perturbation to coexist within the same cells.

If not, the mechanisms may remain biologically separated.

20. Noncoding Loci

The noncoding loci require separate treatment.

PITPNA-AS1 and MIR22HG

Standard bulk RNA-seq can potentially detect these lncRNAs, but lower abundance, transcript annotation, and library-selection differences make cross-platform absence difficult to interpret.

They will therefore be analyzed through:

- eyeIntegration gene/transcript matrices;
- appropriate retinal single-cell datasets where captured;
- developmental datasets.

MIR22

Mature miR-22 is not validly quantified from conventional poly(A) mRNA RNA-seq.

MIR22 will therefore be excluded from the ordinary mRNA-expression ranking and analyzed later using small-RNA/miRNA-specific datasets.

RN7SL105P

Because RN7SL105P is a highly homologous pseudogene, apparent RNA-seq reads may be ambiguously mapped.

It will not receive an expression score unless locus-specific mapping quality can be established.

21. Current Phase 3A Classification

Strong adult retinal expression

PRPF8

SERPINF1

PITPNA

Clear/moderate adult retinal expression

RPA1

INPP5K

TLCD2

Detectable but lower adult retinal expression

RTN4RL1

SMYD4

Low whole-retina expression

SLC43A2

WDR81

Very low whole-retina expression

SERPINF2

RILP

SCARF1

This classification is based only on the current bulk retinal screen and is not a final biological ranking.

22. Hypothesis Status After Phase 3A

H1 — Membrane/endosomal co-localization

Partially supported.

PITPNA, INPP5K, TLCD2, and SERPINF1 are clearly present in adult retina.

WDR81 and RILP are weak in whole-retina bulk and require cell-type testing.

H2 — Direct retinal localization

Partially supported at bulk level.

PRPF8, SERPINF1, and RTN4RL1 are all detectable in adult retina, with PRPF8 and SERPINF1 particularly strong.

H3 — WDR81/RILP neural-trafficking overlap

Unresolved.

Whole-retina abundance is low for both.

H4 — Developmental enrichment

Not yet tested.

H5 — Magnocellular specificity

Not yet tested.

H6 — Regulatory overlap

Not yet tested at cell-type resolution.

23. Main Interpretation

The first in silico expression layer supports the project enough to continue, but it does not simply confirm the Phase 2 model.

Instead, it modifies it.

The strongest retinal molecular set at this stage is:

PRPF8 + SERPINF1 + PITPNA + RPA1 + INPP5K + TLCD2

with RTN4RL1 also clearly detectable.

The specific Phase 2 membrane convergence is narrowed provisionally to:

PITPNA + INPP5K + TLCD2 + SERPINF1

because those four demonstrate substantial adult retinal expression.

WDR81 and RILP remain highly credible neuronal proteins but are not strongly supported as whole-retina adult mechanisms from this first dataset.

That is a meaningful negative/qualifying result rather than a failure of the hypothesis.

24. Phase 3A Working Conclusion

Phase 3A demonstrates that several of the strongest Phase 2 candidate mechanisms

possess clear spatial opportunity within adult human retina.

Most notably:

PITPNA

INPP5K

TLCD2

and:

SERPINF1

are simultaneously represented in adult retinal tissue, supporting the existence of an ocular membrane/lipid-regulatory subnetwork.

PRPF8 is strongly expressed in the same gross tissue, while RPA1 is also substantially expressed and RTN4RL1 is clearly detectable.

Conversely, WDR81, RILP, SCARF1, SERPINF2, and SLC43A2 show low whole-retina expression, emphasizing that molecular importance identified in Phase 2 does not automatically imply adult retinal importance.

The appropriate conclusion is therefore:

Phase 3A provides preliminary spatial support for a subset of the Phase 2 visual-neural model, especially the PITPNA–INPP5K–TLCD2–SERPINF1 membrane/lipid axis, while leaving the WDR81–RILP trafficking component unresolved and weakening direct retinal prioritization of several systemic genes.

The next analysis must move below the tissue level.

Whole-retina RNA cannot determine whether these genes occur in the same biological cells.

The decisive next question is:

Which retinal cell populations—rods, cones, bipolar cells, amacrine cells, retinal ganglion cells, Müller glia, RPE, vascular cells, or immune cells—express each affected locus?

Phase 3B — Human Ocular Bulk Expression

Editorial scope note: Phase 3B was defined in the frozen Phase 3 protocol but was not

executed as a separate results section before Phase 3C. Its initial ocular bulk screen was incorporated into Phase 3A above. The original Phase 3B protocol text is reproduced verbatim here so that the Phase 3 record remains complete without inventing a standalone result.

GTEx alone is insufficient for detailed eye biology.

The primary ocular bulk resource will be eyeIntegration, which integrates healthy human ocular RNA-seq datasets processed through a consistent pipeline and provides gene- and transcript-level expression matrices. ("eyeIntegration")

The ocular analysis should include, where available:

- neural retina;
- retinal pigment epithelium;
- choroid;
- optic nerve;
- cornea;
- other ocular tissues as internal comparisons.

The main objective is to determine whether the Phase 2 candidates are actually enriched in:

neural retina and/or RPE relative to non-neural ocular tissues.

Phase 3C — Adult Human Retinal Single-Cell Localization

1. Objective

Phase 3A established that several affected genes are detectably expressed in adult human retina at the whole-tissue level.

Phase 3C asks the more discriminating question:

Which adult human retinal cell populations actually express the affected loci, and do the Phase 2 mechanistically related genes occupy the same cellular compartments?

This distinction is essential because bulk retinal RNA-seq can generate apparent convergence even when different genes are expressed in completely different cell populations.

The principal cell classes considered are:

rod photoreceptors, cone photoreceptors, bipolar cells, amacrine cells, horizontal cells, retinal ganglion cells, Müller glia, astrocytes, microglia, and retinal pigment epithelial cells.

RPE is treated separately from the neural retina because it is an epithelial support layer rather than a retinal neuron, despite its critical role in photoreceptor physiology.

2. Reference Data

The principal contemporary reference is the 2026 Human Retina Cell Atlas, which integrated approximately 3.9 million cells from 125 human donors and identified more than **130 retinal cell types. The atlas integrates multiple published studies with approximately 2.7**

million previously unpublished observations and includes both transcriptomic and chromatinaccessibility information. (Li et al.)

For replication and interpretation, older independent human retinal datasets remain useful. The 2019 Lukowski atlas profiled 20,009 cells from three healthy adult donors and resolved all major neural retinal populations, including rods, cones, bipolar cells, amacrine cells, retinal ganglion cells, horizontal cells, Müller glia, astrocytes, and microglia. (Lukowski et al.)

The current gene-level screen uses accessible Human Protein Atlas single-cell annotations together with independent retinal studies. Because some HPA classifications vary between releases and because the largest 2026 atlas is distributed primarily through interactive and downloadable single-cell resources, release-dependent signals are classified as preliminary rather than treated as replicated facts.

3. Evidence Classification

Phase 3C uses four expression-evidence levels.

Level

Interpretation

Reproduced across independent human data or supported

3 — Strong

by current single-cell evidence plus orthogonal retinal
evidence

Convincing localization in one major human single-cell

2 — Moderate

resource

Suggestive, release-dependent, low-abundance, or

1 — Preliminary

indirectly localized

0 — Unsupported

No reliable adult retinal cell-type localization identified

An unsupported result does not mean that the gene is biologically absent. Low
expression, technical dropout, library chemistry, or rare-cell localization may prevent detection.

4. Adult Retinal Cell-Type Matrix

Strongest current cell-type

Locus

Evidence

Phase 3C interpretation

signal

RPE-enriched

INPP5K

RPE

phosphoinositide

mechanism

RPE-enhanced lipid-transfer

PITPNA

RPE

mechanism

Possible glial/immune

SLC43A2

Microglial-associated signal

localization; weak direct

neural-retinal role

Supports immune/vascular

Microglial/astrocytic/

SCARF1

rather than neuronal
endothelial bias
localization
No convincing retina-
Adult retinal cell of origin

RILP

specific enrichment; broader
unresolved
neuronal association
Broad retinal rather than
Broad neural retina,

PRPF8

uniquely photoreceptor-
including rods and cones
specific requirement
Possible cone-associated

TLCD2

Cone photoreceptor signal
1–2
membrane-lipid mechanism
No strong adult retinal

WDR81

Microglial-associated signal

neuronal localization yet

Strongest current cell-type**Locus****Evidence****Phase 3C interpretation****signal**

No meaningful retinal

Supports systemic rather

SERPINF2

specialization identified

than retinal mechanism

Strongly replicated

SERPINF1**RPE**

RPE/PEDF localization

Broad regulatory expression

No retinal-specific

SMYD4

rather than specialized

enrichment

retinal role

Broad/low-specificity

Housekeeping genome-

RPA1

cellular expression

maintenance mechanism

Bulk retinal expression

confirmed, but RGC

RTN4RL1

Cell of origin unresolved

localization not

demonstrated

PITPNA-AS1

Unresolved

—

lncRNA capture limitation

lncRNA/host-transcript

MIR22HG

Unresolved

—

capture limitation

Not validly measurable by

MIR22

—

Requires small-RNA data

standard scRNA-seq

Pseudogene mapping

RN7SL105P

Unresolved

—

ambiguity

5. INPP5K Localizes Strongly Toward Retinal Pigment Epithelium

Current Human Protein Atlas single-cell annotation classifies INPP5K as:

group enriched in retinal pigment epithelial cells and late spermatids

and places it in the expression cluster:

Retinal pigment epithelial cells — Retinol metabolism. ("INPP5K Single Cell")

This substantially improves the Phase 3A interpretation.

INPP5K is not merely present somewhere in whole retinal tissue. Its expression appears preferentially associated with the RPE compartment.

That provides cellular opportunity for its Phase 2 phosphoinositide mechanism to operate at the interface responsible for:

photoreceptor support

↓

outer-segment turnover

↓

retinoid cycling

↓ membrane trafficking and metabolic exchange.

The current evidence does not show that INPP5K loss disrupts these functions. It establishes only that the enzyme is preferentially positioned within an ocular compartment where intensive membrane processing occurs.

Phase 3C status

RPE localization: supported. Photoreceptor localization: not established as the principal signal. RGC localization: not established.

6. PITPNA Also Shows RPE Enhancement

PITPNA independently shows cell-type-enhanced expression in retinal pigment epithelial cells, in addition to several nonocular populations. ("PITPNA Single Cell")

This result is important because PITPNA and INPP5K were already independently linked in Phase 2 through phosphatidylinositol biology.

Their molecular functions differ:

PITPNA

→ phosphatidylinositol/phosphatidylcholine binding and transfer

whereas:

INPP5K

→ phosphoinositide 5-phosphate hydrolysis.

Phase 3C now adds a spatial relationship:

both are preferentially represented in RPE-associated single-cell expression.

This is substantially stronger evidence of functional convergence than their mere presence in the same whole-retina sample.

It still does not demonstrate physical interaction or epistasis.

Phase 3C status

INPP5K + PITPNA RPE co-localization: supported at the cell-class level.

The appropriate term is:

RPE-centered functional convergence.

7. SERPINF1 Produces the Strongest RPE Signal

SERPINF1/PEDF is currently classified as group enriched in retinal pigment epithelial cells and decidual stromal cells in the Human Protein Atlas. ("SERPINF1 Single Cell")

Independent human retinal single-cell work also identified an RPE cluster characterized by high SERPINF1 and RLBP1 expression. (Voigt et al.)

This provides replication across independent human evidence.

SERPINF1 therefore receives: Expression Evidence Level 3 — Strong.

The result aligns closely with its established PEDF biology.

Phase 2 showed that PEDF contributes to:

- retinal trophic support;
- antiangiogenic regulation;
- photoreceptor survival;
- extracellular-matrix signaling;
- PEDF-R/PNPLA2-linked lipid metabolism.

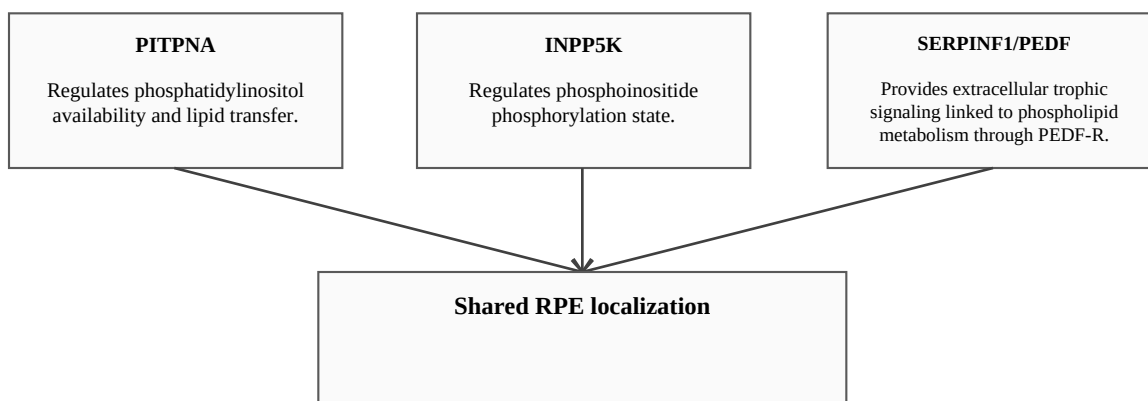
Phase 3C now localizes this mechanism strongly to:

retinal pigment epithelium.

8. An RPE-Centered Molecular Module Now Emerges

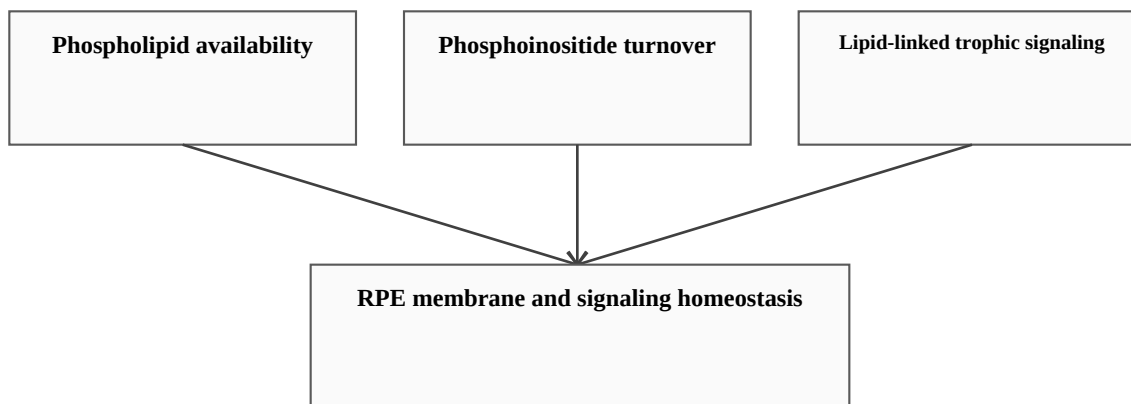
The strongest new Phase 3C result is therefore:

RPE-centered functional convergence



Their functions are chemically different but complementary.

Resulting model



This is currently the strongest cell-type-specific convergence identified anywhere in the project.

Importantly, it differs from the original broad Phase 2 model.

The strongest membrane/lipid convergence in adult retina presently appears to occur primarily at the:

RPE–neural-retina interface

rather than being uniformly distributed throughout retinal neurons.

9. Why the RPE Finding Matters

RPE cells perform unusually membrane-intensive functions.

They support photoreceptors through:

- outer-segment phagocytosis;
- retinoid recycling;
- nutrient transport;
- ion and water regulation;
- growth-factor secretion;
- lipid handling.

Therefore a convergence of lipid and phosphoinositide regulators within RPE is biologically coherent.

However, this observation must not be converted prematurely into:

RPE dysfunction causes the candidate visual phenotype.

At this stage the correct conclusion is narrower:

The deletion simultaneously affects several genes whose normal expression

and molecular functions converge within RPE cells.

That establishes a plausible cellular site of combined dosage perturbation.

10. PRPF8 Is Broadly Expressed Across Neural Retina

Earlier HPA releases classified PRPF8 as enhanced in rod photoreceptor cells and placed it within a photoreceptor/visual-perception expression cluster. ("PRPF8 Single Cell")

However, more recent evidence argues against interpreting PRPF8 as primarily rodspecific.

A 2026 human retinal-organoid study reported PRPF8 expression throughout the neural retina, including both rod and cone photoreceptors, and noted that this distribution is consistent with existing human retinal single-cell data showing broadly uniform retinal expression. (Zimmann et al.)

Therefore the stronger interpretation is:

PRPF8 is broadly required across retinal cell populations, with robust

photoreceptor expression, rather than being a uniquely rod-restricted gene.

This fits its molecular role as a core spliceosomal protein.

Phase 3C status

Adult neural-retinal expression: strong. Rod expression: strong. Cone expression: strong. Strict rod specificity: not supported.

11. PRPF8 Therefore Represents a Different Type of Retinal Risk

The RPE module involves cell-enriched membrane and trophic functions.

PRPF8 is different.

Because it is a core RNA-processing protein expressed broadly through neural retina, reduced PRPF8 dosage—if functionally important—could potentially affect multiple retinal cell populations simultaneously.

Thus:

RPE module

→ spatially concentrated biochemical convergence

whereas:

PRPF8

→ broad retinal information-processing vulnerability.

The two mechanisms could coexist in the same tissue without being part of the same direct pathway.

12. TLCD2 Shows a Possible Cone-Photoreceptor Signal

The Human Protein Atlas v24 classification identifies TLCD2 as cell-type enhanced in **cone photoreceptor cells, along with several nonocular cell populations. ("TLCD2 Single Cell")**

This is potentially important because TLCD2 regulates phospholipid acyl-chain composition and membrane fluidity.

Cone photoreceptors depend on specialized membrane architecture and intense lipid turnover.

However, the TLCD2 signal is currently less secure than the RPE findings because celltype classifications have varied across HPA releases.

Therefore TLCD2 receives:

Expression Evidence Level 1–2: preliminary to moderate.

The current hypothesis is:

TLCD2 dosage loss

↓

possible altered phospholipid composition within cone-associated cells

but not: TLCD2 loss causes cone dysfunction.

Replication in the full 2026 Human Retina Cell Atlas should be required before assigning strong cell-type specificity.

13. The Lipid Cluster Therefore Splits Into at Least Two Adult Retinal Compartments

Phase 3A suggested a broad retinal set:

PITPNA + INPP5K + TLCD2 + SERPINF1.

Phase 3C now indicates that this set is not homogeneous.

The strongest current organization is:

RPE-associated module

PITPNA

INPP5K

SERPINF1

Possible photoreceptor-associated component

TLCD2

particularly:

cone photoreceptors

This distinction is scientifically important.

The deletion may perturb related aspects of membrane biology in adjacent but different retinal compartments rather than through a single intracellular pathway operating in one cell.

14. WDR81 Does Not Show Strong Adult Retinal Neuronal Enrichment

WDR81 is classified by the Human Protein Atlas as cell-type enhanced in microglia, together with several non-neural cell types. ("WDR81 Single Cell")

Phase 3A also found only approximately 0.7 nTPM in whole-retina bulk tissue.

Thus the current adult-retinal evidence does not support strong WDR81 expression in:

- rods;
- cones;
- bipolar neurons;
- amacrine neurons;
- retinal ganglion cells.

The available signal instead suggests a possible microglial association.

This does not weaken WDR81's established neurodevelopmental importance.

It changes where that importance should be investigated.

WDR81 may prove more relevant to:

- developmental neural progenitors;
- brain neurons;
- glial populations;
- dLGN;
- cortical systems

than to mature retinal neurons.

Phase 3C status

Adult retinal-neuronal hypothesis: weakened. Glial/microglial possibility: retained.

Developmental neural hypothesis: still open.

15. RILP Also Fails to Show a Strong Retina-Specific Cell-Type Signal

RILP is associated broadly with a neuronal neurotransmission/signaling expression

cluster, and current HPA data classify it as enhanced in “other brain neurons” among several

non-neural populations. ("RILP Single Cell")

However, no convincing adult-retinal cell-type enrichment emerged from the current screen.

Combined with its very low whole-retina bulk abundance, this argues against making RILP a leading adult retinal mechanism at present.

This is an important negative result.

Phase 2 demonstrated strong neuronal RILP biology.

Phase 3C now suggests that the best place to test that biology may be:

central visual-system neurons rather than mature retina.

Therefore the forthcoming dLGN and V1 analyses become especially important for RILP.

16. The Original WDR81–RILP Retinal Convergence Is Not Supported

Phase 2 predicted that WDR81 and RILP might form a coherent endosomal sequence:

WDR81

→ endosome maturation

↓

RAB7-positive late endosome

↓

RILP

→ organelle transport.

At the molecular level this remains plausible.

At the adult-retina expression level, however, Phase 3C does not currently demonstrate that both genes are strongly expressed in the same retinal neuronal population.

WDR81 shows a microglia-associated signal.

RILP shows a broader neuronal signal without clear retina-specific enrichment.

Therefore:

H3 is not supported in adult retina at this stage.

This does not falsify the molecular pathway.

It means the pathway's spatial opportunity has not yet been demonstrated in mature retinal cells.

It should now be tested in the dLGN, V1, and developmental datasets.

17. RTN4RL1: An Important Negative/Unresolved Finding

Phase 2 identified RTN4RL1 as a strong neural candidate because Nogo-receptor-family

biology directly affects retinal ganglion-cell axons and optic-nerve regeneration.

Phase 3A showed detectable whole-retina expression.

However, current HPA single-cell classification does not identify retinal ganglion cells as an RTN4RL1-enriched population. Instead, the strongest whole-body single-cell specificity appears in spermatogenic cells, while human brain single-nucleus expression shows low overall cell-type specificity. ("RTN4RL1 Single Cell")

Human Protein Atlas protein-level data do support broader neural expression, including neurons, astrocytic/neuropil and microglial/neuropil compartments, but that does not establish a retinal ganglion-cell-specific signal. ("RTN4RL1 HPA")

Therefore the hypothesis:

RTN4RL1 is specifically enriched in adult human retinal ganglion cells

is not currently supported.

This is important because it prevents the Phase 2 optic-nerve literature from being converted into an expression claim that the data do not yet support.

Phase 3C status

Retinal presence: supported by bulk RNA. RGC-specific enrichment: unresolved/not demonstrated. Central neural relevance: still strongly plausible.

18. RPA1 Appears to Behave as a Broad Cellular Maintenance Gene

RPA1 has low overall cell-type specificity in current single-cell resources, with enrichment signals occurring outside the retina, particularly in proliferative or specialized nonretinal populations. Its broader expression pattern is classified as consistent with basic cellular processes. ("RPA1 Single Cell")

This is consistent with RPA1's molecular function.

RPA1 is required for fundamental:

- ssDNA protection;
- replication;
- repair;
- ATR signaling.

Its substantial retinal bulk expression therefore likely reflects a widely distributed **cellular maintenance requirement, rather than specialization within a particular mature retinal**

neuron.

This matters for interpretation.

RPA1 remains the strongest direct heterozygous molecular dosage candidate from Phase 2, but Phase 3C provides no evidence that its dosage vulnerability is retina-specific.

19. SMYD4 Also Shows No Specialized Adult Retinal Localization

SMYD4 currently demonstrates low cell-type specificity overall and no convincing retinal-specific enrichment. Its strongest HPA single-cell enhancement is reported in myonuclei rather than ocular neurons. ("SMYD4 Single Cell")

Thus its Phase 3A retinal bulk expression should be interpreted as:

general expression compatible with a broad regulatory role

rather than:

evidence for a specialized retinal epigenetic mechanism.

The best later opportunity for SMYD4 may be developmental analysis, where epigenetic regulators can have transient effects not apparent in mature tissue.

20. SLC43A2 Shows a Possible Microglial Association

SLC43A2 is not strongly expressed in whole adult retina.

Current single-cell resources nevertheless show an elevated signal involving

microglia/neuropil among several other cell populations, while its overall strongest expression

remains associated with renal and epithelial transport systems. ("SLC43A2 Single Cell")

This supports a limited possibility that SLC43A2 has an immune/glial role within neural tissues.

However, the evidence does not support elevating SLC43A2 into the principal neuralretinal pathway.

Phase 3C status

Photoreceptor role: unsupported. Retinal neuronal role: unsupported. Possible microglial contribution: preliminary.

21. SCARF1 Is Consistent With a Glial/Vascular Rather Than Neuronal Role

SCARF1 shows strong cell-type specificity for several non-neuronal populations, including microglia and astrocytes, and its expression cluster is annotated as brain endothelial cells — blood-brain barrier. ("SCARF1 Single Cell")

This is consistent with its Phase 2 role in:

- scavenger recognition;
- apoptotic-cell clearance;
- immune homeostasis.

Its very low adult-retinal bulk expression makes it unlikely to represent a major photoreceptor or retinal-neuron mechanism.

Instead, if it proves relevant to the visual system, the more plausible route would involve: immune, glial, endothelial, or clearance biology.

22. SERPINF2 Remains Essentially Non-Retinal

SERPINF2 shows strong single-cell enrichment in proximal tubular cells, with liver-associated systemic expression and only very low adult-retinal bulk abundance. ("SERPINF2 Single Cell")

No compelling mature retinal neuronal or RPE cell-type signal emerged.

This strongly reinforces the Phase 2 classification:

SERPINF2 is primarily a systemic hemostatic locus rather than a direct retinal gene.

Its inclusion in later retinal analyses should remain as an unbiased negative control rather than because there is currently a positive retinal hypothesis.

23. Noncoding Loci Require a Separate Assay Strategy

The four noncoding loci cannot be treated like ordinary protein-coding genes.

PITPNA-AS1

As a lncRNA, PITPNA-AS1 may have low abundance and can be poorly captured by droplet-based or 3'-biased scRNA-seq.

Current absence of a strong cell-type annotation is therefore:

unresolved, not evidence of true biological absence.

MIR22HG

The same limitation applies to MIR22HG.

Its host-transcript abundance may be measurable in appropriate total-RNA datasets, but conventional retinal single-cell assays may underrepresent it.

MIR22

Mature miR-22 cannot be validly inferred from ordinary poly(A)-selected scRNA-seq.

Its Phase 3 localization requires: small-RNA sequencing or miRNA-specific expression data.

RN7SL105P

Because RN7SL105P is a homologous pseudogene, short-read mapping may not reliably distinguish it from related 7SL-derived sequences.

No expression score should presently be assigned.

24. Revised Cellular Architecture of the Deletion

Phase 3C now permits the Phase 2/3A model to be reorganized by cellular compartment.

Adult retinal compartment

Most relevant affected loci currently supported

RPE

SERPINF1, INPP5K, PITPNA

Photoreceptors

PRPF8, possible TLCD2 cone enrichment

Broad neural retina

PRPF8, broad RPA1 availability

Microglial/glial-associated

WDR81, SCARF1, possible SLC43A2

Specific RGC enrichment

None demonstrated yet

Adult retinal endosomal WDR81–RILP pair

Not demonstrated

Systemic/non-retinal

SERPINF2 strongly; SLC43A2 predominantly

Unresolved noncoding

PITPNA-AS1, MIR22HG, MIR22, RN7SL105P

This is more biologically informative than the Phase 3A bulk ranking.

25. Hypothesis H1 — Membrane/Endosomal Co-Localization

Original H1 predicted that multiple members of:

INPP5K + PITPNA + TLCD2 + WDR81 + RILP + SERPINF1

would occupy at least one common retinal or visual-neural cell population.

Current result

Partially supported, but substantially revised.

A clear adult RPE module is emerging: INPP5K + PITPNA + SERPINF1.

TLCD2 may instead preferentially involve cones.

WDR81 and RILP are not presently shown to occupy the same adult retinal population.

Therefore the original six-gene same-cell model is:

not supported as a single adult retinal module.

The broader pathway may instead span multiple adjacent cellular compartments.

26. Hypothesis H2 — Direct Retinal Localization

The pre-specified candidates were:

PRPF8

SERPINF1

RTN4RL1.

PRPF8

Strongly supported.

Broad neural-retinal and photoreceptor expression.

SERPINF1

Strongly supported.

Strong replicated RPE localization.

RTN4RL1

Partially supported.

Bulk retinal expression exists, but the specific adult retinal cell population remains unresolved.

Therefore H2 is:

supported for PRPF8 and SERPINF1, unresolved at cell-type level for

RTN4RL1.

27. Hypothesis H3 — WDR81/RILP Neural Trafficking Overlap

In adult retina:

not currently supported.

The two genes do not presently demonstrate convincing shared retinal cell-type enrichment.

This is one of Phase 3C's most important negative results.

The appropriate next test is not to rescue the hypothesis by changing the criterion.

It is to move the same frozen hypothesis into:

human dLGN

and: human V1.

28. Hypothesis H6 — Regulatory Overlap

PRPF8 is broadly present through neural retina and therefore has cellular opportunity to influence transcripts in multiple retinal populations.

The miRNA component cannot yet be evaluated because mature miR-22 requires a different sequencing modality.

Thus: PRPF8 overlap: spatially plausible. MIR22 regulatory overlap: unresolved.

No direct target relationship should yet be inferred.

29. Major New Phase 3C Finding

The principal result is not simply that several deleted genes are expressed in retina. It is that their expression begins to organize into distinct retinal biological compartments.

Strongest current model

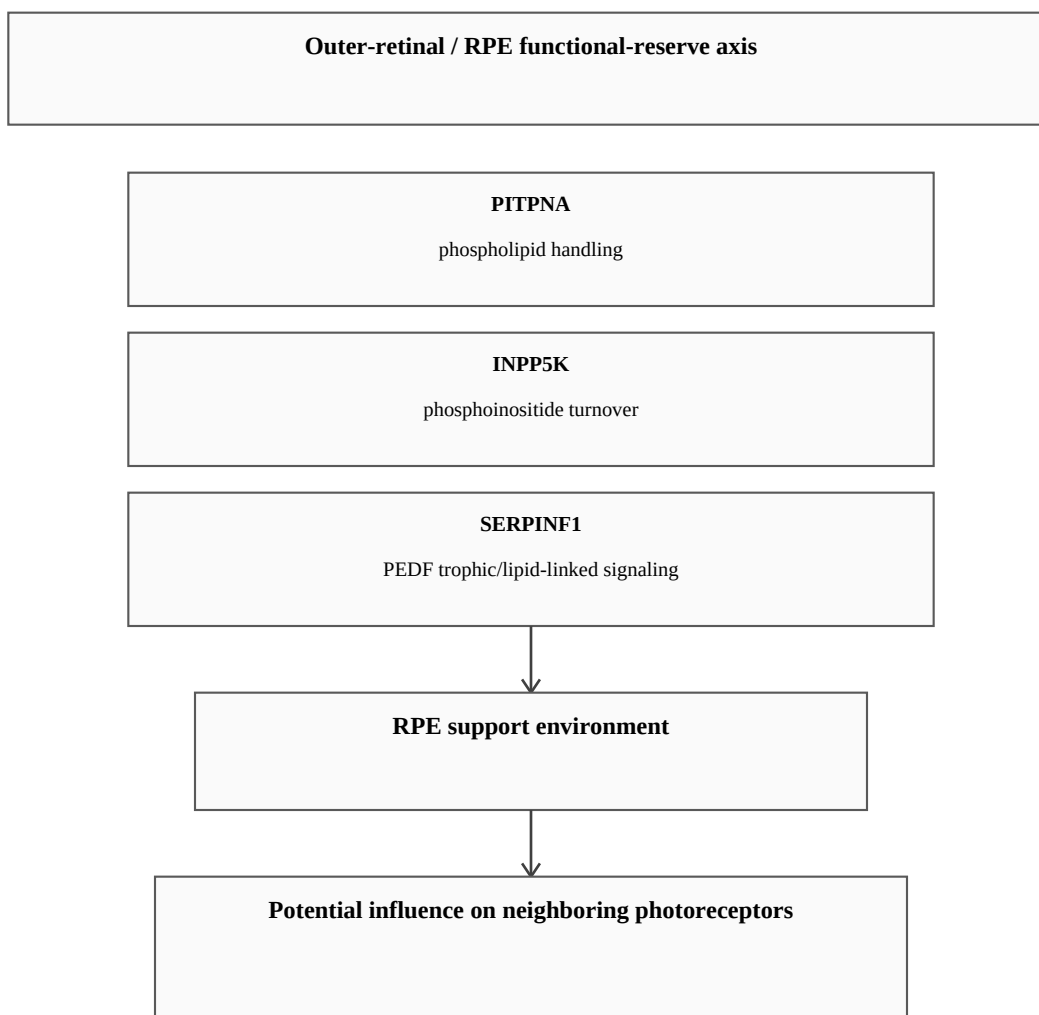
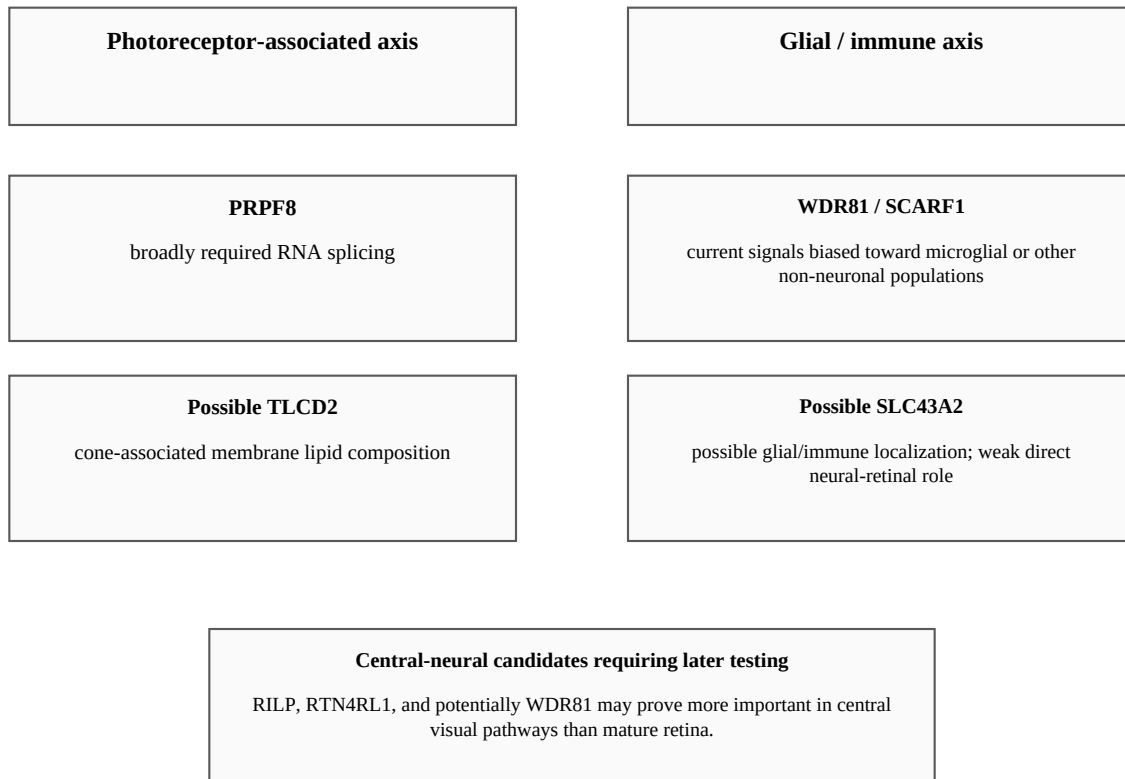


Figure continues on the next page.



WDR81

may prove more important in central visual pathways than mature retina.

30. Implication for the Original Visual-Neurological Question

Phase 3C strengthens the possibility of a retinal biological contribution, but not in the simplistic form initially conceivable.

Current evidence does not suggest:

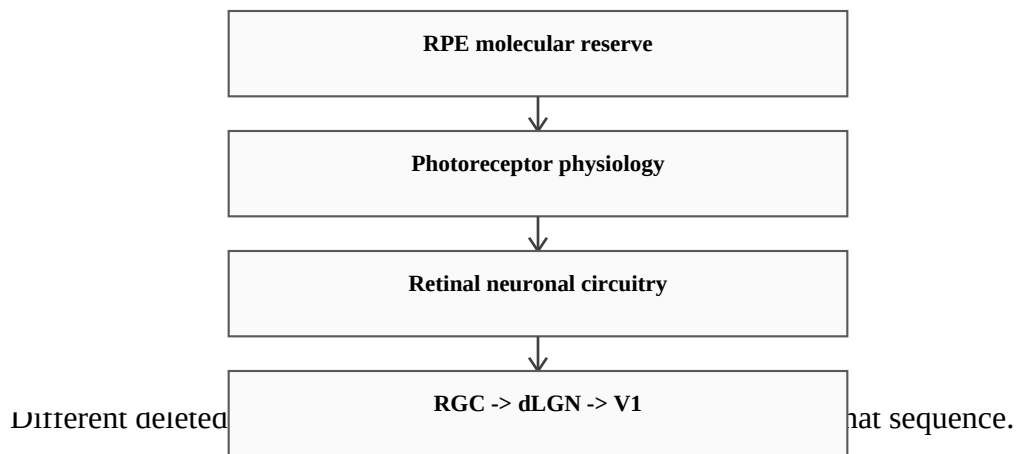
all deleted neural genes are simultaneously weakened inside the same retinal neuron.

Instead it suggests a distributed system:

RPE molecular reserve

↓

Distributed visual-system sequence



This is potentially more biologically realistic than a single-cell, single-pathway model.

It also makes later visual-pathway analysis essential.

31. Important Negative Findings

Phase 3C currently provides no strong evidence for:

RTN4RL1-specific enrichment in retinal ganglion cells.

It provides no strong evidence that: WDR81 and RILP form a coexpressed adult-retinal trafficking module.

It provides no evidence that: SERPINF2 is a significant retinal gene.

It provides no convincing specialized retinal localization for: SMYD4.

And conventional scRNA-seq cannot resolve: mature MIR22 dosage.

These negative results should be retained in the frozen dataset rather than discarded.

32. Current Cell-Type Priority Ranking

For adult retinal biology specifically, the strongest current candidates are:

Rank class**Loci****Reason****SERPINF1, INPP5K, PITPNA,**

Strong cell-type localization plus

Tier A**PRPF8**

established molecular relevance

Relevant expression but weaker retinal

Tier B**TLCD2, RPA1**

specificity or mechanism

Strong prior biology but adult retinal

Tier C**RTN4RL1, WDR81, RILP**

cell localization remains unresolved

Weak/specialized non-neuronal retinal

Tier D**SCARF1, SLC43A2, SMYD4**

evidence

Little evidence for direct adult retinal

Tier E**SERPINF2**

participation

PITPNA-AS1, MIR22HG, MIR22,

Separate assay required

RNA-class/technical limitations

RN7SL105P

This ranking is only for adult retinal localization.

It does not replace the Phase 2 molecular-priority ranking.

33. Evidence Boundary

Phase 3C now permits statements such as:

INPP5K, PITPNA and SERPINF1 show convergent expression within the

RPE compartment.

It permits:

PRPF8 is broadly expressed across human neural retina, including

photoreceptors.

It tentatively permits:

TLCD2 may be enhanced in cone photoreceptors.

It does not permit:

The deletion impairs RPE function.

It does not permit:

The deletion damages cones.

It does not permit:

The RPE module causes visual stress.

And it does not permit:

The genes interact epistatically.

Expression establishes:

cellular opportunity for a mechanism.

It does not establish pathological execution of that mechanism.

34. Phase 3C Working Conclusion

Adult human retinal single-cell analysis substantially refines the bulk-expression model developed in Phase 3A.

The strongest finding is a reproducible RPE-centered convergence involving

SERPINF1, INPP5K, and PITPNA. These genes independently regulate extracellular retinal

trophic signaling, phosphoinositide metabolism, and phospholipid handling, respectively.

Their shared RPE localization provides the first strong evidence that multiple deleted genes with related membrane/lipid functions occupy the same biologically relevant human ocular cell compartment.

PRPF8 represents a second major retinal axis. Rather than being narrowly restricted to rods, current evidence supports broad expression through the neural retina, including rods and cones, consistent with a fundamental spliceosomal function.

TLCD2 provides a preliminary cone-photoreceptor signal that may extend the membranelipid mechanism into neural retina, but replication is required.

Conversely, the expected WDR81–RILP endosomal pair is not currently supported as a shared adult retinal module. WDR81 shows a more microglial-associated expression pattern,

whereas RILP lacks strong retina-specific enrichment and retains a broader neuronal profile. RTN4RL1 also lacks demonstrated retinal ganglion-cell enrichment despite its strong experimental optic-nerve biology.

Phase 3C therefore supports retinal involvement while narrowing and reorganizing

the mechanism.

The strongest present model is no longer:

one six-gene membrane pathway operating uniformly throughout retina.

It is:

an RPE-centered membrane/phosphoinositide/trophic module, a separate photoreceptor-associated RNA/lipid module, and additional neural-trafficking genes whose principal relevance may emerge farther downstream

in the central visual pathway or during development.

That distinction should now be carried forward unchanged into the next analyses.

Phase 3D — Human Dorsal Lateral Geniculate Nucleus Analysis

1. Objective

Phase 3D tests whether the loci affected by the 17p13.3 deletion show preferential

localization within one of the principal human dorsal lateral geniculate nucleus visual relay systems:

magnocellular — M

parvocellular — P

or: koniocellular — K.

This analysis directly addresses one of the project's pre-specified exploratory hypotheses:

H5 — One or more affected loci may show preferential expression in human

magnocellular dLGN neurons.

The analysis is deliberately symmetric.

Evidence favoring P or K neurons would be retained with equal weight, and absence of pathway-specific enrichment is treated as a meaningful negative result.

2. Biological Context

The dorsal lateral geniculate nucleus is the principal thalamic relay between:

retina

↓

dLGN

↓ primary visual cortex — V1.

In primates, visual information has traditionally been divided into magnocellular, parvocellular and koniocellular streams.

These populations differ substantially in:

- morphology;
- receptive-field properties;
- temporal response;
- contrast sensitivity;
- chromatic processing;
- retinal input;
- cortical projection patterns.

The transcriptomic question is whether those physiological distinctions are accompanied by correspondingly strong differences in gene expression.

Current human evidence indicates that the answer is only partly yes.

3. Primary Human Dataset

The principal dataset is the Allen Institute/Bakken et al. Comparative LGN study.

The Allen Institute currently provides processed human LGN expression and metadata for 1,209 nuclei. ("Comparative LGN")

The published analysis used approximately 950 neuronal nuclei from three postmortem

human donors to construct the principal neuronal taxonomy. The study identified:

- three GABAergic neuronal populations;
- two koniocellular populations;
- one combined magnocellular/parvocellular projection-neuron population.

The two koniocellular populations were designated:

Kap

and: Kp.

The remaining major glutamatergic population was designated:

MP. (Bakken et al.)

This point is central to interpretation.

4. Human M and P Neurons Do Not Form Cleanly Separated Transcriptomic Classes

Despite the well-established physiological distinction between magnocellular and parvocellular neurons, human M and P nuclei did not resolve into two clean transcriptomic clusters.

Instead:

M and P neurons largely formed a common MP cluster.

Within that cluster, the investigators identified a continuous M-to-P expression gradient.

Cross-species integration with macaque data confirmed that the human ends of this continuum correspond to more clearly separable macaque M and P populations. (Bakken et al.)

Therefore:

M versus P

in adult human dLGN is better represented transcriptomically as:

a graded molecular distinction

rather than:

two strongly separated gene-expression states.

This places an important constraint on the original magnocellular hypothesis.

5. Published Human M-versus-P Marker Test

The study identified 32 genes showing consistent M-versus-P enrichment across all three human donors.

Validated examples included:

Magnocellular-associated

CRH

SUSD2

with:

ROBO2

also marking the M-directed end of the conserved human/macaque continuum.

Parvocellular-associated

EPHA7

FOXP2.

A pan-M/P marker included:

BTNL9. (Bakken et al.)

The complete published Figure 3 marker set includes genes such as:

- ABCC3
- ADAMTSL3
- AKAP2
- ARHGAP6
- ARHGEF28
- CDH6
- CDH8
- CNKSR3
- COL11A1
- COL21A1
- CRH
- EGFEM1P
- ELL2
- FAM19A5
- FER1L6
- FOXP2
- GALNTL6
- GPC5
- GRID2
- HPSE2
- KCNH7

- KCNJ6
- LUZP2
- MPPED2
- PLCL1
- PRKG1
- PTER
- ROBO2
- RUNX1
- SH3RF2
- SPON1
- SUSP2
- TMTC1
- TOX
- VSNL1
- EPHA7

across the donor-specific and joint M/P comparisons displayed by the study. (Bakken et al.)

6. None of the 17 Deleted Loci Is a Published M/P Marker

The affected loci are:

**INPP5K

PITPNA

SLC43A2

SCARF1

RILP

PRPF8

TLCD2

WDR81

SERPINF2

SERPINF1

SMYD4

RPA1

RTN4RL1

PITPNA-AS1

MIR22HG

MIR22

RN7SL105P.**

None appears among the published genes showing the consistent human M-versus-P marker pattern.

Therefore:

No affected locus currently demonstrates published evidence of preferential

magnocellular transcriptional enrichment in the adult human dLGN.

Equally:

No affected locus emerges as a strong published parvocellular marker.

This is a genuine negative result for the pre-specified analysis.

7. This Does Not Mean the Genes Are Absent From M or P Neurons

The distinction is critical.

A gene can be expressed in:

both M and P neurons

without being: differentially expressed between M and P neurons.

For example, a housekeeping or broadly neuronal protein could be essential to both populations while never appearing in an M-versus-P marker list.

Therefore the result supports:

no demonstrated M-specific enrichment

but does not support:

no expression in magnocellular neurons.

Those are different propositions.

8. Magnocellular Hypothesis Status

The frozen Phase 3 hypothesis was:

One or more affected genes may show preferential expression in human magnocellular dLGN neurons.

The current result is: H5 — Not supported at the published marker level.

None of the affected loci appears among the consistent human M-enriched markers.

Therefore the project should not currently claim that the deletion preferentially targets the magnocellular pathway.

This substantially narrows the original hypothesis.

9. Why This Result Is Particularly Important

A weaker analytical strategy could have proceeded from:

visual stress

↓

historical magnocellular hypotheses

↓

search for any deleted neural gene

↓

claim magnocellular involvement.

Phase 3D does not permit that inference.

When the actual human M/P transcriptomic dataset is examined first, the affected genes do not appear among the genes distinguishing M from P neurons.

Thus:

current transcriptomic evidence does not support a magnocellular-specific

mechanism for this deletion.

This is one of the clearest examples so far of the project's phenotype-neutral design protecting against confirmation bias.

10. Koniocellular Analysis

Human dLGN transcriptomics resolves the K populations more clearly than M and P.

The Bakken et al. study identified:

Kap

and:

Kp

koniocellular types.

The principal reported K-associated markers include:

CALB1

CAMK2A

PRKCG

with:

PENK

selectively identifying the Kp population.

PENK-positive Kp neurons were anatomically confirmed in posterior K1 and K2 layers using RNA in situ hybridization. (Bakken et al.)

11. No Deleted Locus Is a Defining K Marker

None of the 17 affected loci is among the primary K-defining markers reported by the human study.

Thus the current evidence also fails to support:

selective koniocellular involvement.

At the transcriptomic-marker level, the deletion does not map specifically onto:

M

P

or:

K

identity.

12. Gene-by-Gene dLGN Specificity Screen

Published M-specific

Published P-specific

Affected locus

Defining K marker

Current interpretation

marker

marker

No relay-class

INPP5K

No

No

No

specificity demonstrated

No relay-class

PITPNA

No

No

No

specificity demonstrated

No relay-class

SLC43A2

No

No

No

specificity demonstrated

No relay-class

SCARF1

No

No

No

specificity demonstrated

Neural relevance

RILP

No

No

No

retained; M/P/K

specificity unsupported

Likely broader cellular

PRPF8

No

No

No

role rather than relay-

class marker

No relay-class

TLCD2

No

No

No

specificity demonstrated

Neural relevance

WDR81

No

No

No

retained; M/P/K

specificity unsupported

No dLGN specialization

SERPINF2

No

No

No

demonstrated

Strong retinal role does

SERPINF1

No

No

No

not extend to dLGN

subtype specificity

No relay-class

SMYD4

No

No

No

specificity demonstrated

Consistent with broad

RPA1

No

No

No

cellular maintenance

role

Neural signaling

RTN4RL1

No

No

No

relevance retained; no

M/P/K specialization

PITPNA-AS1

No evidence

No evidence

No evidence

Unresolved

MIR22HG

No evidence

No evidence

No evidence

Unresolved

Published M-specific**Published P-specific****Affected locus****Defining K marker****Current interpretation****marker****marker**

Not appropriately

Not appropriately

Not appropriately

MIR22

measured as mature

Separate assay required

measured

measured

miRNA

Mapping/function

RN7SL105P

No evidence

No evidence

No evidence

unresolved

The table should be interpreted as a specificity matrix, not an absolute-expression matrix.

13. RILP

RILP was specifically important entering Phase 3D because Phase 3C failed to establish a strong adult retinal localization despite compelling neuronal trafficking biology.

The dLGN analysis does not rescue RILP as a magnocellular-specific candidate.

RILP is not among the published:

- M markers;
- P markers;
- K markers.

Thus its neuronal trafficking mechanism may still operate in visual-relay neurons, but current evidence does not associate it preferentially with one dLGN visual channel.

This leaves RILP as a:

general neural-trafficking candidate rather than a magnocellular candidate.

14. WDR81

The same conclusion applies to WDR81.

Phase 2 established strong neural-development and endosomal PI3P biology.

Phase 3C suggested that adult retinal expression may be more glial than neuronal.

Phase 3D finds no evidence that WDR81 is a defining:

- magnocellular;
- parvocellular;
- koniocellular

transcriptomic marker.

Therefore:

WDR81 remains a strong neural mechanism but not a visual-relay-class-

specific mechanism.

Its strongest remaining Phase 3 opportunity may lie in:

- developmental brain;
- broader thalamic neurons;
- cortical populations.

15. RTN4RL1

RTN4RL1 was another important candidate because Phase 2 established direct axonal and optic-nerve relevance.

However, RTN4RL1 also fails to appear among the published M/P/K defining gene sets.

Therefore its functions involving:

- axonal growth;
- CSPG recognition;

- synaptic structural regulation

should currently be interpreted as:

general neuronal structural biology

rather than evidence of a magnocellular pathway specialization.

This is consistent with Phase 3C, where retinal ganglion-cell-specific enrichment was also not demonstrated.

Two independent Phase 3 analyses therefore now fail to support a highly specialized RTN4RL1 visual-channel localization.

16. PRPF8

PRPF8 likewise shows no M/P/K marker specificity.

This is biologically unsurprising.

Its function is fundamental spliceosomal RNA processing.

Phase 3C demonstrated broad neural-retinal availability rather than narrow retinal subtype restriction.

Phase 3D now supports the same general interpretation centrally:

PRPF8 is more plausibly a broadly required cellular factor than a transcriptomic marker of a specific visual relay stream.

This does not weaken PRPF8's retinal relevance.

It weakens only a hypothesis of magnocellular-selective PRPF8 vulnerability.

17. PITPNA, INPP5K, TLCD2 and SERPINF1

The Phase 3C retinal lipid module does not reappear as an M-specific dLGN transcriptional module.

None of:

PITPNA

INPP5K

TLCD2

or:

SERPINF1

is among the published human M/P discriminating genes.

Therefore the strongest Phase 3C finding remains principally:

retinal/RPE biology

rather than:

magnocellular thalamic biology.

This begins to spatially separate the two hypotheses.

18. A Critical Property of the Human dLGN Dataset

Failure to find M-specific enrichment should not be exaggerated.

The original study itself found that human M and P neurons are remarkably similar transcriptomically relative to their major physiological and morphological differences.

Only a limited set of genes consistently distinguishes the two.

The human M/P cells remain one MP transcriptomic population with a molecular gradient rather than completely distinct cell types. (Bakken et al.)

Consequently:

physiological specialization need not require large differences in steady-state

gene expression.

A deleted gene expressed equally in M and P neurons could theoretically have different functional consequences if the two neuronal types impose different physiological demands.

That possibility remains open.

However, there is currently no positive transcriptomic evidence supporting it.

19. Independent Primate Evidence Supports the Complexity of LGN Cell Identity

A 2026 reanalysis of the Allen macaque LGN dataset identified additional heterogeneity within the conventional M/P/K categories:

- two magnocellular subpopulations;
- two parvocellular subpopulations;
- three koniocellular subpopulations.

The analysis used 1,309 macaque LGN nuclei and found substantial within-class transcriptomic complexity. (Sun, Renshaw, and Pezaris)

This reinforces an important conceptual point:

M, P and K are themselves not molecularly homogeneous populations.

The recent macaque analysis does not highlight RTN4RL1, RILP, WDR81, PRPF8, PITPNA, INPP5K, SERPINF1, TLCD2, or RPA1 among the principal subtype genes discussed in the article. (Sun, Renshaw, and Pezaris)

This is supporting primate evidence only.

It does not replace the human analysis.

20. Revised Interpretation of the Magnocellular Hypothesis

At the start of the project, one conceivable mechanism was:

17p13.3 deletion

↓

specific disruption of genes important to the magnocellular pathway

↓

altered temporal/contrast processing

↓

candidate visual-stress phenotype.

Phase 3D does not support the critical middle step.

The current evidence instead favors:

17p13.3 deletion

↓

multiple retinal and broadly neuronal molecular perturbations

↓

possible visual-system vulnerability

without evidence of: preferential magnocellular transcriptional targeting.

This is a substantially more conservative and defensible model.

21. Consequence for the Visual-Stress Hypothesis

The historical literature around visual stress has sometimes invoked magnocellular dysfunction.

Phase 3D means that this project should not use that literature as a shortcut to connect

the CNV with visual stress.

At present:

The genomic findings do not independently predict a magnocellular-specific

lesion.

If later physiological evidence were to implicate magnocellular processing, that would require a separate explanation.

The present transcriptomic evidence does not supply it.

22. What Phase 3D Does Support

Phase 3D does support a different conclusion.

Several of the affected genes have substantial:

- neural;
- retinal;
- membrane;
- trafficking;
- signaling

functions.

But those mechanisms appear more likely to operate as:

broad cellular systems

than as molecular identities unique to one thalamic visual channel.

This is especially consistent with:

PRPF8

→ universal spliceosomal machinery

RPA1

→ genome-maintenance machinery

RILP

→ general neuronal endosomal transport

WDR81

→ general endosomal signaling

RTN4RL1

→ broadly neuronal structural signaling.

23. Cross-Phase Comparison

Phase 3 has now produced three distinct spatial findings.

Phase 3A — Whole Retina

Strong expression of:

****PRPF8**

SERPINF1

PITPNA

RPA1

INPP5K

TLCD2**

with weaker adult-retinal signals for several trafficking genes.

Phase 3C — Retinal Cell Types

Strongest convergence:

INPP5K + PITPNA + SERPINF1

within the:

RPE compartment

with PRPF8 broadly available through neural retina and possible TLCD2 cone association.

Phase 3D — dLGN

No affected locus is currently identified as a strong:

M

P

or:

K

marker.

Thus the most specific spatial convergence discovered so far remains:

RPE-centered rather than magnocellular-centered.

24. Hypothesis Status After Phase 3D**H1 — Membrane/endosomal colocalization**

Partially supported, principally through RPE rather than dLGN subtype specialization.

H2 — Direct retinal localization

Supported for PRPF8 and SERPINF1; strong RPE convergence with INPP5K/PITPNA.

H3 — WDR81/RILP neural-trafficking overlap

Not demonstrated in adult retina and not demonstrated as M/P/K-specific in human

dLGN.

Still open in cortical/developmental analyses.

H4 — Developmental enrichment

Not yet tested.

H5 — Magnocellular specificity

Not supported.

This is now a meaningful negative result.

H6 — Regulatory overlap

Still unresolved.

25. Evidence Boundary

Phase 3D permits the statement:

None of the 17 affected loci has emerged as a published defining marker of

adult human magnocellular, parvocellular, or koniocellular dLGN identity.

It also permits:

The available human transcriptomic data do not support preferential

magnocellular targeting by the deletion.

It does not permit:

The affected genes are absent from magnocellular neurons.

Nor does it permit:

Magnocellular physiology cannot be affected.

Those stronger claims would require direct quantitative expression and physiological data.

26. Technical Limitation

The Allen Institute makes the complete processed human LGN expression matrix and metadata publicly downloadable, containing 1,209 nuclei. ("Comparative LGN")

The published paper and associated supplementary analyses provide the marker-level M/P/K results used here, including the 32 reproducible human M/P genes and the defining K markers. The underlying processed dataset is also deposited under GSE182211 and through the Allen Comparative LGN resource. ("GSE182211"; "Comparative LGN")

A final frozen Phase 3 matrix should ultimately include direct numerical extraction of each affected gene's expression across individual M, P and K nuclei.

Therefore the present conclusion is appropriately classified as:

strong marker-level negative evidence for subtype specificity

rather than:

complete quantitative absence of differential expression.

27. Phase 3D Working Conclusion

Human dLGN transcriptomics does not currently support the hypothesis that the 17p13.3 microdeletion preferentially disrupts a magnocellular molecular program.

Human magnocellular and parvocellular relay neurons themselves are transcriptionally much more similar than their well-known physiological differences might suggest. They form a largely shared MP transcriptomic population with a comparatively subtle M-to-P expression gradient. A limited set of genes—including CRH, SUSP2 and ROBO2 toward the magnocellular end and EPHA7 and FOXP2 toward the parvocellular end—defines this gradient. None of the 17 loci affected by the index deletion appears among the published reproducible M/P marker genes. (Bakken et al.)

Likewise, the affected loci do not overlap the principal human koniocellular markers CALB1, CAMK2A, PRKCG and PENK.

Accordingly:

The pre-specified magnocellular-specificity hypothesis is not supported by

current adult human dLGN transcriptomic evidence.

This result does not weaken the broader proposition that the deletion may affect visualneural biology. Instead, it changes the likely mechanism.

The strongest spatial evidence remains the RPE-centered INPP5K–PITPNA–

SERPINF1 convergence, combined with broad retinal PRPF8 expression and additional neural

mechanisms that do not appear restricted to a particular thalamic visual stream.

The current model should therefore shift from:

a magnocellular-specific defect

toward:

a distributed retinal and neural functional-reserve model in which several molecular systems may be altered without preferentially targeting one

classical M/P/K visual channel.

That is a stronger conclusion scientifically because it follows the transcriptomic evidence rather than the original phenotype hypothesis.

Phase 3C - Integrated Expression and Spatiotemporal Analysis

Research Scope and Status

Phase 3 evaluates where and when genes disrupted by the GRCh37/hg19

chr17:1,411,408-1,841,103 microdeletion are expressed, with particular emphasis on the retina, retinal pigment epithelium (RPE), dorsal lateral geniculate nucleus (dLGN), primary visual cortex (V1), and developing human neocortex. The phase does not treat expression as evidence of disease causation. Its purpose is narrower: to determine whether genes already identified as structurally disrupted and mechanistically plausible in Phases 1 and 2 have spatial and temporal opportunity to influence visual-neural biology.

The deletion interval contains thirteen protein-coding loci considered in the Phase 3 expression analysis (INPP5K, PITPNA, SLC43A2, SCARF1, RILP, PRPF8, TLCD2, WDR81, SERPINF2, SERPINF1, SMYD4, RPA1, and RTN4RL1) and four noncoding annotations

(PITPNA-AS1, MIR22HG, MIR22, and RN7SL105P). Phase 3 preserves the structural interpretation established previously: the deletion begins within INPP5K and ends within RTN4RL1, with the latter losing essentially the canonical coding sequence on the affected allele. Those structural facts are background constraints rather than expression results.

Abstract

Phase 3 integrates bulk tissue expression, single-cell and single-nucleus transcriptomics,

retinal cell-type resources, human dLGN transcriptomics, adult human brain atlases, developmental cortical multiomics, and spatial fetal-cortex transcriptomics. The strongest direct cell-localized convergence remains retinal and RPE-centered: INPP5K and PITPNA show RPE-associated expression, while SERPINF1 is strongly associated with retinal pigment epithelial cells and has an established retinal trophic context. Human Protein Atlas data classify INPP5K as group-enriched in retinal pigment epithelial cells and PITPNA as cell-type enhanced in retinal pigment epithelial cells; SERPINF1 is likewise group-enriched in retinal pigment epithelial cells (Human Protein Atlas, “INPP5K”; Human Protein Atlas, “PITPNA”; Human Protein Atlas, “SERPINF1”). PRPF8 is broadly expressed in retina rather than restricted to one visual cell class, TLCD2 has a cone-photoreceptor signal that remains preliminary for mechanistic interpretation, WDR81 is more compatible with microglial/non-specific cellular expression than a mature retinal neuronal module, and RTN4RL1 is detectably expressed in human retina but has not been demonstrated as retinal-ganglion-cell-specific (Human Protein Atlas, “TLCD2”; Human Protein Atlas, “WDR81”; Human Protein Atlas, “RTN4RL1”).

Central visual-system analysis does not support a magnocellular-specific model. Singlecell and single-nucleus profiling of primate dLGN found that human magnocellular and parvocellular projection neurons are transcriptomically similar relative to their marked

physiological differences; the affected 17p13.3 loci do not emerge as defining published M-, P-, or K-cell markers (Bakken et al.). Adult cortical analysis similarly supports distributed neural expression rather than a deletion-wide V1-specific signature. Developmental analysis strengthens evidence that selected genes have temporal opportunity during cortical development, especially WDR81, RTN4RL1, RILP, and SERPINF1, but independent human developmental atlases demonstrate readily detectable V1 specialization without identifying the deletion interval as a conspicuous V1-defining program (Bhaduri et al.; Wang et al.; Qian et al.). The resulting model is therefore one of distributed functional reserve: several deleted genes plausibly constrain membrane, phosphoinositide, endosomal, trophic, RNA-processing, or genome-maintenance systems at different visual-system sites and developmental stages, but the complete causal chain from heterozygous dosage loss to biochemical disturbance to visual-neural phenotype has not been demonstrated.

Methods and Evidentiary Rules

Phase 3 was designed as an expression-opportunity analysis rather than a phenotype-

association screen. Evidence was interpreted in the following order: genomic disruption, known molecular function, tissue or cell-type expression, spatiotemporal overlap, pathway convergence, perturbational evidence, phenotype association, and causality last. Accordingly, detection of a transcript in retina or cortex establishes only that the gene is biologically available in that compartment. It does not establish that the heterozygous deletion changes transcript abundance, protein abundance, enzymatic flux, synaptic physiology, or perception.

Adult tissue-level observations were treated as lower-resolution evidence than cell-resolved data. Single-cell and single-nucleus findings were prioritized when the relevant cell type was represented and assay limitations were understood. Developmental comparisons were

not interpreted as longitudinal trajectories unless the underlying study design supported that inference. Cross-sectional BrainSpan or Allen-derived standardized expression values from different donors and ages were therefore treated as suggestive temporal observations rather than within-subject fold changes. BrainSpan provides RNA-seq and microarray expression data across multiple cortical and subcortical structures and developmental periods, including prenatal laser-microdissection resources (Allen Institute for Brain Science).

Noncoding loci required additional caution. Standard polyadenylated or nuclear RNA workflows are not reliable assays for mature microRNA abundance; consequently, MIR22 cannot be considered absent or low merely because it is poorly represented in ordinary singlenucleus RNA-seq. MIR22HG and MIR22 were also treated as related host/microRNA annotations rather than independent deletion events. PITPNA-AS1 and RN7SL105P remain technically underdetermined in the available visual-system single-cell resources.

Evidence labels used throughout this phase are: established, when supported directly by the relevant human dataset or well-validated cell-resolved resource; suggestive, when supported by cross-sectional, lower-resolution, or indirect evidence; inference, when a mechanistic interpretation combines independent observations without direct experimental demonstration; and unsupported, when the proposed specificity or relationship is not demonstrated by the reviewed evidence.

Phase 3A: Adult Whole-Tissue and Ocular Expression

The initial adult expression screen asked whether the affected genes were detectably expressed in retina and brain before attempting cell-type localization. This screen supported substantial adult retinal expression opportunity for PRPF8, SERPINF1, PITPNA, RPA1, INPP5K, and TLCD2, with lower or less discriminatory whole-retina signals for several other

interval genes. Whole-tissue expression was not treated as evidence that all of these genes act in the same retinal cell. The Human Protein Atlas retina resource illustrates the interpretive problem: the retina expresses a large fraction of the human proteome, so bulk detection alone has limited specificity (Human Protein Atlas, “Retina Proteome”).

RTN4RL1 is a useful example. Current Human Protein Atlas data report detectable retinal RNA, with a GTEx-derived consensus around 7.9 nTPM, supporting retinal opportunity but not assigning that expression to retinal ganglion cells or another specific neuronal population (Human Protein Atlas, “RTN4RL1”). TLCD2 likewise shows retinal RNA in bulk data, while its single-cell annotation includes cone photoreceptor enhancement; this makes cone involvement a reasonable candidate for follow-up, not an established cone-specific deletion mechanism (Human Protein Atlas, “TLCD2”).

Phase 3A therefore established a permissive rather than causal result: multiple affected

genes are present in adult ocular tissue, so retinal mechanisms cannot be excluded on expression grounds. The phase then moved to cell-resolved localization because bulk retinal abundance could not distinguish photoreceptor, RPE, glial, vascular, immune, or inner-retinal contributions.

Phase 3B-3C: Retinal Cell Types and RPE-Centered Convergence

Human retinal single-cell resources provide a substantially stronger anatomical test. Cowan and colleagues sequenced 285,441 single cells from retinal organoids across developmental time points and from adult human peripheral retina, fovea, pigment epithelium, and choroid, demonstrating that retinal disease-related genes can show marked cell-type specificity (Cowan et al.). Independent adult retinal atlases likewise resolve foveal and peripheral

retinal cell populations at single-cell scale (Yan et al.). These datasets make it possible to ask whether mechanistically related deletion genes actually occupy compatible retinal compartments.

The strongest Phase 3 retinal result is the RPE-centered convergence of INPP5K, PITPNA, and SERPINF1. Human Protein Atlas single-cell annotations classify INPP5K as group-enriched in retinal pigment epithelial cells and place it in an RPE-associated expression cluster. PITPNA is cell-type enhanced in RPE, while SERPINF1 is group-enriched in RPE and decidual stromal cells (Human Protein Atlas, “INPP5K”; Human Protein Atlas, “PITPNA”; Human Protein Atlas, “SERPINF1”). This is important because the three genes are not merely co-expressed names: their known molecular functions occupy complementary membrane, lipid, phosphoinositide, and trophic biology. Phase 3 does not claim that they form a single biochemical complex or that their heterozygous loss produces a measured RPE defect. It establishes spatial opportunity for functional convergence.

PRPF8 presents a different pattern. It is broadly expressed in retina and is biologically relevant to retinal disease under established variant mechanisms, but its expression is not restricted to one RPE or photoreceptor subtype. Consequently, PRPF8 is best treated as a broadly available RNA-processing constraint rather than evidence for the RPE membrane module. Its presence may matter if future perturbational work demonstrates altered splicing in affected visual cells, but expression alone cannot supply that missing step.

TLCD2 provides a weaker but potentially informative signal. Current Human Protein Atlas single-cell annotation lists cone photoreceptor cells among TLCD2-enhanced cell types, while bulk retina RNA is detectable (Human Protein Atlas, “TLCD2”). This supports a preliminary cone-associated expression opportunity. It does not establish that TLCD2 dosage

affects cone membrane composition in the deletion carrier, nor that such an effect contributes to photophobia, pattern glare, color processing, or another visual phenotype.

WDR81 did not produce the expected mature retinal neuronal convergence with RILP. Human Protein Atlas annotations describe WDR81 as having low overall cell-type specificity with enhancement in microglia among several cell types (Human Protein Atlas, “WDR81”). RILP, by contrast, is assigned to a neuronal neurotransmission-and-signaling expression cluster in the broader single-cell atlas (Human Protein Atlas, “RILP”). These findings do not demonstrate a shared adult retinal neuronal trafficking module. The WDR81-RILP relationship therefore remained mechanistically plausible from Phase 2 but anatomically unresolved in adult retina.

The retinal conclusion is correspondingly asymmetric. INPP5K-PITPNA-SERPINF1 represents the strongest direct cell-localized convergence in Phase 3. PRPF8 contributes broad retinal RNA-processing opportunity; TLCD2 has preliminary cone-associated support; RTN4RL1 is expressed in retina but lacks demonstrated retinal-ganglion-cell enrichment; and WDR81-RILP co-localization in mature retinal neurons has not been established.

Phase 3D: Human dLGN and the Magnocellular Hypothesis

Phase 3D tested whether the deletion genes preferentially map to the magnocellular

visual pathway. This was a particularly important falsification test because a magnocellular model could otherwise become an attractive but weakly constrained explanation for visual discomfort or contrast-related symptoms. Bakken and colleagues profiled primate dLGN using single-cell and single-nucleus RNA sequencing, including nuclei from anatomically defined human magnocellular, parvocellular, and koniocellular regions. They identified four principal thalamocortical neuron types in primates but found relatively few strong transcriptional

differences between M and P populations despite their well-known morphological and physiological differences (Bakken et al.).

In human dLGN, M and P neurons largely occupy a shared transcriptomic MP cluster with a gradient rather than two completely separate molecular classes. Published M-associated markers include CRH and SUSD2, whereas EPHA7 is among validated P-associated markers; the study also identifies K-cell classes separately (Bakken et al.). None of the affected 17p13.3 loci emerged as a defining published M, P, or principal K marker in this analysis.

This is a negative result with a specific meaning. It does not show that the deletion genes are absent from magnocellular neurons. It shows that the available human dLGN transcriptomic evidence does not support preferential magnocellular enrichment. Because M and P neurons are themselves transcriptomically similar, absence from a marker list cannot be treated as proof of equal functional importance in both populations. Nevertheless, the data fail to support a deletionwide magnocellular-specific mechanism. Hypothesis H5 is therefore classified as unsupported.

Phase 3E: Adult Human V1 and Central Cortical Distribution

Phase 3E extended the analysis from the thalamus to human cerebral cortex, with

particular interest in V1. Large adult human brain atlases demonstrate extensive neuronal and glial diversity across the brain and provide a framework for distinguishing broadly expressed genes from regionally specialized programs. Siletti and colleagues profiled more than three million nuclei from approximately one hundred dissections and identified thousands of transcriptomic subclusters, including area-specific cortical neurons (Siletti et al.).

Within this framework, several affected genes have credible central-neural expression, but the interval does not emerge as a coherent adult V1-specific program. RILP gains support as

a neuronal gene because the Human Protein Atlas places it in a neuronal neurotransmission- and signaling expression cluster (Human Protein Atlas, “RILP”). RTN4RL1 has human cerebral cortex and retinal expression and is classified as tissue enriched in cerebral cortex in current Human Protein Atlas protein-summary metadata, but this does not establish V1 or cortical-layer specificity (Human Protein Atlas, “RTN4RL1”). PITPNA and INPP5K remain broadly available rather than demonstrably V1-selective.

SERPINF1 retains trophic and vascular/glial opportunities in central tissue but does not reproduce the same sharply localized RPE pattern in V1. WDR81 and RILP remain compatible with endosomal and neuronal trafficking biology, yet Phase 3E found no evidence that WDR81-RILP-RTN4RL1 form a synchronized adult V1 cell-type module. PRPF8 and RPA1 are even less regionally discriminatory because their core functions in RNA processing and genome maintenance make broad expression biologically expected.

The adult cortical result therefore favors distributed vulnerability over a simple visual cortex lesion model. The genes can be biologically active in cortex without being selectively enriched in V1, layer 4, intratelencephalic neurons, corticothalamic neurons, or another single cortical class. This distinction became the rationale for Phase 3F: if adult specificity was weak, developmental timing might still reveal a period during which the deleted genes are unusually relevant to visual-cortex formation.

Phase 3F: Human Visual-Cortex Development and Developmental Enrichment

Phase 3F initially centered on the developmental multiomic atlas of Wang and colleagues

and was subsequently broadened to independent cortical arealization and spatial-transcriptomic resources. Wang et al. analyzed 27 brain specimens and 38 biological samples spanning five developmental stages from the first trimester through adolescence. Paired single-

nucleus RNAseq and ATAC-seq were generated from PFC and V1, producing 243,535 nuclei after initial quality control and a final cortical dataset of approximately 232,000 nuclei. The study resolved progenitor, neuronal, glial, immune, and vascular classes and explicitly modeled regional developmental differences (Wang et al.).

The atlas is particularly valuable because it identifies a V1-specific layer-4 intratelencephalic trajectory rather than assuming that all layer-4 neurons are equivalent. Common and V1-specific EN-L4-IT populations share an early trajectory and diverge later in development. The authors reported 1,908 differentially expressed genes distinguishing V1-specific from common EN-L4-IT neurons and highlighted markers such as CUX1 and KCNIP1. Their analysis used sample-level pseudobulk logic with age and dataset effects and Benjamini-Hochberg multiple-testing correction (Wang et al.). None of the affected 17p13.3 loci is among the conspicuous published markers defining that V1-specific branch.

A crucial evidence boundary was maintained throughout the Table 18 investigation. The supplementary archive containing the full differential-expression table is publicly listed, but the exact per-gene rows for all affected loci were not successfully retrieved in the research environment. Consequently, Phase 3 does not label the deletion genes statistically nonsignificant in that comparison. The justified statement is narrower: they are not among the highlighted defining V1-specific EN-L4-IT markers, while their exact effect sizes and adjusted P values remain an archival verification item.

The broadened search materially strengthens the interpretation because independent datasets show that V1 specialization is readily detectable when present. Bhaduri and colleagues profiled ten major brain structures and six neocortical areas during peak neurogenesis and early gliogenesis. They found extensive area-specific differential expression in the second trimester,

including in radial glia, with areal signatures becoming more pronounced as cells differentiate toward excitatory neurons. Their results support strong frontal-versus-occipital molecular patterning rather than a cortex that is transcriptionally homogeneous during development (Bhaduri et al.).

Qian and colleagues provide an even stronger spatial constraint. Using MERFISH and complementary approaches, they constructed a spatial atlas exceeding 18 million fetal cortical cells across eight cortical areas and seven developmental time points. They detected a discrete V1-V2 molecular boundary by approximately gestational week 20 and found early synaptogenesis-related changes in V1-specific layer-4 neurons (Qian et al.). Thus, the available technology and datasets are capable of detecting visual-cortex-specific developmental programs. The failure of the 17p13.3 interval to emerge as a conspicuous defining program cannot simply be attributed to an inability of developmental transcriptomics to resolve V1. Developmental Candidate: WDR81

WDR81 remains one of the strongest early-development candidates in the interval, but the evidence supports developmental opportunity rather than V1 specificity. Cross-sectional developing-brain resources suggest prenatal cortical expression, and its established endosomal biology gives that expression mechanistic relevance. However, WDR81 is not identified as a major V1 arealization determinant in the Bhaduri, Wang, or Qian studies. The Phase 3F classification is therefore: developmental neural relevance, suggestive to moderate; preferential V1 specification, unsupported. Developmental Candidate: RTN4RL1

RTN4RL1 is detectably expressed in human retina and brain and has a neuronal cell surface biology compatible with neurite and circuit maturation. BrainSpan and Allen developmental resources provide temporal expression opportunity, while current Human Protein

Atlas data show cerebral-cortex enrichment at the tissue level (Allen Institute for Brain Science; Human Protein Atlas, “RTN4RL1”). The expression pattern is better described as developmentally dynamic than as a simple prenatal peak. Because RTN4RL1 also suffers nearcomplete canonical coding loss on the affected allele, it remains a high-priority mechanistic candidate. Yet it is not established as a V1-specific arealization gene. Developmental Candidate: RILP

RILP retains a credible neuronal trafficking role but has weaker evidence for early cortical specification. Its neuronal expression-cluster assignment supports central-neural opportunity (Human Protein Atlas, “RILP”), while the developmental atlases do not identify it as a major V1 arealization or layer-4 determinant. The most conservative interpretation is that RILP may matter during neuronal maturation and endolysosomal cargo transport rather than during initial specification of V1 identity. Developmental Candidate: SERPINF1

SERPINF1 bridges the strongest retinal result and the developmental-cortical analysis. Its RPE enrichment is direct, while a 2026 developmental-brain analysis reported a rapid increase in SERPINF1 expression between approximately 24 gestational weeks and four months after birth (Musante et al.). This observation is secondary to that study's primary disease question, so it should not be over-weighted; nevertheless, it supports a developmental trophic opportunity during the late-prenatal to early-postnatal transition. It does not establish V1 specificity.

Functional Ordering without a Demonstrated Molecular Chain

The broadened developmental evidence permits a cautious temporal functional model: WDR81 is most compatible with earlier endosomal and developmental regulation; RTN4RL1 with later structural, neurite, and circuit-maturation opportunity; and RILP with neuronal endolysosomal transport during maturation. This ordering is an inference assembled from

independent functional and expression observations. It is not evidence that one gene activates the next, that the genes form a shared transcriptional module, or that heterozygous loss produces a sequential biochemical cascade.

Wang et al. also inferred hundreds of enhancer-associated regulatory programs from paired RNA and chromatin-accessibility data. The presence of a mapped regulatory landscape makes a shared deletion-specific V1 regulon testable in principle. At present, however, no published result establishes that the affected genes constitute a coordinated V1-specific eRegulon. Regulatory hypothesis H6 therefore remains technically testable but unsupported at the deletion-module level.

Integrated Gene-Level Interpretation

INPP5K. Strong RPE localization opportunity; phosphoinositide biology makes the RPE finding mechanistically coherent. Developmental cortical expression is compatible with a broader role, but V1 specificity is not demonstrated. Evidence status: established RPE expression opportunity; developmental regional specificity unsupported.

PITPNA. RPE-enhanced expression and neuronal/synaptic biology support membranelipid opportunity in retina and brain. The strongest Phase 3 use of PITPNA is as part of the RPE-centered convergence, not as a V1 marker. Evidence status: established RPE opportunity; V1 specificity unsupported.

SLC43A2. Expression opportunity is broad and not strongly visual-system-specific in the reviewed Phase 3 resources. Evidence status: low discriminatory value for Phase 3.

SCARF1. The available expression pattern is more compatible with immune/vascular biology than a specialized retinal-neuronal or V1 program. Evidence status: low support for a direct visual-neuronal localization hypothesis.

RILP. Central neuronal expression is supported, and its trafficking function remains mechanistically relevant. Adult retinal neuronal co-localization with WDR81 and developmental V1 specificity are not demonstrated. Evidence status: moderate neuronal opportunity; specific visual module unresolved.

PRPF8. Broad retinal and neural expression is expected for a core spliceosomal factor. It remains biologically important but spatially nondiscriminatory. Evidence status: strong expression opportunity; low regional specificity.

TLCD2. Bulk retinal expression and cone-photoreceptor enhancement provide a preliminary cell-type lead. Evidence status: suggestive cone-associated opportunity; mechanistic and dosage consequences untested.

WDR81. Developmental and central-neural opportunity is supported, with microglial enhancement in current single-cell annotations. It remains a strong developmental candidate because of independent endosomal biology, not because it is a V1-specific marker. Evidence status: moderate developmental relevance; V1 specificity unsupported.

SERPINF2. Phase 3 provides little evidence for a specialized visual-neural expression pattern. Evidence status: low visual-system discriminatory value.

SERPINF1. Strong RPE enrichment and late-prenatal/early-postnatal developmental expression dynamics make SERPINF1 one of the most spatially informative genes in the interval. Evidence status: strong retinal/RPE opportunity; developmental trophic opportunity supported; V1 specificity unsupported.

SMYD4. Broad/low-specificity expression provides little anatomical discrimination. Evidence status: low Phase 3 specificity.

RPA1. Broad expression is consistent with genome-maintenance function. Its importance is more likely to emerge through dosage-sensitive cellular stress than visual-system localization. Evidence status: strong general cellular opportunity; low visual specificity.

RTN4RL1. Retinal and cortical expression plus severe allelic coding disruption make RTN4RL1 a high-priority neuronal candidate. Cell-surface and circuit-maturation biology are compatible with visual-neural relevance, but retinal-ganglion-cell, V1-layer, and magnocellular specificity remain unproven. Evidence status: moderate-to-high neural opportunity; regional specificity unsupported.

PITPNA-AS1 / MIR22HG / MIR22 / RN7SL105P. Available Phase 3 assays are inadequate for confident visual-system localization of these noncoding loci, especially mature MIR22. Evidence status: technically underdetermined; no negative inference from ordinary snRNA-seq is permitted.

Hypothesis Adjudication

H1: Membrane/endosomal colocalization. Partially supported. The strongest cellocalized evidence is the RPE-centered INPP5K-PITPNA-SERPINF1 configuration. A comparable deletion-wide neuronal module has not been demonstrated in retina, dLGN, or V1.

H2: Direct retinal localization. Supported for selected genes, especially INPP5K, PITPNA, SERPINF1, PRPF8, TLCD2, and RTN4RL1 at differing levels of cellular resolution. Direct localization does not establish pathological effect.

H3: WDR81-RILP trafficking overlap. Mechanistically plausible but anatomically unresolved. Adult retinal neuronal co-localization is not demonstrated; developmental evidence is more compatible with temporally distributed roles.

H4: Developmental enrichment. Partially supported. Several genes have credible developmental cortical opportunity, but the interval is not demonstrated as a V1-specific developmental program.

H5: Magnocellular specificity. Unsupported. Human dLGN transcriptomics does not identify the affected loci as defining magnocellular markers, and no later developmental evidence restores this hypothesis.

H6: Shared V1 regulatory overlap. Presently unsupported. Modern multiomic atlases make the question testable, but no shared affected-gene V1-specific eRegulon has been established.

Interpretive Synthesis: Developmental and Cellular Functional Reserve

The most defensible lay and technical synthesis of Phase 3 is a functional-reserve model. Several genes in the deletion are expressed in biologically relevant visual or neural compartments, and some occupy complementary molecular systems. In RPE, the convergence is unusually concrete because INPP5K, PITPNA, and SERPINF1 share cell-type opportunity while contributing to phosphoinositide, lipid/membrane, and trophic biology. In the developing cortex, WDR81, RTN4RL1, RILP, SERPINF1, PRPF8, and RPA1 provide different forms of developmental or cellular opportunity, but not a single V1-specific transcriptional program.

The term functional reserve is preferable to assuming a fixed fifty-percent loss of function. A heterozygous deletion removes one genomic copy, but the downstream protein or pathway effect depends on transcriptional compensation, protein stability, feedback regulation, redundant pathways, cell type, developmental stage, and physiological demand. Phase 3 therefore supports the proposition that the deletion could reduce reserve in several molecular systems; it does not quantify that reduction.

Likewise, the project has not yet demonstrated a complete chemical or molecular chain linking the deletion to a visual-neural phenotype. Individual biochemical functions are known: INPP5K acts in phosphoinositide metabolism; PITPNA participates in phosphatidylinositol/phosphatidylcholine transfer; WDR81 and RILP participate in endosomal/lysosomal trafficking; SERPINF1/PEDF has trophic and lipid-linked retinal biology; PRPF8 is a core spliceosomal component; RPA1 supports replication and genome maintenance; and RTN4RL1 is a neuronal surface receptor-family protein. What remains unproven is the connecting chain from heterozygous dosage reduction to altered pathway chemistry in the relevant human visual cell, then to altered circuit function, and finally to a reproducible phenotype.

Negative Findings and Falsification Value

Phase 3 generated several negative findings that materially constrain the research. First,

the deletion does not presently support a magnocellular-specific model. Second, adult V1 does not reveal a deletion-wide region- or layer-specific expression program. Third, the mature retinal WDR81-RILP trafficking pair was not demonstrated as a shared neuronal module. Fourth, the developmental interval genes do not emerge among the conspicuous V1-specific layer-4 markers in modern human developmental atlases. Fifth, no shared V1-specific regulatory network among the affected genes has been established.

These negative findings are not failures of the project. They reduce the number of plausible mechanistic models and prevent the analysis from drifting toward a preferred phenotype narrative. The strongest surviving model is more distributed: retinal/RPE membranotropic vulnerability combined with central-neural developmental, trafficking, RNA-

processing, and genome-maintenance constraints. This model is less anatomically simple but better aligned with the evidence.

Limitations

The principal limitation is that expression is not perturbation. None of the reviewed atlases directly measures the molecular state of cells carrying this specific heterozygous deletion. Normal-reference expression establishes opportunity, not dosage response. A gene may be highly expressed yet fully compensated after heterozygous loss, or modestly expressed yet functionally dose-sensitive.

Second, cell-resolved atlases have sampling and technical limitations. Single-nucleus methods can underrepresent cytoplasmic transcripts, rare cell populations can be missed, and regional dissections are imperfect. The dLGN study itself notes practical limitations in isolating thin K layers (Bakken et al.). Spatial assays solve some localization problems but often measure restricted gene panels or snapshots rather than complete developmental trajectories.

Third, developmental datasets are generally cross-sectional across different donors. Apparent age-related changes should not be interpreted as within-person trajectories unless the analysis explicitly models age across samples. The older Allen/BrainSpan signals used in Phase 3F were therefore treated as supportive context, not definitive temporal kinetics.

Fourth, the full gene-level statistics from Wang et al. Supplementary Table 18 remain incompletely extracted for the affected loci. This prevents a formal statement that each gene is statistically nonsignificant in the V1-specific versus common EN-L4-IT comparison. The correct record is that none is among the conspicuous published defining markers and exact per-gene statistics remain pending archival verification.

Finally, the available evidence does not justify attributing Irlen Syndrome, visual stress, photophobia, pattern glare, cortical hyperexcitability, epilepsy, retinal degeneration, or another clinical phenotype to this deletion. If later phases address such phenomena, they should be decomposed into measurable traits and tested against molecular and physiological data rather than treated as a single genetically established syndrome.

Phase 3 Final Conclusion

Phase 3 establishes that the 17p13.3 deletion intersects multiple genes with credible

spatial and temporal opportunity in the human visual system. The strongest direct anatomical convergence is RPE-centered and involves INPP5K, PITPNA, and SERPINF1. Additional genes contribute broader retinal, neuronal, cortical, developmental, or cellular-maintenance opportunity. Human dLGN data do not support preferential magnocellular enrichment. Adult cortical data favor distributed rather than V1-specific expression. Developmental multiomic and spatial atlases demonstrate that V1-specific programs are detectable, yet the affected interval does not presently emerge as a major defining V1 developmental module.

The resulting evidence supports moderate mechanistic plausibility for a distributed reduction in visual-neural developmental and cellular functional reserve. It does not establish a direct dosage-to-chemistry-to-circuit-to-phenotype chain. Phase 4 should therefore shift from expression opportunity toward perturbational evidence: whether heterozygous reduction of the highest-priority genes measurably changes phosphoinositide handling, membrane composition, endosomal trafficking, trophic signaling, RNA processing, genome-stress responses, neurite/circuit maturation, or combinations of these processes in relevant human cellular models.

Phase 3 Frozen Record

Phase 3 may be considered substantively complete subject to one archival refinement:

retrieval of exact affected-gene rows from Wang et al. Supplementary Table 18. That unresolved lookup should remain recorded as a quantitative limitation rather than converted into a negative biological result. The principal conclusions are sufficiently replicated across independent retinal, dLGN, adult-brain, developmental multiomic, and spatial-transcriptomic resources to support transition to Phase 4.

Phase 4

Dosage Sensitivity, Perturbation, Functional Convergence, and Integrated Adjudication

Abstract

This frozen Phase 4 master report consolidates Phases 4A through 4G of the project "Genomic and Molecular Analysis of a 17p13.3 Microdeletion and Its Candidate Visual-Neurological Mechanisms." The structural reference remains GRCh37/hg19 chr17:1,411,408-1,841,103, a nominal 429,696-bp heterozygous deletion affecting at least seventeen annotated loci, including thirteen protein-coding genes. Phase 4 tested whether heterozygous dosage reduction can move from genomic plausibility into experimentally observable molecular, cellular, developmental, visual-system, and seizure-relevant consequences. The phase explicitly separated four genetic states - heterozygous loss, biallelic or complete loss, hypomorphic reduction, and specific pathogenic variants - and treated causality as the final rather than initial inference.

Across the complete audit, the strongest evidence supports a multi-layer functionalreserve model rather than a single deleted "photosensitivity gene." RPA1 provides the clearest direct human haploinsufficiency phenotype, SERPINF1 provides direct heterozygous dosage reduction of PEDF, INPP5K provides one-copy physiological evidence outside the retina and strong phosphoinositide/lysosomal mechanism, and several other genes produce relevant phenotypes under severe or variant-specific perturbation. PRPF8, WDR81, SERPINF1/PEDFassociated biology, and related systems converge on photoreceptor integrity; PITPNA, WDR81, RILP, and INPP5K converge on membrane identity and intracellular material handling; SERPINF1, RPA1, and SCARF1 define a layered protection-repair-clearance architecture; and WDR81, INPP5K, RTN4RL1, and RILP converge on neuronal development

and circuit organization. These convergences are biologically meaningful but do not constitute demonstrated synergy or epistasis.

The retinal evidence favors reduced resilience, impaired maintenance, delayed recovery, and stress vulnerability substantially more strongly than increased photon-response gain. The strongest partial-dose retinal physiology identified during Phase 4 shows reduced electroretinographic output, not amplified output. Complete RTN4RL1 loss in human neurons alters network construction but decreases synapse density and synchronous calcium-network activity, arguing against a simple generalized-hyperexcitability interpretation. MIR22 provides the strongest seizure-relevant locus-level evidence, but the decisive chronic epilepsy phenotype was shown after complete loss and a prior epileptogenic insult, not after heterozygous loss or photic stimulation.

Phase 4F localized the central causal gap. Human photosensitive epilepsy depends on

frequency-sensitive visual-cortical responses, inhibitory containment, and abnormal large-scale network propagation. The deletion has substantial biology at the retinal end and meaningful general neuronal/white-matter biology downstream, but no current experiment demonstrates abnormal lateral geniculate physiology, V1 flicker responses, visual-cortical inhibition, 10-20-Hz entrainment, pathological occipital-to-distributed-network propagation, a photoparoxysmal response, or visually triggered seizures produced by this exact CNV. Phase 4G therefore classifies the deletion as a biologically consequential, mechanistically plausible candidate modifier of visual-neural and epileptogenic reserve, not an established cause of epileptic photosensitivity.

Phase 4 is frozen as Version 1.0 as of 15 September 2026. The frozen interpretation may

be superseded only by explicitly reopened analysis or by new perturbational, experimental, or clinical evidence. Phase 5 is reserved for perturbational validation against independent in silico or existing experimental datasets.

Precedence

1. Structural and Evidentiary Precedence

The structural coordinates used throughout Phase 4 are the original GRCh37/hg19 coordinates chr17:1,411,408-1,841,103. GRCh38 and current annotations may be used to reconcile transcripts or expression resources, but they do not replace the primary CNV definition. The interval includes thirteen protein-coding genes - INPP5K, PITPNA, SLC43A2, SCARF1, RILP, PRPF8, TLCD2, WDR81, SERPINF2, SERPINF1, SMYD4, RPA1, and RTN4RL1 - and four noncoding loci - PITPNA-AS1, MIR22HG, MIR22, and RN7SL105P. INPP5K and RTN4RL1 are boundary genes. The reported-coordinate model removes substantial INPP5K coding sequence and approximately 99 percent of the canonical RTN4RL1 coding sequence on the affected allele, illustrating why genomic overlap percentage cannot be equated with coding disruption percentage.

The controlling evidence hierarchy for Phase 4 was genomic evidence -> molecular mechanism -> expression/spatiotemporal opportunity -> perturbational evidence -> cellular phenotype -> developmental consequence -> visual-system physiology -> phenotype association -> causality. This hierarchy was used because rare-CNV interpretation is especially vulnerable to overreading knockout phenotypes, assuming that one-copy loss produces exactly 50 percent protein function, or transferring a specific dominant-negative or gain-of-function disease mechanism to a simple deletion.

2. Dosage-Sensitivity Precedence

RPA1 provides the strongest direct precedent that a one-copy genomic loss inside this broader chromosomal region can produce a measurable human molecular and cellular phenotype. O'Driscoll and colleagues studied contiguous-gene deletion cell lines in which RPA1 was haploinsufficient and found reduced RPA1 abundance, impaired ATR-dependent DNA-damage signaling, defective checkpoint behavior after ultraviolet exposure, and increased stress-associated nuclear fragmentation. Restoration of RPA1 rescued key abnormalities, supporting a gene-specific haploinsufficiency effect rather than an unspecified contiguous-deletion artifact (O'Driscoll et al.).

SERPINF1 provides a second clinically relevant dosage precedent. Heterozygous carriers of truncating SERPINF1 variants had reduced SERPINF1 transcript levels and significantly lower serum PEDF concentrations without the severe skeletal phenotype of biallelic SERPINF1 disease (Al-Jallad et al.). This establishes that one functional copy can measurably lower the gene product even when a classic Mendelian phenotype is absent.

INPP5K heterozygous mouse data show that one-copy reduction can alter physiological signaling outside the retina. Ijuin and colleagues found increased insulin sensitivity and altered Akt-dependent glucose uptake in heterozygous SKIP/Inpp5k-deficient mice, demonstrating that a one-copy state is not necessarily molecularly silent (Ijuin et al.). By contrast, WDR81 and several other loci show dramatic phenotypes primarily under biallelic or severe loss, while PITPNA displays substantial developmental buffering through the closely related paralog PITPNB. These contrasting precedents required gene-specific rather than uniform dosage assumptions.

SERPINF2 serves as an important methodological positive control: heterozygous alpha2antiplasmin deficiency can produce a clear biochemical dosage phenotype while remaining mechanistically remote from retinal photon detection. Phase 4 therefore treated "dosagesensitive" and "photosensitivity-relevant" as separate axes rather than synonyms.

3. Molecular and Chemical Precedence

3.1 TLCD2 and Membrane Composition

Ruiz and colleagues showed that human TLCD1/2 influence membrane phospholipid fatty-acyl composition and physical fluidity. TLCD2 knockdown altered incorporation of longchain polyunsaturated fatty acids into phosphatidylcholine and phosphatidylethanolamine and protected cells from palmitate-induced membrane rigidification under the tested conditions (Ruiz et al.). The result establishes a real membrane-composition phenotype but also supplies counterevidence against a simple assumption that TLCD2 loss is intrinsically harmful.

Petkevicius and colleagues later showed tissue- and paralog-dependent phosphatidylethanolamine effects in Tlcd-family mouse models, emphasizing context and redundancy.

3.2 SERPINF1/PEDF and PNPLA2/PEDF-R

PEDF is a retinally important secreted protein encoded by SERPINF1. Subramanian and colleagues identified functional PEDF-R/PNPLA2 interactions and linked PEDF signaling to phospholipase/lipase activity and retinal cell survival (Subramanian et al.). Bernardo-Colon and colleagues subsequently showed that Pnpla2 ablation causes photoreceptor degeneration, altered retinal proteins and lipid species, apoptosis, and attenuated retinal electrophysiology; heterozygous animals already demonstrated measurable physiological impairment (Bernardo-Colon et al.). Rebutini and colleagues showed that Serpinf1 deletion produces senescence-like

RPE changes, reduced Pnpla2/PEDF-R expression, and accumulation of rhodopsin and lipids after outer-segment phagocytosis (Rebustini et al.). Hara and colleagues further demonstrated that PNPLA2 mobilizes retinyl esters from RPE retinosomes and that loss delays 11-cis-retinal regeneration and dark adaptation (Hara et al.).

The PEDF-DHA literature supplies an additional stress-protection precedent. Mukherjee, Marcheselli, and Bazan demonstrated production of the DHA-derived mediator neuroprotectin D1 in human RPE cells and showed protection against oxidative-stress-induced apoptosis (Mukherjee et al.). This supports a chemically plausible SERPINF1/PEDF-linked retinal stressdefense layer without establishing that heterozygous SERPINF1 loss alone reproduces the complete downstream phenotype.

3.3 INPP5K

McGrath and colleagues demonstrated that INPP5K hydrolyzes PI(4,5)P₂ to PI4P on autolysosomal membranes and is required for autophagic lysosome reformation. Strong loss produces PI(4,5)P₂ accumulation, PI4P depletion, persistent clathrin-associated reformation tubules, enlarged autolysosomes, lysosome depletion, and impaired autophagic homeostasis; wild-type INPP5K or reduction of PI(4,5)P₂ synthesis can rescue key defects (McGrath et al.). The evidence is mechanistically strong but is principally derived from skeletal-muscle systems rather than retina.

3.4 WDR81

Liu and colleagues identified WDR81/WDR91 as negative regulators of endosomal PI3P and early-to-late endosome conversion. WDR81/WDR91 loss increases Beclin1/VPS34-associated activity, elevates PI3P-rich early-endosomal identity, enlarges endosomes, and impairs receptor degradation (Liu et al.). This places WDR81 upstream in a

membrane-identity and cargo-maturation system distinct from INPP5K's autolysosomal PI(4,5)P₂/PI4P function.

3.5 RILP

RILP is a Rab7A effector and dynein-associated adaptor rather than an enzyme. Yap and colleagues demonstrated that disruption of the RAB7A-RILP interaction decreases retrograde late-endosome transport in dendrites, impairs dendritic cargo clearance, and reduces dendritic arborization even when somatic lysosomal degradation remains comparatively preserved (Yap et al.). This distinguishes physical cargo transport from terminal degradation and supports a mechanochemical rather than catalytic interpretation of RILP.

3.6 PRPF8

PRPF8 is a core U5 spliceosomal scaffold, not the catalytic metal-binding enzyme or ATPase of the spliceosome. Human retinal-organoid models carrying pathogenic heterozygous PRPF8 variants show that retina-specific transcript processing is unusually vulnerable to PRPF8 dysfunction. The H2309P model demonstrates altered splice-site selection, ciliary/polarity abnormalities, and photoreceptor loss, while the Y2334N model preserves early retinal specification but later develops markedly reduced photoreceptor inner/outer-segment structures and retinal splicing abnormalities (Zimmann et al.; Atkinson et al.). These findings strongly establish PRPF8 retinal relevance but do not demonstrate that whole-allele haploinsufficiency behaves like the pathogenic mutant proteins.

3.7 RPA1 and Photochemical Stress

The RPA1 precedent becomes relevant to light injury because retinal cells can experience photochemically generated DNA damage. Sparrow, Zhou, and Cai showed that A2E-containing RPE exposed to 430-nm illumination accumulates oxidative DNA lesions, including 8-oxo-dG,

and that a singlet-oxygen quencher reduces the damage (Sparrow et al.). Combining this independent retinal-stress literature with direct RPA1 haploinsufficiency establishes a credible but still untested retinal-reserve hypothesis: the same light-associated lesion burden could become harder to manage when DNA-damage-response reserve is reduced.

3.8 Secondary Systems

SCARF1 mediates C1q-associated recognition and efferocytosis of apoptotic cells; SCARF1 deficiency impairs clearance and allows apoptotic and secondary-necrotic material to accumulate (Ramirez-Ortiz et al.). SLC43A2/LAT4 is a facilitated-diffusion transporter for neutral essential amino acids, with global or tissue-specific deletion producing strong systemic transport phenotypes but limited direct retinal evidence (Guetg et al.; Rajendran et al.). SMYD4 is a SET/MYND-domain methyltransferase with experimentally demonstrated developmental and chromatin effects, including H3K4-associated regulation and HDAC1 interaction in zebrafish cardiac development, but no established visual-neural haploinsufficient phenotype (Xiao et al.). These systems were therefore retained as secondary reserve mechanisms rather than core photosensitivity drivers.

4. Cellular-Phenotype Precedence

Several constituent-gene perturbations cross the threshold from molecular mechanism into bona fide retinal or neuronal cellular phenotypes. Pathogenic PRPF8 variants produce human retinal-organoid photoreceptor and outer-segment abnormalities. Severe WDR81 dysfunction causes progressive photoreceptor loss in mice, and wild-type WDR81 rescue restores survival. The PEDF-R/PNPLA2 system produces photoreceptor deformation, apoptosis, altered retinal proteins, and reduced ERG output. These independent systems therefore converge on photoreceptor integrity through different upstream mechanisms.

At the RPE level, complete *Serpinf1* loss causes senescence-associated changes, impaired outer-segment processing, and rhodopsin/lipid accumulation, while heterozygous *Serpinf1* loss already reduces downstream *Pnpla2* expression. *PNPLA2* loss further demonstrates delayed retinoid mobilization and dark adaptation. The source literature thus supports genuine RPE support-system dysfunction but does not establish that the complete heterozygous CNV itself has been tested under visible-light challenge.

For neural cells, *RTN4RL1* is especially informative. Wang and colleagues showed that complete loss in human neurons increases axonal and dendritic elaboration but decreases synapse formation, multiple synaptic proteins, and synchronous calcium-network activity (Wang et al.). This provides strong evidence that the gene can reorganize developing human neuronal networks while simultaneously arguing against a simplistic hyperexcitability model.

5. Developmental Precedence

WDR81 provides the strongest early corticogenesis precedent. Carpentieri and colleagues showed that *WDR81* loss alters endosomal EGFR trafficking, reduces MAPK signaling, slows radial-glial proliferation during mid-to-late neurogenesis, and ultimately reduces brain size. A Cyclin D2 manipulation rescues the proliferation phenotype, providing mechanistic continuity from endosomal trafficking to cell-cycle control (Carpentieri et al.). The experiments principally involve strong or complete loss, so transfer to heterozygous CNV dosage remains unresolved.

PITPNA/PITPNB studies provide an explicit compensation precedent: severe combined loss disrupts cortical progenitor polarity, Golgi phosphoinositide signaling, interkinetic nuclear migration, and cortical development, whereas one remaining functional copy of either paralog can support grossly normal brain development. This demonstrates that pathway essentiality does not imply single-gene haploinsufficiency.

RTN4RL1 and RILP occupy later developmental windows. RTN4RL1 loss alters neurite growth and synaptogenesis in human neurons; RILP-dependent transport is required for normal dendritic arborization. The two effects can even point in opposite morphological directions, establishing topological convergence on circuit construction without directional convergence. SERPINF1/PEDF contributes to postnatal retinal maturation and vascular/stress biology, while PRPF8 pathogenic variants show preserved early retinal specification followed by later photoreceptor maturation failure. MIR22 differs again: complete loss leaves baseline brain structure comparatively normal but worsens chronic epilepsy after an epileptogenic insult, placing its strongest effect in activity-dependent remodeling rather than constitutive brain formation (Silva et al.).

6. Human CNV and White-Matter Precedence

The most directly relevant clinical CNV precedent comes from interstitial 17p13.3 deletions located between YWHAE/CRK and PAFAH1B1. Emrick and colleagues reported multifocal white-matter abnormalities in five of six individuals and narrowed a smallest overlapping region to approximately 345 kb containing eleven RefSeq genes, with RTN4RL1 proposed as the strongest candidate (Emrick et al.). This evidence is substantially closer to the present interval than classic 17p13.3 deletion syndromes involving PAFAH1B1 or YWHAE/CRK. It supports neurological consequence and incomplete penetrance but does not provide a replicated photoparoxysmal or visually triggered seizure phenotype.

7. Photosensitive-Epilepsy Precedence

The disease physiology sets a high bar for any proposed genetic mechanism. Photosensitive epilepsy is not simply excessive retinal photon detection. Timar and colleagues found that PSE patients show increased occipital PPR power particularly during 10-20-Hz

stimulation, while ordinary transient and steady-state visual-evoked responses are not uniformly elevated. Propagating PPRs are associated especially with abnormal occipitofrontocentral connectivity (Timar et al.). Benedetto and colleagues further reported greater visually evoked EEG energy but reduced phase coherence and unstable alpha dynamics in photosensitive idiopathic generalized epilepsy (Benedetto et al.). Earlier MEG work found gamma-band phase clustering before transitions into PPRs (Parra et al.).

Pattern-sensitive epilepsy further localizes the critical transformation to organized cortical visual processing. Spatial frequency, orientation, contrast, and stimulus extent alter epileptogenicity, and classic work argued against a trigger confined to the LGN (Soso, Lettich, and Belagum; "Neurophysiological Aspects of Pattern-Sensitive Epilepsy"). TMS studies demonstrate reduced normal inhibitory influence from visual to motor cortex in PPR-positive patients, supporting a propagation/inhibitory-containment mechanism rather than a purely retinal one (Strigaro et al.). Source-localization studies show occipital-to-parietal-to-frontal recruitment, with thalamic participation becoming more prominent in some generalized events ("Representation and Propagation of Epileptic Activity").

CHD2 supplies a useful genetic benchmark. Galizia and colleagues found rare CHD2 variants enriched in photosensitive epilepsy and showed that *chd2* knockdown enhanced photosensitivity in zebrafish (Galizia et al.). No locus within the present 17p13.3 interval currently has comparable combined human-genetic and controlled-photic experimental evidence.

Our Findings

8. Phase 4A - Dosage Sensitivity and Photosensitivity-Centered Gene Audit

Phase 4A audited all thirteen affected protein-coding genes under a five-level evidence

scale: A for direct heterozygous functional evidence; B for direct heterozygous molecular or biochemical dosage evidence; C for strong perturbational evidence principally from severe, biallelic, knockout, or otherwise non-equivalent models; D for mechanistic plausibility without sufficiently relevant dosage perturbation; and E for insufficient meaningful evidence. Plus and minus modifiers indicate position near a tier boundary. The grade measures dosage/perturbation evidence, not photosensitivity relevance itself.

Gene

Core Phase 4A

Grade

Photosensitivity

Principal limitation

mechanism

priority

INPP5K

PI(4,5)P2/PIP3 5-

B+

High

One-copy physiology

phosphatase; membrane

exists outside retina;

and lysosomal
retinal heterozygous PI
phosphoinositide
phenotype untested.
control PITPNA
PI/PC transfer and local
B-/C+
High
Strong pathway biology
phosphatidylinositol
but PITPNB
substrate availability
compensation and weak
retinal heterozygous
evidence. WDR81
VPS34/PI3P regulation
C+
Moderate-high
Strong severe-loss
and endosome
retinal/developmental
maturation
phenotype;

heterozygous retinal

phenotype absent. RILP

Rab7 effector;

C

Moderate

Strong mechanistic

dynein/dynactin late-

transport evidence;

endosome transport

clean RILP+/- visual

phenotype absent. SERPINF1

PEDF trophic/lipid

B+

Very high

Human heterozygous

signaling through

dosage reduction

PEDF-R/PNPLA2

established; local retinal

functional dose-

response incomplete. RTN4RL1

GPI-anchored neuronal

C+

Moderate
Human KO alters
receptor;
networks, but
neurite/synapse
heterozygous visual-
organization
circuit physiology
untested.

Gene

Core Phase 4A

Grade

Photosensitivity

Principal limitation

mechanism

priority

PRPF8
Core U5 spliceosomal
B-/C+
Very high
Dominant retinal
scaffold and retinal
variants strongly

transcript processing

relevant; whole-gene

haploinsufficiency

unresolved. RPA1

ssDNA protection,

A-

High for stress reserve

Direct human

ATR checkpoint, DNA

haploinsufficiency with

repair

functional defect and

rescue; not retina-

specific. TLCD2

PE/PC fatty-acyl

C+

Very high mechanistic

Strong lipid-

composition and

composition effect;

membrane physical

retinal heterozygous

chemistry

direction unknown. SLC43A2

Neutral essential

C-/D+

Low-moderate

Systemic severe-loss

amino-acid transport

effects; retinal substrate

limitation unproven. SCARF1

C1q-associated

C-/D+

Low-moderate

Clear clearance biology

efferocytosis

but not canonical POS

receptor or primary

photon-response

pathway. SERPINF2

Alpha2-antiplasmin;

B+

Low

Strong biochemical

fibrinolysis inhibition

dosage control with

little photosensitivity

proximity. SMYD4

SAM-dependent

D+

Low

Developmental

methyltransferase;

epigenetic biology but

chromatin regulation

no relevant human

retinal HI phenotype.

The Phase 4A prioritization placed TLCD2, PRPF8, SERPINF1, INPP5K, and PITPNA closest to the chemistry required to create, maintain, or regulate retinal light responses. RPA1, WDR81, RILP, and RTN4RL1 were retained as major stress-, maintenance-, or neural-reserve candidates. SCARF1, SLC43A2, SERPINF2, and SMYD4 remained lower-priority modifiers. This ranking did not imply that the highest-priority genes were proven causal; it identified where the shortest plausible path to the phenotype existed.

The most important conceptual result of 4A was the separation of sensory hypersensitivity from photochemical vulnerability. Sensory hypersensitivity requires the same photons to produce a larger physiological or perceptual response. Photochemical vulnerability requires the same photons to produce more molecular injury. RPA1 fits the second category far better than the first. TLCD2 and PRPF8 were retained as more proximal candidates to

phototransduction or photoreceptor membrane behavior, while SERPINF1, INPP5K, PITPNA, WDR81, and RILP primarily supported maintenance and reserve.

9. Phase 4B - Molecular Perturbation, Biochemical Flux, and Chemical Consequences

Phase 4B strengthened the evidentiary standard: gene function alone was no longer

sufficient. Each mechanism was required to identify a perturbation, a measurable molecular endpoint, direction, magnitude when available, cellular compartment, rescue evidence, and proximity to the visual phenotype. This converted the project from an annotation exercise into a perturbational chemistry audit.

9.1 TLCD2

TLCD2 knockdown changes incorporation of selected long-chain polyunsaturated fatty acids into PC and PE and changes membrane fluidity under palmitate challenge. The key inference was not "TLCD2 loss makes membranes rigid." In the primary experiment, reduction could protect against saturated-fat-induced rigidification. The correct Phase 4B conclusion was that TLCD2 dosage can change membrane composition and therefore potentially alter the highly specialized physical environment of photoreceptor outer-segment membranes, but the direction in a DHA-rich human retina remains unpredictable.

9.2 SERPINF1/PEDF and PNPLA2

The SERPINF1 arm produced one of the strongest molecular chains in the interval: human heterozygous SERPINF1 loss lowers PEDF; PEDF stimulates PEDF-R/PNPLA2-associated phospholipase/lipase biology; PNPLA2 perturbation alters retinal phospholipid species, outer-segment processing, retinal proteins, visual-cycle retinoid mobilization, and ERG physiology. PEDF/DHA signaling additionally connects to neuroprotectin D1 and oxidative stress protection. The project therefore retained SERPINF1 as a

strong RPE/photoreceptor support candidate while explicitly refusing to equate *Pnpla2* heterozygosity with *Serpinf1* heterozygosity.

9.3 PITPNA and INPP5K

PITPNA and INPP5K were found to occupy complementary but nonidentical phosphoinositide roles. PITPNA supports local phosphatidylinositol/phosphatidylcholine transfer and PI availability. INPP5K directly removes the 5-phosphate from PI(4,5)P₂ and PIP₃, with strong-loss experiments demonstrating altered autolysosomal PI(4,5)P₂/PI4P, persistent clathrin-associated tubules, and impaired lysosome regeneration. Phase 4B emphasized that a steady-state lipid concentration could appear normal even if turnover flux falls. Therefore a combined PITPNA/INPP5K dosage defect could theoretically reduce phosphoinositide regulatory reserve without producing a simple monotonic change in total PIP₂.

9.4 WDR81 and RILP

WDR81 and RILP form a sequential cellular-material-flow architecture rather than a single biochemical pathway. WDR81 regulates early-endosomal PI3P and early-to-late maturation; RILP couples Rab7-positive late endosomes to dynein-driven movement. The critical distinction is that endosomal identity, physical transport, terminal degradation, and lysosome recycling are separate control points. Phase 4B therefore described WDR81 → RILP → INPP5K as functional adjacency, not as a proven molecular complex or epistatic chain.

9.5 PRPF8

PRPF8 was treated as a production-side mechanism for photoreceptor machinery. Rather than claiming that PRPF8 is itself phototransduction chemistry, Phase 4B positioned it upstream of the transcript and protein products required for photoreceptor cilia, outer segments, and

signaling. Human retinal models of pathogenic PRPF8 variants strongly support retina-specific splicing vulnerability; however, deletion-level haploinsufficiency remains unresolved.

9.6 RPA1

RPA1 remained the strongest direct heterozygous molecular evidence in the entire interval. Its Phase 4B role was narrowed to stress reserve: reduced RPA1 lowers capacity to manage replication-associated and UV-associated damage in human cells. Independent RPE studies show that blue-light photochemistry can generate oxidative DNA lesions. The project therefore proposed a reserve model in which light-associated lesion production increases demand while RPA1 haploinsufficiency lowers repair/checkpoint capacity. This bridge remains inferential because no retinal RPA1[±] light-challenge experiment exists.

9.7 Lower-Priority Systems

SCARF1 was retained as a post-injury clearance mechanism, not a canonical photoreceptor-outer-segment receptor. SLC43A2 was retained for possible methionine/SAM, glutathione, leucine/mTOR, and other metabolic reserve routes, but no retinal substrate limitation was established. SMYD4 was retained as an epigenetic reserve candidate without retinal targets. SERPINF2 remained a biochemical positive control with little direct photosensitivity relevance. RTN4RL1 was restored to the Phase 4B analysis after an initial omission and was placed in a central-neural branch involving neurite, synaptic, and network organization rather than retinal chemistry.

System**Measured molecular****Observed perturbational****Phase 4B relevance****variable****direction**

TLCD2

PC/PE acyl composition;

Composition changes;

Very high mechanistic,

membrane fluidity

protective under palmitate in

uncertain sign

tested cells SERPINF1/PEDF

PEDF abundance;

Heterozygous PEDF down;

High

PNPLA2/PEDF-R lipid

downstream lipid/retinoid

signaling

effects PITPNA

Local PI/PC transfer and PI

Partial reduction alters lipid

High-moderate

flux

metabolism; developmental

buffering

System

Measured molecular

Observed perturbational

Phase 4B relevance

variable

direction

INPP5K

PI(4,5)P₂ → PI4P; lysosome

PIP₂ up, PI4P down in strong

High

reformation

loss; rescue demonstrated WDR81

VPS34/PI3P early-endosome

PI3P-rich enlarged early

Moderate-high

regulation

endosomes; delayed

maturation RILP

Rab7/dynein late-endosome

Reduced transport and
Moderate
transport
dendritic cargo clearance PRPF8
Retinal transcript splicing
Pathogenic variants alter
Very high
retinal/ciliary splicing RPA1
ATR/DNA-damage response
Direct human HI defect with
Very high for vulnerability
rescue RTN4RL1
Synapse/arborization/network
KO remodels human neuronal
Moderate neural branch
organization
networks SCARF1
Apoptotic-cell clearance
Loss impairs efferocytosis
Moderate-low SLC43A2
Essential-AA transport
Severe-loss systemic transport
Low-moderate

defects SMYD4

Chromatin methylation/HDAC

Developmental epigenetic

Low

interaction

perturbation SERPINF2

Fibrinolytic inhibition

Strong heterozygous

Very low visual

biochemical dosage

10. Phase 4C - Cellular Phenotype and Stress-Response Analysis

Phase 4C asked what the molecular abnormalities actually make retinal or neural cells do

differently. Each claim was graded by genetic state, cell type, phenotype, baseline versus stress dependence, rescue, and proximity to either sensory hypersensitivity or photochemical vulnerability.

10.1 Photoreceptor Integrity

The strongest cellular convergence was photoreceptor integrity. Pathogenic PRPF8 variants produce human retinal-organoid ciliary, outer-segment, and photoreceptor abnormalities. Severe WDR81 dysfunction produces progressive photoreceptor death that is rescued by wild-type WDR81. The SERPINF1-associated PEDF-R/PNPLA2 pathway produces photoreceptor deformation, apoptosis, loss of retinal proteins, and measurable ERG impairment. These mechanisms are independent, so their shared endpoint constitutes phenotypic convergence rather than a single common upstream pathway.

The direction of the physiology mattered. The strongest partial-dose retinal experiment heterozygous *Pnpla2* - showed decreased dark-adapted retinal output, not increased gain. Complete PNPLA2 loss also delayed recovery after photobleaching. Therefore Phase 4C materially weakened a simple "overresponsive retina" hypothesis.

Quantitatively, heterozygous *Pnpla2* mice had retinal thickness of approximately 177.9 $\pm$ 2.156 micrometers at three months versus 209.2 $\pm$ 9.12 micrometers in wild type ($P < .0075$), and 175.5 $\pm$ 5.31 micrometers at seven months versus 203.7 $\pm$ 4.60 micrometers in wild type ($P < .007$). At seven months, the dark-adapted b-wave averaged approximately 215.5 $\pm$ 16.5 microvolts in heterozygotes versus 279.1 $\pm$ 8.91 microvolts in wild type ($P = .0023$). Photopic light-adapted ERG responses were not significantly different at the tested ages. Heterozygous retinas also showed reduced rhodopsin and opsin abundance and substantially reduced overlap between rod-bipolar and synaptic markers. These measurements reinforce that partial PEDF-R/PNPLA2 pathway reduction can impair retinal structure and scotopic physiology without demonstrating increased light-response gain (Bernardo-Colón et al.).

10.2 RPE Support and Outer-Segment Processing

SERPINF1/PEDF became the experimentally strongest RPE arm. *Serpinf1*-null RPE develops senescence-like changes and accumulates rhodopsin and lipids; one-copy loss measurably reduces downstream *Pnpla2* transcript abundance. PNPLA2 deficiency independently impairs retinyl-ester mobilization and visual-cycle recovery. PRPF8 pathogenic variants can also disturb RPE polarity and ciliary biology. By contrast, the Phase 3 RPE spatial convergence of INPP5K + PITPNA + SERPINF1 was only partly validated at the cellular level: direct heterozygous RPE phenotypes for INPP5K and PITPNA remain absent.

10.3 Endosomal/Lysosomal Cellular Phenotype

WDR81 deficiency enlarges early endosomes and delays cargo progression; in the neural-progenitor analysis, mean EEA1-positive early-endosome size increased by approximately 63 percent under WDR81 deficiency. RILP-pathway disruption reduces dendritic late-endosome movement and alters neuronal morphology. INPP5K loss prevents efficient autolysosomal membrane recycling and lysosome replenishment. Together these data support a serial material-handling architecture. They do not prove that heterozygous reduction of all three genes in the same retinal cell produces an additive phenotype.

10.4 Stress Challenge

The stress branch produced a coherent but incomplete protection-repair-clearance model. PEDF/NPD1 biology can protect RPE from oxidative stress; RPA1 haploinsufficiency reduces DNA-damage-response reserve; SCARF1 participates after cell death by clearing apoptotic material. The critical full-CNV experiment - identical visible-light challenge in wild-type versus heterozygous deletion retinal models - has not been performed.

10.5 Central Neural Phenotype

RTN4RL1 became the strongest direct human-neuronal cellular candidate. Complete loss increases neurite growth but decreases synaptic puncta by approximately 30-40 percent; synchronous calcium peaks fall by approximately 40-80 percent and peak amplitudes by approximately 50-70 percent across clones. RILP and WDR81 independently alter dendritic or progenitor phenotypes. MIR22 complete loss worsens later spontaneous epilepsy after status epilepticus: knockout mice averaged roughly 16 seizures per day versus 2.47 in wild type, with mean seizure duration approximately 18.4 seconds versus 11.9 seconds. Chronic heterozygous animals were not followed in the decisive seizure phase, and no photic stimulation was used. The

combined interpretation was altered neural reserve and network construction, not demonstrated hyperexcitability (Wang et al.; Silva et al.).

Locus/system

Model

Genetic state

Cellular phenotype

Phase 4C interpretation

PRPF8

Human retinal organoid

Heterozygous pathogenic

Photoreceptor/ciliary/

High; genotype mismatch

variant

outer-segment defect WDR81

Mouse retina

Severe/homozygous

Photoreceptor death;

High-moderate;

rescue

heterozygous negative SERPINF1/PEDF

Mouse RPE / pathway

+/- and -/-; downstream

RPE processing, retinal

High-moderate

models

PNPLA2

support PNPLA2/PEDF-R

Mouse retina/RPE

+/- and -/-

ERG reduction, retinal

High pathway evidence;

thinning, delayed

gene not deleted

adaptation RPA1

Human LBLs

Heterozygous genomic

Stress-response/

Very high dosage

loss

checkpoint failure; rescue

evidence; nonretinal RTN4RL1

Human neurons

Complete KO

Neurite overgrowth,

High cellular relevance;

synapse loss, lower

not HI

synchronous Ca²⁺ RILP

Cultured neurons

Rab7-RILP disruption

Reduced late-endosome

Moderate

transport, dendrite

defects INPP5K

Myoblast/muscle

Strong loss/KD

Autolysosome

Moderate; nonretinal

enlargement, lysosome

depletion TLCD2

HEK293

siRNA

Membrane-composition

Uncertain direction

change; protection under

palmitate MIR22

Mouse brain

Complete deletion after

Accelerated/worsened

High seizure relevance;

SE

chronic epilepsy

not heterozygous or

photic

11. Phase 4D - Developmental Perturbation and Critical Windows

Phase 4D tested whether affected genes change how retinal and neural systems are built,

rather than only how mature cells function. The key result was distributed timing. No single developmental "hit" explained the interval. Instead, different loci act at progenitor, differentiation, neurite, synaptic, retinal-maturation, and post-insult remodeling stages.

WDR81 was the strongest early corticogenesis candidate. Severe loss reduces mid-to-late radial-glial proliferation and later-born cortical-neuron production/positioning through altered endosomal growth-factor signaling. PITPNA-associated developmental biology operates in a similar broad window but is strongly buffered by PITPNB, preventing transfer of the severe combined-loss phenotype to simple PITPNA heterozygosity.

INPP5K reduction affects neuronal process elaboration and strong-loss ocular development. RTN4RL1 and RILP act later on axon/dendrite and synapse construction, with partly opposing morphological directions. SERPINF1/PEDF contributes to postnatal retinal maturation, vascular development, trophic support, and stress resilience. PRPF8 pathogenic organoids show relatively intact early neural-retina specification followed by late photoreceptor maturation failure. MIR22 has little evidence for gross constitutive brain malformation; its strongest role appears during epileptogenic remodeling after injury.

The developmental phase also strengthened the concept of compensation. PITPNA/PITPNB redundancy, PEDF developmental catch-up in some vascular measures, preserved early PRPF8 retinal specification, and normal baseline MIR22-null anatomy all demonstrate that normal baseline structure does not imply normal reserve. Phase 4D therefore favored a distributed critical-window vulnerability model over a congenital V1-specific or magnocellular lesion.

The noncoding loci were also constrained explicitly. MIR22 is embedded within MIR22HG, so MIR22HG and MIR22 cannot be counted as two independent deletion events supporting the same mechanism. MIR22HG has no independently validated visualdevelopmental mechanism sufficient to add separate weight. PITPNA-AS1 remains dominated by context-dependent cancer-cell literature rather than normal retinal or neural dosage biology. RN7SL105P has no validated locus-specific function that warrants mechanistic weight. These negative assignments are retained in the frozen report to prevent artificial inflation of the multigene convergence model.

Locus

Critical window

Best developmental

4D grade

Principal limitation

endpoint

WDR81

Mid/late corticogenesis

Radial-glial

C+

Severe loss dominates

proliferation;

EGFR/MAPK; late-

born neuron positioning

Locus

Critical window

Best developmental

4D grade

Principal limitation

endpoint

RTN4RL1

Neurite/synapse

Axon/dendrite growth;

C+

Complete KO, not HI

maturation

synapse formation;

network activity SERPINF1

Postnatal retinal

Retinal survival,

C+

Null-heavy evidence

maturation

vascular trajectory,

stress sensitivity PRPF8

Late retinal maturation

Outer-segment and

B-/C+

Pathogenic variant !=

photoreceptor

deletion

specialization INPP5K

Neuronal

Neurite length; ocular

C+

HI unknown

differentiation/eye

morphogenesis under

development

strong loss RILP

Dendritic maturation

Late-endosome

C

Not direct RILP+/-

transport supports

arborization PITPNA

Corticogenesis

Golgi PI signaling,

C-/D+

PITPNB redundancy

polarity, nuclear

migration RPA1

Proliferative reserve

Genome stability and

D+

No direct neural HI

repair

developmental

phenotype MIR22

Post-insult remodeling

Epileptogenic plasticity

E baseline / C+

Chronic heterozygous

after status epilepticus

remodeling

effect untested

12. Phase 4E - Multi-Gene Functional Convergence

Phase 4E tested whether the independently supported effects actually meet. Five levels of

convergence were distinguished: spatial co-localization, molecular-system convergence, sequential-process convergence, phenotypic convergence, and direct combinatorial convergence. The first four were observed to varying degrees; the fifth was not. No relevant primary study jointly reduced two loci from this exact interval and measured a retinal or photosensitivity-related outcome. Accordingly, Phase 4E used "functional convergence" and explicitly withheld "synergy" and "epistasis."

12.1 Membrane Identity and Material Flow

PITPNA, WDR81, and INPP5K converge on phosphoinositide-dependent membrane identity but regulate different lipid pools in different compartments. PITPNA supports PI/PC transfer and PI4P-generating signaling, WDR81 restrains VPS34-dependent PI3P on endosomes, and INPP5K turns over PI(4,5)P₂ on autolysosomal membranes. RILP then links late-endosomal identity to physical transport. This architecture is distributed, not a linear PI → PI3P → PIP₂ cascade.

The most coherent serial cellular architecture was WDR81 → RILP → INPP5K, representing endosome maturation → late-endosome transport → autolysosomal/lysosomal recycling. A serial system can be limited by its weakest stage, so multi-gene effects need not be additive. One defect can mask another, multiple partial defects can remain silent until cargo demand rises, or the system can display threshold behavior.

12.2 Photoreceptor Endpoint Convergence

PRPF8, WDR81, and the SERPINF1/PEDF-R system independently converge on photoreceptor structural and functional integrity through RNA processing, intracellular

maintenance, and trophic/lipid support. TLCD2 adds a plausible membrane-composition route without direct retinal cellular validation. This was the strongest phenotypic convergence identified in Phase 4.

12.3 Stress-Defense Convergence

SERPINF1, RPA1, and SCARF1 map onto different layers of stress response: damage tolerance/protection, DNA-damage response/repair reserve, and post-injury clearance. The architecture is coherent because all three affect how tissue withstands or resolves injury, but they have not been jointly tested in one retinal model.

12.4 Circuit-Construction Convergence

WDR81, INPP5K, RTN4RL1, and RILP converge topologically on neural development and network construction. Directional coherence is weaker. RTN4RL1 knockout increases neurite branching while RILP-pathway disruption reduces dendritic arborization; PITPNA is buffered by PITPNB; and RTN4RL1 knockout decreases synchronous activity. Phase 4E therefore concluded that the interval perturbs multiple controls of network topology without demonstrating a unified hyperexcitable direction.

Convergence module

Principal loci

Topological

Directional coherence

4E grade

convergence

Photoreceptor integrity

PRPF8, WDR81,

Very strong endpoint

Mostly impairment

B

SERPINF1 pathway,

convergence

TLCD2 candidate Endosomal/lysosomal

WDR81, RILP, INPP5K

Very strong sequential

Compatible but not scalar

B-/C+ material flow

convergence RPE outer-segment

SERPINF1;

Strong

Likely impaired

C+ processing

WDR81/RILP/INPP5K

anatomical/mechanistic

candidates Phosphoinositide

PITPNA, WDR81,

Strong molecular-system

Mixed compartment-

C+ membrane identity

INPP5K

specific directions Membrane lipid

TLCD2,

Moderate-strong

Mixed/unknown

C+ composition

SERPINF1/PEDF-R,

PITPNA Photochemical stress

SERPINF1, RPA1,

Strong conceptual

Coherent toward lower

C+/Bdefense

SCARF1

layered reserve

reserve Developmental circuit

WDR81, INPP5K,

Very strong topological

Mixed/opposing

Btopology

RTN4RL1, RILP,

PITPNA Direct retinal sensory

Candidate retinal genes

Weak

Not demonstrated

D/E amplification Epileptogenic reserve

MIR22 plus network-

Moderate opportunity

No consistent

D+/C-

development genes

hyperexcitable direction

13. Phase 4F - Visual-System Relevance and Signal Propagation

Phase 4F mapped the evidence node by node from photoreceptor to photoparoxysmal

response. The major result was asymmetry: the deletion has its strongest biological evidence before the optic nerve and in general network-development systems, while the most phenotypespecific stages - V1 flicker physiology and distributed PPR propagation - remain untested.

13.1 Photoreceptor and RPE Nodes

PRPF8, SERPINF1/PEDF, WDR81, TLCD2, PITPNA, and INPP5K provide meaningful retinal mechanisms. The evidence overwhelmingly concerns structural integrity, adaptation, trafficking, lipid support, or stress tolerance rather than increased photon gain. The strongest partial-dose physiology shows reduced ERG response; PNPLA2 loss slows dark adaptation. Therefore primary retinal amplification was downgraded to weak/unsupported, while altered retinal temporal coding remained plausible but untested.

13.2 RGC, Optic Nerve, and White Matter

PEDF directly supports RGC survival in injury models. RTN4RL1 is expressed in retinal ganglion cells and participates in growth-inhibitory/plasticity systems, but Nogo-receptor family

redundancy is substantial: even combined family loss can leave baseline retinal stratification, optic-nerve myelination, and central RGC projections grossly normal. This weakens a simple optic-pathway miswiring hypothesis.

Human overlapping-CNV evidence nevertheless introduces a credible white-matter branch. The Emrick series demonstrates multifocal leukoencephalopathy in small interstitial 17p13.3 deletions excluding YWHAE and PAFAH1B1, with RTN4RL1 proposed as a candidate. White-matter abnormalities could affect conduction timing and long-range synchronization, but the reported lesions were not shown to selectively involve optic radiations or occipitofrontal pathways and no photic EEG phenotype was reported.

13.3 LGN and V1

Earlier expression analysis found no affected interval gene among the principal defining markers of human magnocellular, parvocellular, or koniocellular dLGN populations. Phase 4 therefore did not support a selective magnocellular mechanism. No deletion-specific LGN hyperexcitability, temporal gain, or thalamocortical entrainment phenotype has been demonstrated.

V1 is the central evidence gap. Human PSE physiology strongly implicates occipital cortex, contrast/gain control, temporal entrainment, inhibition, and network connectivity. The deletion contains cortical and developmental genes, but none has been shown in the heterozygous CNV state to produce abnormal V1 responses to flicker. There is no direct evidence of altered P100, SSVEP, gamma synchronization, alpha stabilization, or V1 inhibitory containment attributable to this CNV.

13.4 Distributed Propagation and Seizure Threshold

PSE often involves occipital-to-parietal-to-frontal spread and abnormal visuomotor connectivity. RTN4RL1, RILP, WDR81, INPP5K, and white-matter findings make altered network topology biologically plausible. Yet no affected-locus or CNV model has demonstrated pathological propagation during visual stimulation. MIR22 is best positioned downstream as a possible modifier of epileptogenic threshold or remodeling rather than a light-trigger mechanism.

Visual node

PSE importance

Deleted-locus evidence

Observed direction

4F status

Photoreceptor

High

PRPF8, SERPINF1

Mostly impaired

Moderate-strong

pathway, WDR81,

structure/function; no

TLCD2 candidate

gain increase RPE

High support role

SERPINF1 strongest;

Reduced support,

Strong support-system

INPP5K/PITPNA

delayed recovery

relevance

candidates Inner retina

Temporal/contrast

PEDF-R pathway;

Reduced ERG / structural

Moderate

filtering

PRPF8 indirect

changes RGC

Final retinal output

PEDF; PRPF8;

Survival/development

Moderate

RTN4RL1 biology

effects Optic nerve/white matter

Timing/conduction

RTN4RL1; overlapping

Structural vulnerability

Moderate

CNV white matter LGN

Relay/modulator

No specific M/P/K

Unknown

Weak

signature V1

Major PPR initiation

General cortical

No deletion-flicker

Major evidence gap

node

mechanisms only

physiology Parietal/frontal network

PPR propagation

RTN4RL1, RILP,

Mixed circuit effects

Plausible, untested

WDR81, white matter Cortical inhibition

Important PSE

No direct CNV E/I

Unknown

Weak

mechanism

phenotype PPR transition

Defining EEG phenotype

No affected-gene/CNV

Unknown

Absent

model General epileptogenic

Seizure conversion

MIR22 severe loss

Lower reserve after insult

Moderate conceptual reserve Clinical photic seizure

Final phenotype

No exact-CNV evidence

Unknown

Unresolved

14. Phase 4G - Integrated Adjudication

Phase 4G formally separated biological consequence from phenotype attribution. The

CNV is very likely biologically non-neutral: it includes directly dosage-responsive genes, multiple retinal pathways, strong severe-loss or pathogenic-variant mechanisms, neuraldevelopment genes, and a clinically relevant overlapping-deletion white-matter phenotype. However, biological consequence is not equivalent to epileptic-photosensitivity causation.

Broad 17p13.3 epilepsy literature cannot be transferred wholesale because many published deletion syndromes include PAFAH1B1 and/or YWHAE/CRK, which are outside this interval. The closest human CNV precedent supports white-matter vulnerability rather than a

defined photosensitive-epilepsy syndrome. Likewise, no locus in this interval meets the CHD2 benchmark of human photosensitivity association plus controlled photic experimental validation.

The integrated causal ladder therefore stops before the disease-specific physiological steps. Genomic lesion and dosage alteration are established. Molecular consequences are established for selected genes. Relevant cellular phenotypes are established in constituent-gene models. Multi-gene functional convergence is supported. A full-CNV retinal phenotype is not established. Abnormal visually evoked cortical physiology is not established. A PPR and a photosensitive seizure caused by the exact CNV are not established.

Claim

Phase 4G adjudication

CNV biologically non-neutral

Supported Molecular reserve altered

Strongly supported for selected loci Retinal mechanisms present

Strongly supported Neurodevelopmental mechanisms present

Strongly supported Multi-gene functional convergence

Supported Multi-gene synergy/epistasis

Not demonstrated Retinal photon-response amplification

Unsupported Retinal light-injury vulnerability

Plausible; not directly demonstrated Abnormal retinal temporal coding

Open hypothesis Long-range network modification

Plausible; partially supported by overlapping CNVs V1 hyperexcitability from CNV

Not demonstrated Abnormal visually driven cortical propagation

Plausible; not demonstrated General epileptogenic reserve reduction

Plausible; strongest evidence through MIR22 complete loss Photoparoxysmal response caused by exact CNV

Not demonstrated Epileptic photosensitivity caused by exact CNV

Insufficient evidence / unresolved Best current classification

Mechanistically plausible candidate modifier of visual-neural and epileptogenic reserve

15. Frozen Phase 4 Mechanistic Model

The strongest integrated model after 4A-4G is a multi-layer visual-neural functionalreserve lesion. It does not require every deleted gene to be individually haploinsufficient, nor does it require every perturbation to move in the same direction. Instead, the interval can lower or alter reserve at several control layers while paralogs, developmental compensation, homeostatic adaptation, and opposite-direction effects preserve baseline function. 17p13.3 deletion -> retinal maintenance reserve down + membrane/endosomal reserve down + stress-response reserve down + developmental/network organization altered

Under this model, baseline demand can remain below the reduced reserve of the heterozygous system, allowing apparently normal anatomy or physiology. Increased demand intense or repetitive visual stimulation, oxidative stress, developmental maturation, metabolic stress, illness, or a separate epileptogenic influence - may expose deficits that are otherwise compensated.

If the CNV ultimately contributes to epileptic photosensitivity, Phase 4 indicates that the most plausible route is not excessive photon capture. The stronger hypothesis is altered retinal timing/adaptation and/or altered circuit construction or white-matter/network organization, combined with reduced epileptogenic reserve, producing a lower threshold for abnormal

propagation of patterned visual activity. This remains a testable hypothesis rather than a demonstrated mechanism.

16. Hypotheses Retained, Downgraded, or Rejected

16.1 Retained as Best-Supported

- The deletion is biologically consequential and can reduce molecular/cellular reserve.
- Retinal and RPE maintenance pathways are genuinely represented in the interval.
- Photoreceptor structural integrity is the strongest phenotype-level convergence.
- Endosomal/lysosomal material handling is the strongest serial mechanistic convergence.
- Distributed developmental and network-organization effects are plausible and experimentally supported under strong perturbation.
- The deletion is a plausible candidate modifier of visual-neural and epileptogenic reserve.

16.2 Retained but Unproven

- Increased retinal vulnerability to the same light dose.
- Abnormal retinal temporal coding or adaptation at provocative flicker frequencies.
- Altered long-range visual-network propagation because of circuit-development or white-matter changes.
- A modest lowering of epileptogenic threshold in the heterozygous CNV state.

16.3 Downgraded or Unsupported

- A simple intrinsically hyperresponsive retina.
- A selective magnocellular lesion.
- A deletion-wide V1-specific developmental program.
- Generalized RTN4RL1-driven hyperexcitability.
- Uniformly harmful TLCD2-mediated membrane rigidification.

- Proven synergy or epistasis among deleted genes.
- Established direct causation of PPR or photosensitive seizures by this exact CNV.

17. Definitive Experiments Implied by Phase 4

The single highest-value experimental strategy is an isogenic reconstruction of the actual heterozygous deletion rather than a collection of unrelated null models. Wild-type, full-CNV heterozygous, corrected/rescued, and selected single-gene heterozygous lines should be differentiated into retinal organoids, RPE, and connected neural systems. The engineered interval should reproduce the reported hg19 coordinates as closely as technically possible, particularly the boundary structures of INPP5K and RTN4RL1.

The retinal arm should measure photoreceptor structure, outer-segment formation, retinal/RPE lipidomics, phosphoinositide pools, outer-segment phagocytosis and degradation, visual-cycle retinoids, dark adaptation, ERG-like responses, oxidative/DNA damage, and recovery after calibrated light exposure. The neural arm should measure progenitor behavior, neurite morphology, synapse density, E/I balance, calcium dynamics, MEA firing and bursting, frequency entrainment, phase locking, and long-range propagation.

For photosensitivity specifically, patterned visual stimulation must be treated as a transfer-function problem. Identical stimuli should be delivered across relevant flicker frequencies, contrasts, wavelengths, and spatial structures while measuring the transformation from retinal output through thalamic and V1-like networks into distributed activity. Only demonstration of genotype-dependent pathological propagation, followed by correction or rescue, would close the major causal gap identified in Phase 4.

Summary

Phase 4 converts the 17p13.3 interval from a list of deleted genes into a disciplined

mechanistic model. The interval contains several genes with real dosage or perturbational consequences, and those consequences repeatedly touch retinal support, photoreceptor integrity, membrane identity, intracellular trafficking, DNA-damage response, neuronal development, synaptic organization, and network plasticity. The biological signal is therefore nonrandom and medically relevant.

At the same time, the phase narrows rather than broadens the photosensitivity claim. The data do not show that the retina is intrinsically overresponsive to individual photons. The best retinal physiology instead shows reduced response or delayed recovery. The best centralneural RTN4RL1 model shows network remodeling with reduced synchronous activity. The strongest seizure-related locus, MIR22, modifies post-insult epileptogenesis under complete loss but has not been shown to create a photic phenotype in heterozygotes.

The strongest surviving interpretation is reduced or altered functional reserve across multiple layers. Photoreceptor/RPE systems may be less resilient; endosomal and lysosomal material handling may have less capacity; damage-response reserve may be reduced; and circuit construction or long-range communication may be subtly altered. Compensation and redundancy can keep baseline function apparently normal. The model predicts that differences may emerge under increased demand or repeated challenge rather than as an obligatory congenital deficit.

For epileptic photosensitivity, the unresolved stage is now precisely located. Human disease physiology requires the transformation of patterned visual input into abnormal occipital and distributed cortical activity. The present CNV has not been shown to alter V1 flicker physiology, cortical inhibition, 10-20-Hz entrainment, long-range visual-network propagation, a

photoparoxysmal response, or visually triggered seizures. Therefore the exact CNV cannot currently be classified as an established cause of epileptic photosensitivity.

The frozen Phase 4 classification is: biologically consequential CNV; high retinal/visual mechanistic relevance; moderate-to-high support for reduced functional reserve; low support for direct sensory amplification; moderate support for developmental/network modification; limited-to-moderate evidence for general epileptogenic modification; insufficient evidence for abnormal visually driven cortical propagation; and insufficient evidence for direct epileptic photosensitivity causation. The best current label is "mechanistically plausible candidate modifier of visual-neural and epileptogenic reserve."

Vocabulary

Adaptation. Adjustment of retinal or neural responsiveness during or after sustained/repeated

stimulation; dark adaptation specifically refers to recovery of visual sensitivity after photobleaching.

Autolysosomal reformation (ALR). Membrane-recycling process that regenerates lysosomes

from autolysosomes after prolonged autophagy; INPP5K is required for normal ALR.

Biallelic loss. Disruption of both gene copies. Biallelic loss is not equivalent to the heterozygous

one-copy loss produced by this CNV.

Candidate modifier. A factor plausibly capable of changing severity, threshold, penetrance, or

expression of a phenotype without being established as a sufficient primary cause.

Cellular phenotype. A measurable change in cell structure, survival, trafficking, signaling, electrophysiology, or stress response following perturbation.

CNV. Copy-number variant; a genomic deletion or duplication altering the number of copies of one or more loci.

Directional convergence. Multiple perturbations pushing a shared endpoint in the same direction. This is distinct from merely affecting the same system.

Dosage sensitivity. The degree to which changing gene copy number alters transcript, protein, molecular function, cell behavior, or clinical phenotype.

Efferocytosis. Recognition and engulfment of apoptotic cells or cellular debris by phagocytic cells; SCARF1 participates in C1q-associated efferocytosis.

Electroretinography (ERG). Measurement of retinal electrical responses to light. The a-wave primarily reflects photoreceptor activity; the b-wave reflects downstream inner-retinal responses.

Endolysosomal system. The linked endosome-to-lysosome network that sorts, transports, degrades, and recycles cellular cargo.

Epileptogenic reserve. Conceptual resistance of a neural network to transitions into persistent or

recurrent epileptic activity; lower reserve implies easier pathological transition under challenge. Epistasis. A genetic interaction in which the effect of one locus depends on the state of another.

Phase 4 did not demonstrate epistasis among deleted loci.

Functional convergence. Independent genes or mechanisms affecting the same molecular
system, cellular process, anatomical compartment, developmental operation, or
phenotype endpoint.

Functional reserve. Difference between a system's maximal or available biological capacity and

the physiological demand placed on that system.

Gain. Ratio between change in output and change in input. Increased sensory gain means a
larger physiological response to the same stimulus.

Haploinsufficiency. A condition in which one functional gene copy is insufficient for normal
molecular, cellular, or clinical function.

Heterozygous loss. Loss or disruption of one of two gene copies. It does not automatically
imply

exactly 50 percent protein abundance or function. Hypomorph. An allele that retains
partial gene function rather than producing complete loss.

Isogenic. Genetically matched experimental lines differing only at the engineered locus or
CNV

of interest, minimizing background-genome confounding.

Koniocellular, magnocellular, parvocellular. Major subdivisions of primate visual-thalamic organization. Phase 4 did not identify selective magnocellular involvement by the deletion.

Lateral geniculate nucleus (LGN). Principal visual relay nucleus of the thalamus between retinal ganglion cells and primary visual cortex.

MIR22. MicroRNA locus within the deletion. Complete loss worsens chronic epilepsy after an

epileptogenic insult in mice; heterozygous photic effects remain unknown.

Neuroprotectin D1 (NPD1). DHA-derived lipid mediator with anti-inflammatory and anti-apoptotic effects in RPE under oxidative stress.

Outer segment. Specialized photoreceptor compartment containing stacked photopigment-rich

membranes where photon absorption and phototransduction begin.

PEDF. Pigment epithelium-derived factor, encoded by SERPINF1; a secreted retinal trophic,

antiangiogenic, lipid-linked, and cytoprotective protein.

Photochemical vulnerability. Greater molecular or cellular injury from the same photon exposure, distinct from sensory hypersensitivity.

Photoparoxysmal response (PPR). Abnormal epileptiform EEG response evoked by intermittent photic or pattern stimulation; it can occur with or without a clinical seizure.

Photosensitive epilepsy (PSE). Epilepsy in which visual stimulation can provoke abnormal EEG activity and, in some patients, clinical seizures.

Phototransduction. Conversion of absorbed photons into an electrical response in

photoreceptors through rhodopsin/opsin, transducin, PDE6, cGMP, and cyclic-nucleotide-gated channel signaling.

Phosphoinositide. Phosphorylated phosphatidylinositol lipid that helps define membrane

identity, trafficking, signaling, and protein recruitment.

PI3P. Phosphatidylinositol 3-phosphate, enriched on early endosomes and produced by VPS34;

WDR81/WDR91 negatively regulate this system.

PI4P. Phosphatidylinositol 4-phosphate, a membrane signaling lipid and product of INPP5K-

mediated PI(4,5)P₂ dephosphorylation.

PI(4,5)P₂. Phosphatidylinositol 4,5-bisphosphate, a membrane signaling lipid whose 5-

phosphate is removed by INPP5K.

Retinal ganglion cell (RGC). Output neuron of the retina whose axons form the optic nerve and

project to visual brain targets.

Retinal pigment epithelium (RPE). Support epithelium adjacent to photoreceptor outer

segments; essential for phagocytosis, retinoid cycling, lipid handling, trophic support, and metabolic homeostasis.

Sensory hypersensitivity. An abnormally large perceptual or physiological response to the same

light stimulus.

Spliceosome. Large ribonucleoprotein machinery that removes introns from pre-mRNA;
PRPF8

is a central U5 spliceosomal scaffold.

Steady-state visual evoked potential (SSVEP). Stimulus-frequency-locked cortical response
 produced by periodic visual stimulation; useful for evaluating visual entrainment and
 temporal stability.

Synergy. A combined effect greater than expected under a specified additive model. Phase
 4 did

not demonstrate synergy among deleted genes.

Temporal functional convergence. Different loci influencing the same biological endpoint at
 different developmental or physiological stages rather than acting simultaneously.

Topological convergence. Different perturbations affecting the same biological network or
 structural endpoint even if they push that endpoint in different directions.

V1. Primary visual cortex, the principal cortical recipient of LGN visual input and a major
 initiation node for photoparoxysmal responses.

Visual-neural reserve. Integrated capacity of retinal, visual-pathway, and cortical systems
 to

maintain stable function under physiological or pathological demand.

Reproducible Formulas

The formulas below are separated into biochemical reactions, quantitative definitions, and conceptual models. Biochemical reactions summarize experimentally established chemistry. Quantitative definitions can be applied directly to future datasets. Conceptual equations are explicitly labeled and should not be mistaken for validated physiological laws.

Reproducible Formulas

Formula / definition	Expression	Interpretation
INPP5K 5-phosphatase reaction	$\text{PI(4,5)P}_2 + \text{H}_2\text{O} \rightarrow \text{PI4P} + \text{P}_i$	Established biochemical reaction.
INPP5K PIP3 reaction	$\text{PI(3,4,5)P}_3 + \text{H}_2\text{O} \rightarrow \text{PI(3,4)P}_2 + \text{P}_i$	Established substrate reaction; do not infer in vivo flux from K_m alone.
VPS34 reaction	$\text{PI} + \text{ATP} \rightarrow \text{PI3P} + \text{ADP}$	Established kinase chemistry; WDR81/WDR91 regulate the associated endosomal PI3K system.
PLC branch	$\text{PI(4,5)P}_2 \rightarrow \text{DAG} + \text{IP}_3$	Canonical signaling branch relevant to calcium-regulatory context.
Dynein/ATPase work	$\text{ATP} + \text{H}_2\text{O} \rightarrow \text{ADP} + \text{P}_i + \text{mechanical work}$	Represents ATP-powered motor/remodeling work; RILP recruits dynein rather than catalyzing ATP hydrolysis itself.

Reproducible Formulas (continued)

Formula / definition	Expression	Interpretation
PEDF-R/PNPLA2 phospholipase	phospholipid + H ₂ O -> lysophospholipid + free fatty acid	Simplified representation of phospholipase/lipase chemistry.
RPA binding	ssDNA + RPA <-> RPA-coated ssDNA	Binding equilibrium underlying ATR recruitment and DNA-damage response.
Rhodopsin conformational equilibrium	MI <-> MII	Simplified metarhodopsin I/metarhodopsin II equilibrium influenced by membrane environment.
Lipid-radical initiation	LH + radical -> L radical	Simplified initiation of lipid peroxidation.
Lipid-peroxyl propagation	L radical + O ₂ -> LOO radical	Simplified oxygen addition step in lipid peroxidation.

Reproducible Formulas (continued)

Formula / definition	Expression	Interpretation
Methionine activation	Met + ATP → SAM	Simplified methionine-to-S-adenosylmethionine activation route.
SAM-dependent lysine methylation	Lys + SAM → methyl-Lys + SAH	Simplified methyltransferase reaction relevant to SMYD4.
Steady-state flux balance	$dX/dt = J_{in} - J_{out}$	At steady state, $dX/dt = 0$ even when turnover flux is high; a normal concentration does not prove normal flux.
Functional-reserve condition	phenotype risk rises when $D(t) > C_{max}$	Conceptual model: $D(t)$ is physiological demand and C_{max} is available system capacity.
Light-injury function	$I(H)$	I is measured injury at photon/light exposure H . A vulnerability phenotype requires $I_{CNV}(H) > I_{WT}(H)$ for the same H .

Reproducible Formulas (continued)

Formula / definition	Expression	Interpretation
Light-injury threshold	H_{crit} = exposure producing a predefined injury criterion	Direct vulnerability would be supported if $H_{crit,CNV} < H_{crit,WT}$.
Neural response gain	$G = \Delta \text{output} / \Delta \text{input}$	Operational definition. Sensory/neural amplification requires $G_{CNV} > G_{WT}$ under matched input.
Retinal temporal gain	$G_R(f) = R(f) / S(f) $	$R(f)$ is retinal output amplitude at frequency f ; $S(f)$ is stimulus amplitude. Useful for testing flicker-frequency sensitivity.
Serial bottleneck capacity	J_{max} approximately $\min(C1, C2, ..., Cn)$	Conceptual model for sequential endosomal/lysosomal stages; overall throughput can be limited by the weakest stage.
Layered lesion burden	$dL/dt = D - P - R$	Conceptual model: D = damage generation, P = prevention/tolerance, R = repair capacity. Used only as a qualitative reserve representation.

Reproducible Formulas (continued)

Formula / definition	Expression	Interpretation
Factorial interaction model	$Y = \text{beta0} + \text{betaA} * A + \text{betaB} * B + \text{betaAB} * (A * B) + \text{epsilon}$	Reproducible test for non-additive interaction. betaAB tests departure from the specified additive model.
Developmental timing model	$P = f(D_{\text{early}}, D_{\text{late}}, D_{\text{adult}})$	Conceptual decomposition of mature phenotype into early developmental, late maturation, and adult dosage effects.
Photosensitive-network threshold	$G_N * I > \text{Theta}_E$	Conceptual model: I = visual input, G_N = network propagation gain, Theta_E = threshold for pathological synchronization.

Reproducibility rule for future Phase 5 and Phase 6 work: formulas that represent measured quantities must be parameterized from independent data, uncertainty intervals must be propagated, and no conceptual equation should be fitted as if it were a validated physiological law without an explicit calibration dataset.

Phase 5

Deterministic Evidence Integration and Reproducibility Analysis - Available Frozen Summary

v1.0

Phase 5 - Casual Reader Summary

What the genome research found, without the technical detail. FROZEN SCIENTIFIC RECORD - VERSION 1.0. Prepared by Zachariah P. Laing, 29 September 2026.

This is a plain-language companion to the frozen Phase 5 scientific record. It simplifies terminology but does not change the scientific conclusion.

The short version

The deletion does not look biologically meaningless. It removes or disrupts a group of genes that have real roles in the retina, brain development, nerve-cell communication, membrane biology, cellular cleanup, and stress responses. The evidence does not point cleanly to one single gene as the explanation, and it also does not support treating every deleted gene as an equal 50% loss.

The best overall explanation is that the deletion may reduce biological "reserve" across several systems at once. In this model, many individual effects can be mild, partly compensated, or only become important under stress. That fits the available research better than a single-gene explanation or a simple sum of seventeen harmful effects.

The original question involved photosensitivity and visually triggered seizures. Phase 5 does not prove that connection. Instead, it shows that broader eye and brain phenotypes are currently better supported than photosensitivity itself.

What appears most worth paying attention to

1. Retinal/RPE functional reserve - This is the strongest eye-related signal. Several deleted genes affect photoreceptors, retinal support cells, cilia, phagocytosis, or retinal maintenance.
2. Brain development and neural-network organization - Several genes affect neurites, synapses, neuronal survival, trafficking, or coordinated network activity.
3. Stress-dependent vulnerability - A recurring pattern is that cells may look relatively normal at baseline but become less resilient under injury, oxidative stress, heavy activity, aging, or other challenge.
4. Seizure susceptibility / epileptogenesis - There is some direct epilepsy-related evidence, especially around MIR22, but it does not show that this exact one-copy deletion causes epilepsy.
5. Retinal timing / visual signal processing - Plausible, but not directly measured in the exact CNV.
6. Cortical synchronization / flicker entrainment - Mechanistically relevant, but currently a missing link rather than a demonstrated phenotype.
7. Photosensitivity / photoparoxysmal response - Still plausible, but it sits furthest downstream and currently has the least direct evidence of the major candidate outcomes.

Why one gene is probably not the whole story

Some genes in the deletion have very strong eye or brain effects when they are completely knocked out or heavily damaged. The problem is that the actual CNV is heterozygous: one copy is affected and one copy remains. For several genes, one copy is enough to avoid the severe phenotype seen in full knockouts.

PRPF8 is a good example: reduced RNA does not necessarily mean reduced protein. PITPNA and WDR81 show threshold or recessive behavior, where the severe phenotype appears mainly after much greater loss of function. That makes it risky to claim that any one of these genes automatically explains the whole CNV phenotype.

Why "all the genes add up" is also too simple

Not every deleted gene pushes biology in the same direction. In some experiments, losing TLCD2, MIR22HG, or SERPINF2 can actually be protective under specific stress conditions. RTN4RL1 loss can reduce coordinated neural activity rather than increase it. These findings do not make the genes irrelevant; they show that their effects depend on context.

That is why the leading model is a distributed reserve model rather than a simple arithmetic total of gene losses.

What the simulations actually contributed

The simulations were not used to invent biological evidence. They were used to answer methodological questions: how many biological replicates would be needed, how easily small effects could be missed, how often frequency scans generate false positives, and whether a distributed mechanism could be mistaken for a single-gene mechanism.

- The final standardized simulation suite used 20,000 replicates per scenario/cell where applicable, totaling about 1.88 million primary stochastic draws/resamples.
- With that many runs, Monte Carlo sampling error became very small; the larger uncertainty now comes from biology, not from the number of simulations.
- Small one-copy effects can be difficult to detect with too few independent cell lines or clones.
- Using many cells from the same clone does not replace independent biological replication.

- Scanning many flicker frequencies without correction can produce apparently significant results by chance surprisingly often.
- Small datasets can make a distributed mechanism look like a simpler single-gene explanation.

So what does Phase 5 actually conclude?

The deletion is best viewed as a candidate distributed visual-neural reserve lesion, not as a proven single-gene photosensitivity mutation.

The strongest candidate consequences are subtle retinal/RPE dysfunction, altered neural development or network organization, and reduced resilience under stress. There is also moderate support for an effect on seizure susceptibility or epileptogenesis. Photosensitivity remains possible, but the key physiological chain from the retina through the thalamus and visual cortex to a photoparoxysmal response has not been directly measured for this exact CNV.

What has NOT been proven

- That every deleted gene loses exactly half of its function.
- That the genes act synergistically.
- That the exact CNV causes a measurable retinal abnormality.
- That the CNV causes abnormal visual-cortex excitation or flicker entrainment.
- That the CNV causes a photoparoxysmal response or visually triggered seizure.
- That one specific gene is the sole driver.

Why the result is still useful

The project started with a broad question about photosensitivity. Phase 5 narrowed the problem. It now appears more defensible to study the CNV as a broader visual-neural reserve problem, with retinal maintenance, neural-network organization, and stress response as the

strongest candidate phenotype domains. That is more specific than simply saying "we do not know," while still respecting the fact that the missing physiological measurements have not been made.

The next project stage, Phase 6 - Conclusion, integrates this frozen Phase 5 record with the earlier genomic, mechanistic, and expression phases. Phase 6 is a separate master conclusion and does not alter the frozen Phase 5 findings.

Phase 6

Project Conclusion, Critical Appraisal, Forward Research Paths, and Frozen Record

Fixed structural reference: GRCh37/hg19 chr17:1,411,408-1,841,103 | heterozygous deletion | nominal span 429,696 bp | 17 affected annotated loci | INPP5K and RTN4RL1 boundary-intersected

1. Purpose and Phase 6 Status

Phase 6 concludes the first-pass in silico research program. It does not reopen or silently revise the frozen findings of Phases 1-5. Its role is to consolidate what the project established, identify what remains unsupported or unresolved, critically assess the methodology and inferential limits, define two legitimate continuation paths, and establish a version-controlled stopping point from which future work can proceed.

The project began with a broad visual-neurological question, including possible photosensitivity and visually triggered seizure mechanisms. The accumulated evidence narrowed that question. The strongest defensible interpretation is now a distributed visual-neural functional-reserve model involving retinal/RPE support, membrane and phosphoinositide biology, endosomal/lysosomal trafficking, genome-stress response, neural development, and network organization. The exact heterozygous CNV has not been shown to cause abnormal visual-cortical flicker processing, a photoparoxysmal response, or visually triggered seizures.

2. Project Facts Preserved at Freeze

Structural reference remains GRCh37/hg19 chr17:1,411,408-1,841,103, nominally 429,696 bp, heterozygous.

The interval affects at least 17 annotated loci: 13 protein-coding loci and 4 noncoding loci.

Eleven protein-coding genes are fully encompassed: PITPNA, SLC43A2, SCARF1, RILP, PRPF8, TLCD2, WDR81, SERPINF2, SERPINF1, SMYD4, and RPA1.

INPP5K and RTN4RL1 are boundary-intersected. Under the reported-coordinate model, approximately 49.6% of INPP5K MANE amino-acid-coding sequence and approximately 99% of RTN4RL1 CCDS sequence on the affected allele lie within the deletion.

The noncoding loci PITPNA-AS1, MIR22HG, MIR22, and RN7SL105P are included in the frozen interval.

The reported array coordinates are operational CNV boundaries, not demonstrated nucleotide-exact chromosome junctions.

3. Integrated Findings of the First-Pass Research

3.1 Structural genomics

Phase 1 converted a generic 17p13.3 label into a reproducible locus-specific substrate. The main structural result was that apparent genomic-span overlap can substantially understate coding disruption at the two boundary genes. The project therefore correctly anchored all later interpretation to the exact reported interval rather than importing phenotypes from broader 17p13.3 deletion syndromes.

3.2 Molecular and biochemical architecture

Phase 2 showed that the interval is not biologically featureless. Multiple affected genes occupy mechanistically relevant systems: phosphoinositide metabolism, phospholipid transfer and membrane composition, late-endosome and lysosome biology, intracellular transport, retinal trophic support, RNA splicing, DNA replication and repair, neuronal structural signaling, amino-acid transport, immune clearance, and fibrinolysis. The strongest mechanistic convergence involved INPP5K, PITPNA, TLCD2, WDR81, RILP, and SERPINF1, with RTN4RL1 adding

neuronal surface and structural signaling. This was evidence of functional adjacency and convergence, not proof of multigene synergy or epistasis.

3.3 Spatial and developmental opportunity

Phase 3 established where affected mechanisms could plausibly operate. The clearest direct cell-localized convergence was RPE-centered, particularly INPP5K, PITPNA, and SERPINF1. PRPF8 showed broad retinal relevance, and other loci provided distributed retinal, neuronal, cortical, and developmental opportunity. The negative results were equally informative: the affected interval did not emerge as a preferential magnocellular program in human dLGN data, and adult/developmental cortical analyses did not support a simple V1-specific expression model. Expression was explicitly treated as opportunity rather than dysfunction.

3.4 Dosage, perturbation, and phenotype-level evidence

Phase 4 moved beyond expression to perturbational evidence and genotype-state discipline. RPA1 provided the clearest direct human haploinsufficiency precedent; SERPINF1 showed direct heterozygous molecular dosage reduction of PEDF; INPP5K provided one-copy physiological evidence outside the retina together with strong phosphoinositide/lysosomal mechanism. Other loci supplied strong but less genotype-equivalent evidence from biallelic loss, knockout, hypomorphs, or variant-specific disease mechanisms. The project repeatedly preserved the distinction between severe loss and heterozygous deletion.

At the systems level, the retinal evidence favored impaired maintenance, reduced resilience, delayed recovery, or stress vulnerability more strongly than increased sensory gain. The best partial-dose retinal physiology identified in the project was reduced, not amplified, electrophysiological response. Likewise, RTN4RL1 complete-loss neural models altered network

construction but did not support a generic hyperexcitability narrative. MIR22 remained relevant to epileptogenesis, but its strongest evidence came from complete loss after an epileptogenic insult, not from heterozygous loss under photic stimulation.

3.5 Phase 5 deterministic integration and simulation

Phase 5 formalized the earlier record into a layered evidence model and treated missing downstream physiology as missing rather than as zero or as implied support. Its modeling and simulation work did not create biological evidence; it examined methodological behavior. The standardized simulation suite used 20,000 replicates per scenario/cell where applicable, approximately 1.88 million primary stochastic draws/resamples. It showed that modest heterozygous effects can be difficult to detect with too few independent biological replicates, that pseudoreplication cannot substitute for independent clones or donors, that broad frequency scans can generate false-positive signals without correction, and that limited data can make a distributed mechanism appear spuriously single-gene.

The integrated Phase 5 conclusion remained conservative: the deletion is best treated as a candidate distributed visual-neural reserve lesion, not a proven single-gene photosensitivity mutation. Broader retinal/RPE dysfunction, developmental/network modification, and stress-dependent vulnerability are better supported than photic seizure causation.

4. Final Evidence Adjudication

Claim	Phase 6 status	Evidence boundary
The CNV is structurally real and biologically non-neutral	Supported	Genomic lesion established; multiple constituent loci have dosage/perturbational consequences.
Retinal/RPE mechanisms are	Strongly supported	Mechanism, expression opportunity,

relevant		and constituent-gene perturbation converge; exact-CNV retinal phenotype remains unmeasured.
Membrane/endosomal and stress-response reserve may be reduced	Supported as a mechanistic model	Multiple loci converge, but combined heterozygous function has not been directly measured.
Neurodevelopment/network organization may be altered	Plausible and partially supported	Strong-model evidence exists; heterozygous full-CNV circuit phenotype is not established.
Multigene synergy or epistasis is present	Not demonstrated	Convergence is not equivalent to interaction, synergy, or epistasis.
The retina has increased photon-response gain	Unsupported by current record	Available physiology more often points toward reduced output, slower recovery, or vulnerability.
The CNV causes abnormal V1 flicker entrainment or cortical propagation	Unresolved / not demonstrated	Direct genotype-specific visual-cortical physiology is absent.
The CNV causes a photoparoxysmal response or visually triggered seizures	Insufficient evidence	The causal chain stops before the disease-specific physiological endpoint.

5. Critical Assessment of the Work

5.1 What the project did well

It froze a precise genomic reference and resisted syndrome-level overgeneralization from larger 17p13.3 deletions containing different genes.

It enforced an evidence hierarchy: deletion -> dosage -> molecular perturbation -> spatial opportunity -> cellular/developmental effect -> visual physiology -> phenotype -> causality.

It distinguished genotype states rather than treating knockout, biallelic loss, pathogenic variants, and heterozygous deletion as interchangeable.

It preserved counterevidence and null findings, including compensation, normal heterozygous states, reduced rather than increased response, and failure to identify magnocellular or V1-specific signatures.

It progressively narrowed the working hypothesis instead of escalating claim strength as more literature accumulated.

Phase 5 explicitly addressed reproducibility hazards such as low independent replicate count, pseudoreplication, multiple-frequency testing, and model simplification.

5.2 Where the methodology remains vulnerable

Literature dependence. Much of the mechanistic evidence is assembled across different species, tissues, ages, perturbations, and assay types. Biological coherence can be real while still failing to transfer quantitatively to this exact CNV.

Sparse exact-genotype evidence. Most loci lack direct measurements in cells carrying the complete heterozygous deletion. This is the largest limitation in converting mechanism into CNV-specific biology.

Potential ascertainment and confirmation bias. The project began from a visual-neurological question, which creates a risk that visual or seizure-relevant literature was more deeply sampled than unrelated phenotypic domains. The counterevidence framework reduces but cannot fully remove that risk.

Pathway and network inflation. Gene-dense CNVs can appear convergent because broadly expressed genes participate in common cellular systems. Computational convergence cannot substitute for direct combinatorial perturbation.

Transcript and breakpoint uncertainty. The array-defined boundaries do not determine the exact chromosome junction, transcript fate, nonsense-mediated decay, promoter/enhancer effects, or allele-specific compensation for INPP5K and RTN4RL1.

Expression-resource limitations. Single-cell and single-nucleus atlases have dropout, donor, sampling, batch, regional-dissection, developmental-stage, and annotation constraints; expression cannot establish dysfunction.

Model dependence. Even disciplined ordinal or deterministic evidence models depend on chosen weights, equivalence penalties, and missingness rules. Robustness analyses help identify sensitivity but cannot convert sparse biology into calibrated causal probability.

5.3 Adversarial interpretation

A skeptical reading can explain much of the current record without invoking photosensitivity: the CNV may be biologically consequential because it affects several broadly important cellular systems, while the apparent visual-neural convergence may reflect a mixture of genuine retinal relevance, broad neuronal expression, and literature availability. That null/weak-specificity alternative has not been eliminated. Conversely, the persistence of retinal/RPE, trafficking, stress-response, and network-development signals across independent evidence types argues against treating the interval as biologically irrelevant. The balanced conclusion is therefore neither “causal” nor “nothing”: the deletion remains a credible candidate reserve lesion whose disease-specific physiological consequences are unmeasured.

6. Evidence Boundary and Falsifiability

The highest supported causal level is: structurally defined heterozygous CNV with selected demonstrated dosage consequences, mechanistically relevant retinal/neural pathways, spatiotemporal opportunity, constituent-gene molecular/cellular phenotypes, and distributed functional convergence. The project does not cross the required bridge into exact-CNV retinal physiology, thalamic transfer, V1 flicker physiology, cortical inhibitory containment, pathological frequency entrainment, long-range propagation, PPR, or visually triggered seizure causation.

Evidence that would materially weaken the functional-reserve model includes: normal molecular and physiological behavior of an isogenic full-CNV heterozygous system across the relevant retinal/neural assays under adequate power; complete compensation of predicted dosage-sensitive nodes; absence of stress-dependent differences after calibrated challenge; or demonstration that matched genomic control intervals show equal or stronger domain-specific convergence under the same analytical pipeline. Evidence that would strengthen the model requires exact-genotype perturbation and, ideally, rescue.

7. Path A - Continue In Silico Research with Mentorship and Peer Exposure

This path treats the first-pass record as a reproducible research portfolio to be stress-tested by domain experts rather than as a finished causal claim. The near-term objective is methodological exposure: place the frozen pipeline, evidence tables, assumptions, and negative findings in front of mentors or peers in human genetics, retinal biology, neurophysiology, bioinformatics, or epilepsy research and invite attempts to reproduce, criticize, or break the model.

Convert the Phase 1-5 evidence base into a clean public or mentor-reviewable repository with frozen inputs, source manifest, scripts, environment specification, and explicit expected outputs.

Ask a mentor or peer to independently audit a bounded component first - for example the CNV structural map, RPE convergence, genotype-equivalence coding, dLGN/V1 interpretation, or Phase 5 statistical assumptions.

Reanalyze one or two public datasets from raw or minimally processed data under preregistered questions, rather than adding more narrative literature synthesis.

Seek external review of the null model and the phenotype-ascertainment problem: whether an unrelated gene-dense CNV would appear similarly “convergent” under the same workflow.

Present negative and contradictory findings alongside positive results. The project should gain credibility by surviving criticism, not by maximizing the number of supportive mechanisms.

Use mentor feedback to define a Version 2 research protocol. Do not alter this frozen Version 1 record; any changes should be explicit amendments or a new project pass.

A useful endpoint for Path A would be a mentor-reviewed methods note, reproducible notebook/repository, poster, undergraduate research presentation, or small methods manuscript that focuses on evidence integration and rare-CNV hypothesis discipline rather than claiming a new clinical syndrome.

8. Path B - Transition Toward Wet-Lab Research

8.1 Brief research proposal

Provisional title: Functional Validation of a Heterozygous 17p13.3 Microdeletion as a Distributed Retinal-Neural Reserve Lesion.

Central question: Does the exact heterozygous chr17:1,411,408-1,841,103 deletion produce measurable, rescue-sensitive deficits in retinal/RPE maintenance, membrane/endosomal biology, stress response, neural development, or visually driven network dynamics?

Primary hypothesis: The CNV produces modest, distributed deficits in retinal/RPE and neural functional reserve that may be compensated at baseline but become measurable under controlled physiological or stress challenge. Photosensitivity is a downstream exploratory phenotype, not the primary assumed outcome.

Alternative/null hypotheses: (1) the heterozygous CNV is substantially compensated and produces no reproducible phenotype under the tested conditions; (2) one locus dominates the phenotype; (3) phenotypes are developmental/network-specific without retinal involvement; or (4) retinal abnormalities occur without abnormal cortical flicker propagation.

8.2 Specific aims

Create or obtain isogenic wild-type, heterozygous full-CNV, corrected/rescued, and selected single-gene heterozygous human cellular lines. Confirm CNV structure and transcript/protein dosage before phenotyping.

Test retinal/RPE reserve using retinal organoids and/or RPE models. Priority endpoints: phosphoinositide pools, lipid composition, outer-segment handling, lysosomal/endosomal function, oxidative/DNA-damage response, dark-adaptation or electrophysiologic analogues, and recovery after calibrated challenge.

Test neural/network reserve in relevant human neuronal systems. Priority endpoints: progenitor behavior, neurite morphology, synapse density, E/I-associated measures, calcium dynamics, multielectrode-array firing/bursting, frequency entrainment, phase locking, and long-range propagation.

Use genotype correction or targeted rescue to distinguish CNV-specific effects from background variation and to identify whether observed phenotypes are distributed or locus-dominant.

8.3 Minimal experimental design principles

Use independent biological clones/donors; do not treat many cells from one clone as independent replication.

Predefine primary endpoints and flicker-frequency analyses; control family-wise or false-discovery error across frequency scans and multiple phenotypes.

Blind image/trace quantification where practicable and preserve raw data.

Report null results and directionally contradictory effects.

Power studies for modest heterozygous effects rather than knockout-scale effects.

Treat rescue as a major causal discriminator.

8.4 Funding and fellowship strategy

The most realistic transition is mentor- and institution-based. A trainee should first join or collaborate with a laboratory that already has the required cell culture, retinal organoid/RPE, electrophysiology, imaging, or network-analysis infrastructure. The project can then be adapted to the host laboratory rather than attempting to acquire an entire platform de novo.

NIH/NEI training mechanisms such as F31 or F32 are appropriate once the investigator is in the corresponding predoctoral or postdoctoral training stage and working with an eligible mentor/institution.

NIH R15/AREA mechanisms can support small-scale biomedical projects at eligible undergraduate-focused institutions, but the principal investigator must satisfy institutional and

appointment requirements; this is better viewed as a mentor/institutional route than an immediate independent application.

Vision-focused foundations can be relevant when the proposal is framed around retinal cell biology or inherited retinal mechanisms; epilepsy foundations become more appropriate only if the project directly tests seizure/network physiology rather than relying on indirect mechanistic relevance.

The grant pitch should emphasize the unusual strength of the preliminary in silico work as a hypothesis-narrowing and assay-prioritization package, while explicitly stating that the central physiological claims remain unvalidated. The strongest proposal is one that funds the missing measurements, not one that presents the missing measurements as already solved.

9. Recommended Decision Point

The two paths are complementary rather than mutually exclusive. The immediate low-cost continuation is Path A: external methodological scrutiny, replication, and mentorship. Path B becomes appropriate when a host laboratory, mentor, and assay platform can support exact-genotype experiments. The first wet-lab study should be deliberately narrower than the full project and should target one or two mechanistic bottlenecks with high discriminating value.

A particularly efficient first experiment would compare isogenic wild-type, full-CNV heterozygous, and corrected cells in a retinal/RPE model under baseline and calibrated stress. If no reproducible molecular or physiological deficit is detected with adequate power, the distributed reserve hypothesis should be weakened. If a deficit is detected and rescued, the project gains a direct CNV-specific causal foothold before moving toward more complex thalamic/cortical photic models.

10. Final Project Conclusion

The first-pass in silico program establishes a structurally precise, biologically meaningful 17p13.3 heterozygous deletion containing multiple loci with credible dosage, retinal, membrane, trafficking, stress-response, and neurodevelopmental relevance. Across the project, the evidence repeatedly converged on reduced or altered functional reserve rather than on a single-gene or simple hyperexcitable-retina mechanism. The strongest phenotype-level support concerns retinal/RPE maintenance, photoreceptor integrity, intracellular material handling, stress resilience, and neural-development/network organization. General epileptogenic modification remains plausible but incompletely genotype-matched. Epileptic photosensitivity remains downstream of an unmeasured physiological gap.

Accordingly, the project does not conclude that the CNV causes photosensitivity, a photoparoxysmal response, or visually triggered seizures. It concludes that the CNV is a credible candidate distributed visual-neural reserve lesion whose most important unresolved questions now require either independent methodological scrutiny or direct experimental perturbation. This narrowing is the principal scientific accomplishment of the first pass.

11. Phase 6 Freeze Declaration

PHASE 6 - COMPLETE

The first-pass in silico Genome Research project is hereby frozen as Version 1.0 as of 30 September 2026. The controlling structural reference remains GRCh37/hg19 chr17:1,411,408-1,841,103, nominally 429,696 bp, heterozygous. The frozen interpretive record is that the deletion is biologically consequential and mechanistically consistent with a distributed visual-neural functional-reserve model, while direct causation of epileptic photosensitivity by the exact CNV remains unresolved.

Future research must not silently modify this record. New evidence should be handled by one of three mechanisms: (1) an erratum correcting a demonstrable factual or computational error; (2) a formally reopened analysis that states which frozen conclusion is being reconsidered and why; or (3) a Version 2 research program that preserves Version 1 and documents all changes in data, assumptions, methods, and interpretation.

Null findings, contradictory evidence, failed hypotheses, and unresolved measurements are part of the frozen scientific record and must remain preserved.

12. Current Forward-Looking Funding Notes (30 September 2026)

These items are examples of funding structures relevant to future continuation and are not claims of present applicant eligibility. Eligibility must be checked at the time of application.

National Eye Institute: F31 predoctoral and F32 postdoctoral fellowship mechanisms support mentored vision research training; F31 support can extend up to five years and F32 generally supports two to three years.

NIH R15/AREA: designed for eligible undergraduate-focused institutions; current NIH guidance states that the PD/PI must hold a primary appointment at the eligible organization and that projects may run up to three years. Current NIH pages list total direct-cost limits up to \$375,000 for the applicable AREA mechanism.

Foundation Fighting Blindness: the FY27 schedule lists an Individual Investigator Research Award LOI deadline of 8 October 2026, a Clinical/Research Fellowship deadline of 15 October 2026, a Career Development Award deadline of 28 January 2027, and a Postdoctoral Fellowship LOI deadline of 8 October 2026. These programs are oriented toward retinal disease research and require program-specific eligibility and scope fit.

CURE Epilepsy: Taking Flight supports early-career epilepsy investigators and requires defined postdoctoral-level eligibility and mentor support. Any application from this project should wait until the work contains a direct epilepsy/network physiology aim rather than using retinal or molecular plausibility as a surrogate.

Record of Source Documents

Phase 1: Locus Characterization and Structural Definition of the 17p13.3 Microdeletion - Deletion Map v1.0.

Phase 2: Molecular and Chemical Characterization of an Interstitial 17p13.3 Microdeletion - Comprehensive Synthesis.

Phase 3: Expression, Cellular Localization, Visual-System Distribution, and Developmental Timing; associated Phase 3 research and methods records.

Phase 4: Dosage Sensitivity, Molecular Perturbation, Cellular Phenotype, Developmental Perturbation, Functional Convergence, Visual-System Relevance, and Integrated Adjudication - Frozen Master Report v1.0.

Phase 5: Deterministic Evidence Integration, In-Silico Gap Resolution, Mechanistic Modeling, and Reproducibility Analysis - Frozen scientific record and casual-reader summary v1.0.

Funding Sources Consulted for Forward-Looking Notes

National Eye Institute, Funding for Training and Career Development, accessed 30 September 2026.

NIH Grants & Funding, Research Enhancement Awards (R15), accessed 30 September 2026.

Foundation Fighting Blindness, Scientific Grants and Awards Programs / FY27 schedule, accessed 30 September 2026.

CURE Epilepsy, Grants Program and Opportunities for Epilepsy Researchers, accessed 30 September 2026.

Vocabulary and Formulas Appendix

Reader-facing terminology, calculation register, and symbol index

Purpose and scope

This appendix is a reader-facing reference for terminology, variables, and formulas used across Phases 1–6 of the project “Genomic and Molecular Analysis of a 17p13.3 Microdeletion and Its Candidate Visual-Neurological Mechanisms.” It preserves the frozen project’s terminology and inferential boundaries. Definitions describe how a term is used in this project, which may be narrower than a general textbook definition.

Formula classification rule

Formulas are labeled as structural calculations, statistical definitions, biochemical reactions, operational definitions, conceptual models, or Phase 5 modeling constructs. A conceptual equation is not a validated physiological law. An ordinal score is not a biological effect size or probability. Where the frozen record named a standard statistical method but did not preserve its explicit equation, this appendix supplies the canonical mathematical definition and labels it as an explanatory addition rather than silently attributing it to the original calculation pipeline.

Fixed structural reference

GRCh37/hg19 chr17:1,411,408–1,841,103; nominal span 429,696 bp; heterozygous deletion. The frozen locus set contains 13 protein-coding and 4 noncoding loci. INPP5K and RTN4RL1 are boundary-intersected. GRCh38 may be used for modern annotation only with documented build conversion and must not silently replace the frozen coordinates.

Appendix A — Vocabulary

This glossary contains 123 project terms. It emphasizes terms that affect interpretation, reproducibility, genotype-state equivalence, expression analysis, visual physiology, and causal claims.

Term	Project-specific description	Primary phase(s)
Adaptation	Adjustment of retinal or neural responsiveness during or after sustained or repeated stimulation. In this project, dark adaptation specifically means recovery of visual sensitivity after photobleaching.	Phase 4; retained in Phase 6
Allele	One of the copies or sequence states of a locus. For this CNV, interpretation generally concerns the affected allele versus the homologous chromosome carrying the remaining copy.	Phases 1, 4
Amino-acid-coding sequence	The CDS nucleotides that encode amino acids, excluding the stop codon when the project reports the INPP5K coding-loss percentage.	Phase 1
Annotation release	A defined version of a genome/gene annotation resource. The project records releases because gene boundaries, transcripts, and labels	Phases 1, 3, 5

	can change without the underlying CNV changing.	
Autolysosomal reformation (ALR)	Membrane-recycling process that regenerates lysosomes from autolysosomes after prolonged autophagy. INPP5K is implicated in normal ALR.	Phases 2, 4
BED coordinate convention	A 0-based, half-open interval convention commonly used by BED/BEDTools. The project warns that GFF3 coordinates must be converted correctly before overlap calculations.	Phase 1
Biallelic loss	Disruption of both gene copies. It is not genotype-equivalent to the one-copy heterozygous loss produced by the index CNV.	Phases 4, 5
Biological replicate	An independently derived biological sample, donor, line, or clone that supports inference beyond technical repeat measurements. Many cells from one clone do not replace independent biological replication.	Phases 5, 6
Boundary-intersected gene	A gene crossed by a reported CNV boundary rather than fully encompassed by the interval.	Phase 1

	INPP5K and RTN4RL1 are the two boundary-intersected protein-coding loci in the frozen map.	
Breakpoint	The location at which a structural variant transitions between retained and deleted sequence. The array-reported coordinates are operational boundaries, not nucleotide-exact junction sequences.	Phases 1, 6
Bulk RNA-seq	RNA sequencing performed on mixed populations of cells from a tissue. It measures aggregate expression but cannot by itself identify which constituent cell types produce the signal.	Phase 3
Candidate modifier	A factor plausibly capable of changing phenotype severity, threshold, penetrance, or expression without being established as a sufficient primary cause.	Phase 4
Causal chain	The ordered evidentiary/biological sequence from genomic disruption through dosage, molecular perturbation, spatial opportunity, cellular physiology, systems physiology, and phenotype. Missing	Phases 4, 5, 6

	downstream links cannot be supplied by strong upstream evidence.	
CCDS	Consensus Coding Sequence. In Phase 1 it is used as a standardized coding reference, including for the RTN4RL1 coding-overlap calculation.	Phase 1
Cell-type specificity	The degree to which expression or another molecular signal is concentrated in one cell class relative to others. It is evidence of biological opportunity, not evidence of dysfunction.	Phases 3, 5
Cellular phenotype	A measurable change in cell structure, survival, trafficking, signaling, electrophysiology, or stress response after perturbation.	Phase 4
CNV (copy-number variant)	A genomic deletion or duplication that alters the number of copies of one or more loci. The index CNV is a heterozygous deletion.	All phases
Coding sequence (CDS)	The nucleotide sequence of a transcript that encodes protein. The Phase 1 overlap calculations operate on CDS segments rather than total	Phase 1

	genomic gene span.	
Coexpression	Correlated or overlapping expression of genes across samples, cells, or biological units. Coexpression does not establish molecular interaction, epistasis, synergy, or causality.	Phases 3, 5
Compensation	Biological buffering that reduces the functional impact of a genomic perturbation, including intact-allele expression, paralogs, developmental adaptation, or homeostatic responses.	Phases 4–6
Complete knockout	Experimental removal or inactivation of all functional copies of a gene. Knockout evidence is not treated as equivalent to heterozygous deletion evidence.	Phases 4, 5
Confidence interval / uncertainty interval	A quantitative interval representing uncertainty around an estimate. Phase 5 requires uncertainty to be retained where calculable rather than replaced by point estimates alone.	Phase 5
Constraint / dosage characteristic	Genomic or gene-level evidence relevant to tolerance of loss or copy-number change. In Phase 5, such characteristics may be used in	Phase 5

	matched null intervals when feasible.	
Counterevidence	Evidence that contradicts or weakens a proposed mechanism, such as normal heterozygous phenotypes, compensation, reduced rather than increased physiological response, or protective effects under specific conditions.	Phases 4–6
Cryptographic hash	A deterministic digital fingerprint proposed for freezing Phase 5 inputs so later users can verify that the analyzed dataset has not silently changed.	Phase 5
Dark adaptation	Recovery of visual sensitivity after photobleaching or sustained light exposure; treated as a possible retinal reserve endpoint.	Phases 4–6
Detection fraction	The fraction of cells/nuclei or samples in which a gene is detectably measured. In single-cell data it is influenced by both biology and technical dropout.	Phases 3, 5
Developmental trajectory	The pattern of a gene's expression or biological effect across developmental stages. Phase 3 specifies within-gene normalization	Phase 3

	to identify relative prenatal, neonatal, childhood, or adult peaks.	
Directional convergence	Multiple perturbations pushing a shared endpoint in the same direction. This is stronger than merely affecting the same system, but still does not itself prove synergy.	Phase 4
dLGN / lateral geniculate nucleus (LGN)	The dorsal lateral geniculate nucleus, the principal thalamic relay between retinal ganglion cells and primary visual cortex.	Phases 3–6
Dominant-negative variant	A variant whose product interferes with the normal product. Its mechanism is not assumed to be equivalent to a simple heterozygous deletion.	Phases 4, 5
Dosage consequence	A demonstrated change in RNA, protein, molecular function, or cellular behavior attributable to copy-number/genotype state. Deletion alone does not establish the size of the dosage effect.	Phases 4–6
Dosage sensitivity	The degree to which changing gene copy number alters transcript, protein, molecular function, cell	Phases 1, 2, 4–6

	behavior, or phenotype.	
Dropout	Failure to detect a transcript in single-cell/single-nucleus sequencing even when biologically present, especially for low-abundance transcripts. Zero counts in a small cell population are therefore not automatically biological absence.	Phase 3
Efferocytosis	Recognition and engulfment of apoptotic cells or debris by phagocytic cells; SCARF1 participates in C1q-associated efferocytosis.	Phases 2, 4
Electroretinography (ERG)	Measurement of retinal electrical responses to light. The a-wave primarily reflects photoreceptor activity; the b-wave reflects downstream inner-retinal activity.	Phases 4, 6
Endolysosomal system	The connected endosome-to-lysosome network that sorts, transports, degrades, and recycles cellular cargo.	Phases 2, 4–6
Enrichment	A signal occurring more strongly or frequently than a defined comparison or null expectation. Phase 5 requires matched null analysis before treating	Phases 3, 5

	convergence as unusually specific.	
Entrapment of missingness / missing $\neq$ zero	Project rule that an unmeasured causal layer must remain missing/unresolved rather than being encoded as zero or inferred from neighboring layers. Zero is reserved for defined null or evidence-of-absence states.	Phase 5
Epileptogenesis	The biological process by which a network develops a persistent tendency to generate seizures. MIR22 evidence in the project is most directly related to post-insult epileptogenesis rather than photic seizure causation.	Phases 4–6
Epileptogenic reserve	Conceptual resistance of a neural network to transitions into persistent or recurrent epileptic activity; lower reserve implies easier pathological transition under challenge.	Phases 4–6
Epistasis	A genetic interaction in which the effect of one locus depends on the state of another. The project did not demonstrate epistasis among deleted loci.	Phases 2, 4–6
Evidence hierarchy	The ordered project rule: genomic	Phases 4–6

	evidence → molecular mechanism → expression/spatiotemporal opportunity → perturbational evidence → cellular phenotype → developmental consequence → visual-system physiology → phenotype association → causality.	
Expression prevalence	The proportion of relevant biological units showing detectable expression; closely related to detection fraction and distinct from expression magnitude.	Phase 5
False discovery rate (FDR)	Expected proportion of false discoveries among findings declared significant. Phase 3 specifies Benjamini-Hochberg control with $q < 0.05$ for planned coexpression testing.	Phase 3
Functional convergence	Independent genes or mechanisms affecting the same molecular system, cellular process, anatomical compartment, developmental operation, or phenotype endpoint.	Phases 2–6
Functional reserve	Difference between a system's available/maximal biological capacity and the physiological	Phases 4–6

	demand placed on that system. The project's leading model proposes distributed reduction or alteration of reserve rather than obligatory baseline failure.	
Gain	Ratio between change in output and change in input. Increased sensory gain means a larger physiological response to the same matched stimulus.	Phase 4
Gain-of-function variant	A variant that increases or changes activity in a way not equivalent to simple loss of one copy.	Phases 4, 5
Genome assembly	A defined coordinate reference for the genome. The frozen structural reference is GRCh37/hg19; GRCh38 may be used for modern annotation only with documented conversion.	All phases
Genotype-state discipline	Project rule requiring explicit distinction among heterozygous deletion, partial deletion, hypomorph, biallelic loss, complete knockout, dominant-negative/gain-of-function variants, and pharmacologic perturbation.	Phases 4–6
GFF3 coordinate convention	A 1-based, inclusive annotation	Phase 1

	coordinate convention used in the Phase 1 computational overlap workflow.	
Haploinsufficiency	A state in which one functional gene copy is insufficient for normal molecular, cellular, or clinical function. It must be demonstrated rather than presumed from deletion.	Phases 2, 4–6
Heterozygous deletion / heterozygous loss	Loss or disruption of one of two gene copies. It does not automatically imply exactly 50% RNA, protein, or function.	All phases
Hypomorph	An allele retaining partial gene function rather than complete loss.	Phases 4, 5
In silico	Analysis conducted computationally using genomic, transcriptomic, literature-derived, or simulated data rather than direct wet-lab perturbation of the exact CNV.	Phases 5, 6
Independent replication	Reproduction of an observation in independent donors, lines, datasets, studies, or experiments rather than repeated measurements of the same underlying sample.	Phases 3, 5, 6
Isogenic	Genetically matched experimental	Phases 4, 6

	lines differing only at the engineered locus/CNV of interest, minimizing background-genome confounding.	
Koniocellular (K), magnocellular (M), parvocellular (P)	Major subdivisions of primate visual-thalamic organization. Phase 3 did not identify the affected loci as a preferential adult human magnocellular transcriptional program.	Phases 3, 4
Leave-one-locus-out analysis	Sensitivity procedure in which each affected locus is removed in turn and the model is recomputed to determine whether conclusions are distributed or driven by one/small set of loci.	Phase 5
Leave-one-source-out analysis	Sensitivity procedure that removes an evidence source/study and recomputes conclusions to test dependence on individual publications or datasets.	Phase 5
lncRNA (long noncoding RNA)	A longer non-protein-coding RNA transcript. PITPNA-AS1 and MIR22HG are treated as potentially detectable but often low-abundance or assay-sensitive.	Phases 1, 3
Locus	A defined genomic location. The	All phases

	frozen deletion affects at least 17 annotated loci.	
MANE Select	Matched Annotation from NCBI and EMBL-EBI representative transcript used as the preferred standardized transcript reference when available.	Phase 1
Marker gene	A gene reproducibly enriched in or characteristic of a cell class/subtype. Failure of an affected locus to appear in a marker list is not the same as proof of absence.	Phase 3
Mechanistic domain	A project-defined biological subsystem into which locus-level evidence can be grouped, such as retinal reserve, membrane/endosomal reserve, damage-response reserve, or neural/network reserve.	Phase 5
Mechanistically plausible	Supported by a coherent chain of biological evidence but not established as causal for the final phenotype.	Phases 4–6
MicroRNA (miRNA)	A small regulatory RNA. MIR22 is within the deletion; conventional poly(A) RNA-seq is not a valid direct measurement of mature	Phases 1, 3–6

	miR-22 abundance.	
Monte Carlo simulation	Repeated stochastic sampling used in Phase 5 to examine methodological behavior such as power, false-positive behavior, and model identification. It did not generate biological evidence for the CNV.	Phase 5
Multiple-testing correction	Adjustment for testing many hypotheses/frequencies so nominally significant findings are not overcounted. Phase 3 specifies BH-FDR for coexpression; Phase 6 requires correction for future frequency scans.	Phases 3, 5, 6
Neural/network reserve	Operational Phase 5 domain concerning neurite development, synapse density, network organization, calcium dynamics, and white-matter/conduction-related evidence.	Phase 5
Neuroprotectin D1 (NPD1)	A DHA-derived lipid mediator with anti-inflammatory and anti-apoptotic actions in retinal pigment epithelium under oxidative stress.	Phases 2, 4
Normalized expression	Expression transformed or scaled according to a defined pipeline to	Phases 3, 5

	permit within-dataset comparison. Values from unrelated sequencing platforms should not be compared as if directly interchangeable.	
nTPM	Normalized transcripts per million as reported by Human Protein Atlas resources used in Phase 3 screening. It is a source-specific normalized expression measure and not an absolute molecule count.	Phase 3
Null / weak-specificity model	Competing model in which the deleted genes have real biology but the apparent visual-neurological convergence is not specific evidence for the index phenotype.	Phases 5, 6
Null result	A measured result that does not support the tested effect under specified conditions. It is retained as evidence rather than discarded or redefined.	All phases
Operational CNV boundary	The reported array-coordinate edge used for calculations. It is not asserted to be the nucleotide-exact physical chromosome junction.	Phases 1, 6
Outer segment	Specialized photoreceptor compartment containing	Phases 4, 6

	photopigment-rich membranes where photon absorption and phototransduction begin.	
Pairwise coexpression	Coexpression evaluated between two genes, typically across independent biological units. It is distinct from multi-gene coexpression.	Phase 5
Paralog compensation	Functional buffering by a related gene with overlapping function. It is a potential reason why severe knockout phenotypes may not appear after one-copy loss.	Phases 4–6
Parameter sensitivity	Dependence of a model conclusion on defensible changes to weights, mappings, penalties, or missing-data rules. A conclusion is called robust only if it survives reasonable variation.	Phase 5
Partial gene deletion	A deletion affecting only part of a gene/locus. The project treats INPP5K and RTN4RL1 as partial genomic deletions with substantial coding disruption on the affected allele.	Phases 1, 5
PEDF	Pigment epithelium-derived factor, encoded by SERPINF1; a secreted	Phases 2, 4–6

	retinal trophic, antiangiogenic, lipid-linked, and cytoprotective protein.	
Perturbation equivalence	The degree to which an experimental perturbation matches the index CNV genotype state. Phase 5 requires separate weighting/mapping rather than treating knockout and heterozygous deletion as equivalent.	Phase 5
Phase locking	Consistency of response phase relative to a periodic stimulus or oscillation; proposed as an endpoint for future flicker/network experiments.	Phases 4, 5, 6
Phenotype association	Observed statistical or clinical relationship between genotype and phenotype. Association is downstream of molecular evidence but is not automatically causation.	Phases 4–6
Phosphoinositide	A phosphorylated phosphatidylinositol lipid involved in membrane identity, trafficking, signaling, and protein recruitment.	Phases 2, 4–6
Photochemical vulnerability	Greater molecular or cellular injury from the same photon/light exposure, distinct from an abnormally large	Phase 4

	sensory response.	
Photoparoxysmal response (PPR)	An abnormal epileptiform EEG response evoked by intermittent photic or pattern stimulation; it may occur with or without a clinical seizure.	Phases 4–6
Photosensitive epilepsy (PSE)	Epilepsy in which visual stimulation can provoke abnormal EEG activity and, in some patients, clinical seizures.	Phases 4–6
Phototransduction	Conversion of absorbed photons into an electrical photoreceptor response through opsin/rhodopsin, transducin, PDE6, cGMP, and cyclic-nucleotide-gated-channel signaling.	Phase 4
PI(3,4,5)P3 / PIP3	Phosphatidylinositol 3,4,5-trisphosphate, another substrate represented in the INPP5K reaction set.	Phase 4
PI(4,5)P2	Phosphatidylinositol 4,5-bisphosphate, a membrane signaling lipid whose 5-phosphate can be removed by INPP5K.	Phases 2, 4
PI3P	Phosphatidylinositol 3-phosphate, enriched on early endosomes; the	Phases 2, 4

	WDR81/WDR91-associated system regulates endosomal PI3P.	
PI4P	Phosphatidylinositol 4-phosphate, a signaling lipid produced by INPP5K-mediated dephosphorylation of PI(4,5)P2.	Phases 2, 4
Primary visual cortex (V1)	Principal cortical recipient of LGN visual input and a major node relevant to visually driven cortical responses and photoparoxysmal propagation.	Phases 3–6
Pseudobulk	Aggregation of single-cell observations into biologically meaningful units such as donor $\times$ cell type $\times$ region before gene-level statistical analysis, reducing pseudoreplication.	Phase 3
Pseudogene	A gene-like sequence generally lacking canonical protein-coding function. RN7SL105P is retained as an affected pseudogene but requires mapping caution.	Phases 1, 3
Pseudoreplication	Treating non-independent observations (for example many cells from one clone) as independent biological replicates, artificially	Phases 3, 5, 6

	inflating apparent precision.	
q-value	FDR-adjusted significance measure. Phase 3 specifies $q < 0.05$ as the primary criterion for planned coexpression analyses where sample size permits.	Phase 3
Random-interval null model	Phase 5 control strategy comparing the index CNV with matched genomic intervals processed through the same convergence-scoring pipeline.	Phase 5
Rescue	Correction or restoration of a gene/CNV state or pathway that reverses a phenotype. Rescue provides stronger causal specificity than association alone.	Phases 4–6
Retinal ganglion cell (RGC)	Output neuron of the retina whose axons form the optic nerve and project to visual brain targets.	Phases 3, 4
Retinal pigment epithelium (RPE)	Support epithelium adjacent to photoreceptor outer segments; critical for phagocytosis, retinoid cycling, lipid handling, trophic support, and metabolic homeostasis.	Phases 2–6
Retinal/photoreceptor reserve	Operational Phase 5 domain	Phase 5

	including photoreceptor survival, outer-segment integrity, ERG amplitude, dark adaptation, recovery after light challenge, and oxidative-stress response.	
Robust conclusion	In Phase 5 terminology, a conclusion that remains stable across reasonable parameterizations/sensitivity analyses.	Phase 5
Sensory hypersensitivity	An abnormally large perceptual or physiological response to the same light stimulus. This is distinct from photochemical vulnerability.	Phase 4
Sequential-process convergence	Different genes acting at distinct stages of the same biological process. Phase 5 distinguishes this from direct physical interaction or synergy.	Phase 5
Single-cell RNA-seq (scRNA-seq)	RNA sequencing of individual cells. It provides cell-type localization but is subject to dropout, donor non-independence, batch effects, and assay-specific capture biases.	Phase 3
Single-nucleus RNA-seq (snRNA-seq)	RNA sequencing of isolated nuclei. Useful for human brain tissue but may differ from whole-cell RNA-seq	Phase 3

	in transcript capture and abundance patterns.	
Spatial co-localization	Presence of relevant genes/mechanisms in the same tissue or cell type, providing opportunity for interaction but not proving it.	Phases 3, 5
Spatiotemporal opportunity	Evidence that a gene/mechanism is expressed in the relevant tissue/cell type and developmental or physiological window. Expression creates opportunity, not dysfunction.	Phases 3–6
Spearman rank correlation	Nonparametric rank-based measure specified in Phase 3 for coexpression when sample size permits, avoiding an assumption of linear relationships.	Phase 3
Spliceosome	Large ribonucleoprotein machinery that removes introns from pre-mRNA; PRPF8 is a central U5 spliceosomal scaffold.	Phases 2, 4
Steady-state visual evoked potential (SSVEP)	Stimulus-frequency-locked cortical response produced by periodic visual stimulation; relevant to entrainment and temporal-stability testing.	Phase 4
Stress-dependent vulnerability	A phenotype that may be weak/absent at baseline but emerges	Phases 4–6

	under oxidative stress, injury, repeated activity, aging, metabolic demand, or other challenge.	
Synergy	A combined effect greater than expected under a specified additive model. The project does not claim demonstrated synergy among deleted loci.	Phases 2, 4–6
Temporal functional convergence	Different loci affecting the same endpoint at different developmental or physiological stages rather than acting simultaneously.	Phase 4
Topological convergence	Different perturbations affecting the same biological network or structural endpoint even if they push that endpoint in different directions.	Phase 4
TPM (transcripts per million)	RNA-seq abundance normalization used in Phase 3 tissue-expression resources. Cross-platform absolute TPM values are not assumed directly comparable.	Phase 3
Visual-neural reserve	Integrated capacity of retinal, visual-pathway, and cortical systems to maintain stable function under physiological or pathological	Phases 4–6

	demand.	
Wet-lab validation	Direct experimental testing in engineered/biological systems. Phase 6 proposes exact-CNV isogenic systems with rescue/correction rather than inferring exact-genotype physiology from literature alone.	Phase 6
Within-gene z-score	A standardized value proposed in Phase 3 to show whether a gene is relatively high or low compared with its own expression across developmental stages, not to compare absolute expression between different genes.	Phase 3

Appendix B — Formula and Calculation Register

This register contains 45 formulas, reactions, quantitative definitions, and model expressions. Each entry states what the expression means and whether it was an actual project calculation, a specified analysis method, or a conceptual/future-test model.

B.1 Nominal interval span

$$L = D_e - D_s + 1$$

Variables / definitions: D_s = reported inclusive start coordinate; D_e = reported inclusive end coordinate.

Interpretation: For chr17:1,411,408–1,841,103, this yields 429,696 bp.

Status / limitation: Structural calculation; exact for the reported inclusive coordinates, not the physical junction length.

Project provenance: Phase 1

B.2 Per-CDS-segment overlap

$$L_{\text{overlap}} = \max(0, \min(C_{\text{e}}, D_{\text{e}}) - \max(C_{\text{s}}, D_{\text{s}}) + 1)$$

Variables / definitions: $C_{\text{s}}/C_{\text{e}}$ = inclusive CDS-segment start/end; $D_{\text{s}}/D_{\text{e}}$ = inclusive deletion start/end.

Interpretation: Counts the number of deleted bases within one CDS segment under 1-based inclusive coordinates.

Status / limitation: Structural calculation used by the replication-ready Phase 1 method.

Project provenance: Phase 1

B.3 Total deleted coding length

$$L_{\text{deleted}} = \sum_i L_{\text{overlap},i}$$

Variables / definitions: i indexes CDS segments.

Interpretation: Sums deleted bases across all CDS segments of the reference transcript.

Status / limitation: Structural calculation.

Project provenance: Phase 1

B.4 Total CDS length

$$L_{\text{CDS}} = \sum_i (C_{\text{e},i} - C_{\text{s},i} + 1)$$

Variables / definitions: i indexes CDS segments.

Interpretation: Total coding length of the transcript under inclusive coordinates.

Status / limitation: Structural calculation.

Project provenance: Phase 1

B.5 Percent CDS deleted

$$\% \text{CDS deleted} = (L_{\text{deleted}} / L_{\text{CDS}}) \times 100$$

Variables / definitions: L_deleted = deleted coding bases; L_CDS = total coding bases.

Interpretation: Quantifies the fraction of coding sequence intersected by the CNV. Stop-codon handling must be stated.

Status / limitation: Structural calculation.

Project provenance: Phase 1

B.6 INPP5K amino-acid coding denominator

$$448 \text{ aa} \times 3 \text{ nt/aa} = 1,344 \text{ coding nt}$$

Variables / definitions: 448 = protein length used in Phase 1; stop codon excluded for amino-acid-coding calculation.

Interpretation: Provides the denominator for the INPP5K amino-acid-coding loss estimate.

Status / limitation: Worked structural calculation.

Project provenance: Phase 1

B.7 INPP5K coding-loss estimate

$$666 / 1,344 = 0.4955 \approx 49.6\%$$

Variables / definitions: 666 = approximate deleted coding nucleotides under the reported-coordinate model.

Interpretation: Approximate fraction of MANE amino-acid-coding sequence lying within the deleted interval on the affected allele.

Status / limitation: Worked structural estimate; dependent on reported CNV boundaries and transcript model.

Project provenance: Phase 1

B.8 RTN4RL1 CCDS-loss estimate

$$1,313 / 1,326 \times 100 \approx 99.0\%$$

Variables / definitions: 1,313 = approximate CCDS nucleotides on deleted side; 1,326 = total CCDS nucleotide count stated in Phase 1.

Interpretation: Shows that small genomic-span intersection can correspond to near-total coding disruption.

Status / limitation: Worked structural estimate; coordinate-model dependent.

Project provenance: Phase 1

B.9 BED conversion for an inclusive deletion interval

$$\text{BED_start} = D_s - 1; \text{BED_end} = D_e$$

Variables / definitions: D_s/D_e are 1-based inclusive coordinates.

Interpretation: Converts the deletion to 0-based half-open BED form; Phase 1 gives chr17 1411407 1841103.

Status / limitation: Coordinate-system conversion rule.

Project provenance: Phase 1

B.10 Developmental within-gene z-score (canonical definition)

$$z_{g,t} = (x_{g,t} - \text{mean}_g) / \text{SD}_g$$

Variables / definitions: $x_{g,t}$ = expression for gene g at stage t ; mean_g and SD_g are calculated across the stages included for that gene.

Interpretation: Implements the Phase 3 concept of expressing each developmental value relative to that gene's own trajectory.

Status / limitation: Canonical mathematical definition supplied here for reader clarity; the frozen Phase 3 text specifies a within-gene z-score but does not preserve an explicit equation or exact preprocessing pipeline.

Project provenance: Phase 3

B.11 Detection fraction

$$p_{\text{detect}} = n_{\text{detected}} / n_{\text{total}}$$

Variables / definitions: n_{detected} = cells/nuclei or biological units passing the defined detection threshold; n_{total} = units tested.

Interpretation: Quantifies expression prevalence. Threshold and unit of independence must be declared.

Status / limitation: Canonical implementation of the Phase 3/5 metric; threshold was dataset-dependent.

Project provenance: Phases 3, 5

B.12 Spearman rank correlation

$$\rho_s = \text{corr}(\text{rank}(X), \text{rank}(Y))$$

Variables / definitions: X, Y = expression values across independent biological units.

Interpretation: Rank-based coexpression measure specified because linearity is not assumed.

Status / limitation: Statistical method specified in Phase 3; exact handling of ties/missing data must follow the software implementation.

Project provenance: Phase 3

B.13 Benjamini-Hochberg FDR adjustment

Sort $p_{(1)} \leq \dots \leq p_{(m)}$; compare $p_{(i)}$ to $(i/m)Q$, or compute adjusted q-values monotonically from $m \cdot p_{(i)} / i$

Variables / definitions: m = number of tests; i = ordered test rank; Q = target FDR (Phase 3 primary criterion $q < 0.05$).

Interpretation: Controls expected false-discovery proportion across multiple coexpression tests.

Status / limitation: Canonical algorithmic definition supplied for reader clarity; project specifies BH-FDR, not a custom implementation.

Project provenance: Phase 3

B.14 INPP5K 5-phosphatase reaction



Variables / definitions: P_i = inorganic phosphate.

Interpretation: Represents established INPP5K 5-phosphatase chemistry.

Status / limitation: Biochemical reaction; not a quantitative in vivo flux model.

Project provenance: Phase 4

B.15 INPP5K PIP3 reaction



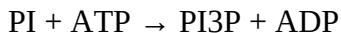
Variables / definitions: Substrate/product phosphoinositides; P_i = inorganic phosphate.

Interpretation: Represents an established substrate reaction of INPP5K.

Status / limitation: Biochemical reaction; do not infer in vivo flux from kinetic constants alone.

Project provenance: Phase 4

B.16 VPS34 reaction



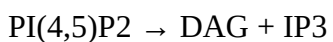
Variables / definitions: PI = phosphatidylinositol.

Interpretation: Represents kinase chemistry associated with endosomal PI3P production.

Status / limitation: Biochemical reaction.

Project provenance: Phase 4

B.17 PLC branch



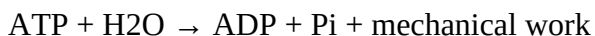
Variables / definitions: DAG = diacylglycerol; IP₃ = inositol trisphosphate.

Interpretation: Canonical signaling branch relevant to calcium-regulatory context.

Status / limitation: Biochemical pathway representation.

Project provenance: Phase 4

B.18 Dynein/ATPase work



Variables / definitions: P_i = inorganic phosphate.

Interpretation: Represents ATP-powered motor/remodeling work; RILP recruits dynein but does not catalyze ATP hydrolysis itself.

Status / limitation: Biochemical/energetic representation.

Project provenance: Phase 4

B.19 PEDF-R/PNPLA2 phospholipase representation



Variables / definitions: Generic phospholipid substrate/products.

Interpretation: Simplified representation of phospholipase/lipase chemistry.

Status / limitation: Biochemical simplification, not a complete enzyme mechanism.

Project provenance: Phase 4

B.20 RPA binding equilibrium



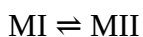
Variables / definitions: ssDNA = single-stranded DNA.

Interpretation: Binding equilibrium underlying RPA-mediated ATR recruitment and DNA-damage signaling.

Status / limitation: Conceptual biochemical equilibrium.

Project provenance: Phase 4

B.21 Rhodopsin conformational equilibrium



Variables / definitions: MI/MII = metarhodopsin I/II states.

Interpretation: Simplified photopigment conformational equilibrium influenced by membrane environment.

Status / limitation: Biochemical/biophysical representation.

Project provenance: Phase 4

B.22 Lipid-radical initiation



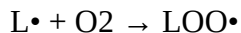
Variables / definitions: LH = lipid; $\text{L}^\bullet$ = lipid radical.

Interpretation: Simplified initiation step in lipid peroxidation.

Status / limitation: Biochemical representation.

Project provenance: Phase 4

B.23 Lipid-peroxyl propagation



Variables / definitions: $L\bullet$ = lipid radical; $LOO\bullet$ = lipid peroxyl radical.

Interpretation: Simplified oxygen-addition step in lipid peroxidation.

Status / limitation: Biochemical representation.

Project provenance: Phase 4

B.24 Methionine activation



Variables / definitions: Met = methionine; SAM = S-adenosylmethionine.

Interpretation: Simplified activation route generating methyl donor SAM.

Status / limitation: Biochemical representation.

Project provenance: Phase 4

B.25 SAM-dependent lysine methylation



Variables / definitions: SAH = S-adenosylhomocysteine.

Interpretation: Simplified methyltransferase reaction relevant to SMYD4.

Status / limitation: Biochemical representation.

Project provenance: Phase 4

B.26 Steady-state flux balance

$$dX/dt = J_{in} - J_{out}$$

Variables / definitions: X = pool/concentration; J_{in}/J_{out} = input/output fluxes.

Interpretation: At steady state $dX/dt = 0$ even when turnover is high; normal concentration does not prove normal flux.

Status / limitation: Quantitative definition/conceptual mass balance.

Project provenance: Phase 4

B.27 Functional-reserve condition

phenotype risk rises when $D(t) > C_{\text{max}}$

Variables / definitions: $D(t)$ = physiological demand at time t ; C_{max} = available system capacity.

Interpretation: Formalizes the reserve concept: phenotype may emerge when demand exceeds capacity.

Status / limitation: Conceptual model; not a calibrated probability law.

Project provenance: Phase 4

B.28 Light-injury function

$I_{\text{CNV}}(H) > I_{\text{WT}}(H)$ at matched H indicates greater vulnerability

Variables / definitions: $I(H)$ = measured injury; H = light/photon exposure.

Interpretation: A direct vulnerability phenotype requires greater injury in CNV versus WT at the same exposure.

Status / limitation: Operational definition for future testing.

Project provenance: Phase 4

B.29 Light-injury threshold

H_{crit} = exposure producing a predefined injury criterion; vulnerability if $H_{\text{crit,CNV}} < H_{\text{crit,WT}}$

Variables / definitions: H_{crit} = challenge dose at a prespecified injury endpoint.

Interpretation: Lower threshold in CNV would support greater photochemical vulnerability.

Status / limitation: Operational definition.

Project provenance: Phase 4

B.30 Neural response gain

$$G = \Delta \text{output} / \Delta \text{input}$$

Variables / definitions: Input/output must be explicitly defined and measured under matched conditions.

Interpretation: Sensory/neural amplification requires $G_{\text{CNV}} > G_{\text{WT}}$ for the same input domain.

Status / limitation: Operational quantitative definition.

Project provenance: Phase 4

B.31 Retinal temporal gain

$$G_{\text{R}}(f) = |R(f)| / |S(f)|$$

Variables / definitions: $R(f)$ = retinal output amplitude at frequency f ; $S(f)$ = stimulus amplitude at f .

Interpretation: Quantifies frequency-dependent retinal transfer gain.

Status / limitation: Operational quantitative definition for future flicker testing.

Project provenance: Phase 4

B.32 Serial bottleneck capacity

$$J_{\text{max}} \approx \min(C_1, C_2, \dots, C_n)$$

Variables / definitions: C_i = capacity of sequential processing stage i .

Interpretation: Represents a serial process whose throughput is limited by the weakest stage.

Status / limitation: Conceptual model; approximation.

Project provenance: Phase 4

B.33 Layered lesion burden

$$dL/dt = D - P - R$$

Variables / definitions: L = lesion/damage burden; D = damage generation; P = prevention/tolerance; R = repair capacity.

Interpretation: Qualitative reserve representation balancing damage and protective/reparative processes.

Status / limitation: Conceptual model; not a validated physiological differential equation.

Project provenance: Phase 4

B.34 Factorial interaction model

$$Y = \beta_0 + \beta_A \cdot A + \beta_B \cdot B + \beta_{AB} \cdot (A \times B) + \varepsilon$$

Variables / definitions: Y = outcome; A/B = perturbation indicators or doses; β_{AB} = interaction coefficient; ε = residual.

Interpretation: β_{AB} tests departure from the specified additive model; useful for testing non-additive interaction/synergy claims.

Status / limitation: Standard statistical model proposed for reproducible interaction testing.

Project provenance: Phase 4

B.35 Developmental timing model

$$P = f(D_{\text{early}}, D_{\text{late}}, D_{\text{adult}})$$

Variables / definitions: P = mature phenotype; D terms = dosage effects during developmental windows.

Interpretation: Decomposes mature phenotype conceptually into early, late, and adult contributions.

Status / limitation: Conceptual model.

Project provenance: Phase 4

B.36 Photosensitive-network threshold

$$G_N \times I > \Theta_E$$

Variables / definitions: I = visual input; G_N = network propagation gain; Θ_E = threshold for pathological synchronization.

Interpretation: Conceptual condition under which visually driven network activity could cross a pathological threshold.

Status / limitation: Conceptual model; not calibrated to the index CNV.

Project provenance: Phase 4

B.37 Locus-level evidence function

$$E_{i,k} = f(D_i, M_i, S_i, P_i, V_i, N_i, R_i, C_i)$$

Variables / definitions: D_i = dosage evidence; M_i = molecular perturbation; S_i = spatiotemporal opportunity; P_i = perturbation equivalence; V_i = visual relevance; N_i = neural/network relevance; R_i = rescue/causal specificity; C_i = contradiction penalty.

Interpretation: Defines support from locus i to mechanistic domain k without assuming simple additivity.

Status / limitation: Phase 5 modeling framework; f was intentionally not fixed to a biologically unjustified numerical law.

Project provenance: Phase 5

B.38 Domain-level evidence aggregation

$$F_k = g(E_{1,k}, E_{2,k}, \dots, E_{n,k})$$

Variables / definitions: $E_{i,k}$ = locus-level contributions to domain k .

Interpretation: Aggregates locus evidence while requiring attention to correlated evidence and double-counting.

Status / limitation: Phase 5 modeling framework; g may be weighted, gated, minimum-link, geometric, or other justified form.

Project provenance: Phase 5

B.39 Causal-layer sequence

$$G \rightarrow D \rightarrow M \rightarrow S \rightarrow F_R \rightarrow F_N \rightarrow F_C \rightarrow F_P$$

Variables / definitions: G = genomic disruption; D = dosage; M = molecular/cellular perturbation; S = spatiotemporal opportunity; F_R = retinal consequence; F_N = neural/circuit consequence; F_C = cortical/network transformation; F_P = photic phenotype relevance.

Interpretation: Represents evidence propagation through necessary biological layers.

Status / limitation: Phase 5 causal architecture.

Project provenance: Phase 5

B.40 Bottleneck score example

$$B = \min(G, D, M, S, F_R, F_N, F_C, F_P)$$

Variables / definitions: Terms are causal-layer support measures on a common defined scale.

Interpretation: Prevents strong early-layer evidence from compensating numerically for an absent necessary downstream layer.

Status / limitation: Candidate Phase 5 formulation; explicitly not to be adopted automatically.

Project provenance: Phase 5

B.41 Convergence matrix

K_{ij}

Variables / definitions: K_{ij} records evidence that loci i and j converge biologically and the reason for the edge.

Interpretation: Encodes pairwise convergence while distinguishing pathway co-membership, direct interaction, sequential dependency, and demonstrated interaction.

Status / limitation: Phase 5 model object; not inherently a causal or synergy matrix.

Project provenance: Phase 5

B.42 Directional consistency coding

$\text{direction} \in \{+1, 0, -1\}$

Variables / definitions: +1 = supports proposed mechanism; 0 = neutral/noninformative; -1 = contradicts.

Interpretation: Prevents counting any phenotype as supportive merely because a perturbation produces an effect.

Status / limitation: Phase 5 deterministic coding rule.

Project provenance: Phase 5

B.43 Ordinal evidence example

0=no relevant evidence; 1=weak/indirect; 2=moderate; 3=strong/direct; 4=replicated/direct/rescue-supported

Variables / definitions: Scale is illustrative unless formally frozen in the implementation.

Interpretation: Provides an interpretable ordinal mapping when continuous quantification is not justified.

Status / limitation: Phase 5 example scale; ordinal score is not a biological effect size or probability.

Project provenance: Phase 5

B.44 Empirical null-tail estimate (canonical)

$$p_{\text{emp}} = (1 + \#\{T_{\text{null}} \geq T_{\text{index}}\}) / (N_{\text{null}} + 1)$$

Variables / definitions: T_{index} = index-interval statistic; T_{null} = same statistic in matched null intervals; N_{null} = number of null intervals.

Interpretation: A standard way to quantify how extreme the index statistic is versus a simulated/matched null distribution.

Status / limitation: Canonical formula included for reader clarity because Phase 5 calls for empirical enrichment statistics but the accessible frozen protocol does not preserve a unique implemented formula.

Project provenance: Phase 5

B.45 Monte Carlo standard error for an estimated probability (canonical)

$$\text{MCSE}(\hat{p}) = \sqrt{\hat{p}(1-\hat{p})/N}$$

Variables / definitions: $\hat{p}$ = simulated proportion; N = number of Monte Carlo replicates.

Interpretation: Shows why 20,000 replicates per scenario/cell can make stochastic sampling error small relative to biological uncertainty.

Status / limitation: Canonical simulation formula supplied for reader clarity; the Phase 5 summary reports replicate counts but not a stored MCSE equation.

Project provenance: Phase 5

Appendix C — Variable and Symbol Index

Symbol	Meaning in this project
A, B	Perturbation indicators or doses in the factorial interaction model.
ADP	Adenosine diphosphate.
ATP	Adenosine triphosphate.
B	Example Phase 5 bottleneck score.
C _i	Capacity of sequential process stage i.
C _i (Phase 5 locus vector)	Counterevidence/contradiction penalty for locus i; context distinguishes this from capacity C _i .
C _{max}	Available/maximal biological capacity in the functional-reserve condition.
C _s , C _e	Inclusive start/end coordinates of a CDS segment.
D	Damage generation in the layered-lesion equation; elsewhere “dosage consequence” in the Phase 5 causal chain.
D _i	Dosage evidence for locus i in the Phase 5 evidence vector.
D _s , D _e	Inclusive start/end coordinates of the deletion.
D _{early} , D _{late} , D _{adult}	Conceptual dosage effects in different developmental windows.
D(t)	Physiological demand as a function of time.

$E_{i,k}$	Evidence support contributed by locus i to mechanistic domain k .
ε	Residual/error term in the factorial interaction model.
F_C	Cortical/network transformation evidence in the Phase 5 causal chain.
F_D	Damage-response reserve domain.
F_E	Epileptogenic containment reserve domain.
F_k	Aggregated evidence for mechanistic domain k .
F_M	Membrane/endosomal handling reserve domain.
F_N	Neuronal developmental/network reserve; in the causal chain, neural/circuit consequence.
F_P	Photic phenotype relevance.
F_R	Retinal/photoreceptor reserve or retinal/visual-system consequence, depending on context.
f	Generic mapping function; also used as stimulus frequency when written $G_R(f)$.
G	Genomic disruption in the Phase 5 causal chain; response gain in the Phase 4 gain definition, depending on context.
G_N	Network propagation gain.
$G_R(f)$	Retinal temporal gain at stimulus frequency f .
H	Photon/light exposure or dose.

H_crit	Exposure producing a predefined injury criterion.
I	Visual input in the photosensitive-network threshold; I(H) denotes measured injury as a function of exposure H in the light-injury model.
J_in, J_out	Input and output fluxes in the steady-state balance equation.
J_max	Approximate maximum throughput of a serial/bottleneck system.
K_ij	Pairwise convergence-matrix entry for loci i and j.
L	Nominal genomic interval length or lesion burden when written L(t), depending on context.
L_CDS	Total CDS length.
L_deleted	Total coding length overlapped/deleted.
L_overlap	Coding overlap length for one segment.
m	Number of hypothesis tests in BH-FDR notation.
M	Molecular/cellular perturbation in the causal chain.
M_i	Molecular perturbation evidence for locus i.
N_i	Neural/network relevance for locus i.
N_null	Number of null simulations/intervals in an empirical-tail estimate.
P	Prevention/tolerance in $dL/dt = D - P - R$; also mature phenotype in the developmental timing model; context

	is required.
P_i	Perturbation-equivalence evidence/weight for locus i .
$\hat{p}$	Estimated probability/proportion from Monte Carlo simulations.
Q	Target false discovery rate in BH procedure.
q	FDR-adjusted significance value.
R	Repair capacity in the lesion-burden equation; $R(f)$ = retinal response amplitude at frequency f .
R_i	Rescue/causal-specificity evidence for locus i .
S	Spatiotemporal opportunity in the causal chain; $S(f)$ = stimulus amplitude at frequency f .
S_i	Spatiotemporal opportunity for locus i .
SD_g	Within-gene standard deviation across developmental stages.
T_{index}	Index-CNV test/convergence statistic in a null comparison.
T_{null}	Null-interval statistic generated under the same pipeline.
V_i	Visual-system relevance for locus i .
X	Generic molecular pool/concentration in the flux-balance equation.
Y	Outcome in the factorial interaction model.
β_0	Intercept in the factorial model.

β_A, β_B	Main-effect coefficients for perturbations A and B.
β_{AB}	Interaction coefficient; tests departure from the specified additive model.
Θ_E	Threshold for pathological synchronization in the conceptual photosensitive-network model.
ρ_s	Spearman rank-correlation coefficient.
Σ	Summation across indexed segments/loci.
Δ	Change in a measured quantity.

Appendix D — Interpretation and Reproducibility Rules

Genomic deletion does not by itself establish RNA, protein, or functional dosage.

Expression establishes biological opportunity, not dysfunction or causal effect.

Knockout, biallelic loss, hypomorph, pathogenic missense/dominant-negative/gain-of-function states, and pharmacologic inhibition are not interchangeable with heterozygous deletion.

Coexpression, pathway membership, network proximity, sequential dependence, biochemical interaction, genetic interaction, epistasis, and synergy are distinct evidence classes.

An ordinal evidence score is not a biological effect size, probability, or percent causality.

Missing evidence must remain missing. It must not be recoded as zero merely to complete a matrix.

Zero should represent a defined null result or evidence of absence under stated conditions.

Conceptual equations require independent calibration data before they can be interpreted quantitatively.

Frequency scans and high-dimensional tests require prespecified correction for multiple comparisons.

Cells from the same clone/donor are not independent biological replicates unless the experimental design explicitly supports that assumption.

Null findings, contradictory effects, and failed hypotheses remain part of the frozen record.

GRCh37/hg19 remains the controlling CNV coordinate system unless the project is formally versioned; GRCh38 annotations require documented conversion and cannot silently replace the fixed interval.

The project's strongest current interpretation is a candidate distributed visual-neural functional-reserve lesion; direct causation of PPR or visually triggered seizures by the exact heterozygous CNV remains unresolved.

Appendix E — Provenance Note

This appendix was compiled from the frozen Phase 1 structural report, Phase 2 molecular synthesis, Phase 3 expression/spatiotemporal research records, Phase 4 frozen master report, the Phase 5 deterministic modeling protocol and frozen summary available in the project record, and the Phase 6 frozen conclusion. It does not reopen prior conclusions. Where a canonical formula was added because the frozen record named a standard method but did not print its equation (for example Spearman correlation, Benjamini-Hochberg correction, within-gene z-score, empirical null-tail probability, or Monte Carlo standard error), the entry is explicitly labeled as an explanatory mathematical definition rather than a recovered hidden calculation.

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Consolidated APA 7-style bibliography

Bibliographic Note

This bibliography consolidates the source lists preserved across Phase 1; Phase 2; the Phase 3 research-through-3D, abstract/methods, and integrated expression/spatiotemporal records; Phase 4; the available frozen Phase 5 summary; and Phase 6. Clear duplicates were merged by DOI, URL, or title where the identity was unambiguous. The available Phase 5 casual-reader summary contains no standalone Works Cited section, so no unique Phase 5 external reference has been invented; its scientific evidence base is inherited from the earlier frozen phases and the Phase 5 evidence model. Several frozen reports preserved author fields as “et al.” rather than complete author lists. Those omissions are retained rather than reconstructed from outside sources. Accordingly, the entries below follow APA 7 ordering, date placement, retrieval-date conventions, and hanging-indent layout to the extent permitted by the metadata preserved in the project record.

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